

# Interpretation of Electrolyte and Acid-Base Parameters in Blood and Urine

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## CHAPTER OUTLINE

WATER AND SODIUM, 758  
POTASSIUM, 772

METABOLIC ALKALOSIS, 783  
METABOLIC ACIDOSIS, 786

An analysis of laboratory data from samples of blood and urine is essential to make accurate diagnoses and to design optimal therapy for patients with disturbances of water, sodium ( $\text{Na}^+$ ), potassium ( $\text{K}^+$ ), and acid-base homeostasis.<sup>1</sup> Our clinical approach and interpretation of these tests rely heavily on an understanding of the basic concepts of the physiology of the renal regulation of water, electrolyte, and acid-base homeostasis. Hence, each section in this chapter begins with a discussion of physiologic concepts that help focus on the important factor(s) in the regulation of the homeostasis of the substances in question.

Based on this understanding of physiology, we discuss the laboratory data used to help determine the underlying pathophysiology of the disturbance. This information is then used to construct our approach to patients with each of these disorders. At the end of each section, clinical cases are presented succinctly to illustrate how this approach is used at the bedside.

We emphasize that there are no normal values for the urinary excretion of water and electrolytes because subjects in the steady state excrete all ions that are consumed and not lost by nonrenal routes. Hence, data should be interpreted in the context of the prevailing stimulus and the expected renal response.

## WATER AND SODIUM

In this section, we illustrate how we use information about the volume and composition of the urine in the clinical approach to patients with disorders causing polyuria, those causing a decreased effective arterial blood volume (EABV), and those causing hyponatremia.

## POLYURIA

There are two definitions of polyuria. The conventional definition of polyuria is a urine volume that is more than

2.5 L/day. This is an arbitrary definition based on comparing the 24-hour urine volume in the patient with the usual values of 24-hour urine volumes observed in individuals who consume a typical Western diet.

We prefer a physiology-based definition. Polyuria is present if the urine flow rate is higher than what is expected in a specific setting. In the presence of vasopressin action, urine volume is determined by the rate of excretion of effective osmoles and the effective osmolality of the medullary interstitial compartment. Hence, polyuria is present if the urine volume is higher than what is expected for the rate of excretion of effective osmoles, even if the urine volume does not exceed 2.5 L/day. In contrast, if the concentration of  $\text{Na}^+$  in plasma ( $\text{P}_{\text{Na}}$ ) is less than 135 mmol/L, the release of vasopressin should be inhibited, and the urine flow rate could be as high as 10 to 15 mL/min in an adult subject (which extrapolates to around 14 to 22 L/day). A lower urine volume in this setting, although still higher than 2.5 L/day, represents oliguria and not polyuria.

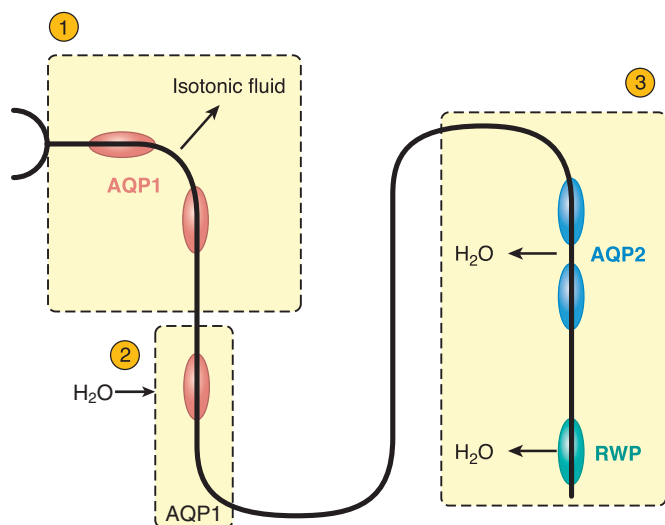
There are two categories of polyuria, a water diuresis and an osmotic diuresis.

## WATER DIURESIS

### Concept 1

To move water across a membrane, there must be a channel that allows water to cross that membrane (an aquaporin [AQP]) and a driving force for the movement of water—a difference in the concentrations of effective osmoles or a difference in the hydrostatic pressures across that membrane.

**Water Channels.** There are two critically important aquaporin water channels in the luminal membranes of cells in the kidney, AQP1 and AQP2 (Fig. 24.1). AQP1 channels are nonregulated water channels that are constitutively present in the luminal membrane of proximal tubule (PT) cells. AQP1 channels are also constitutively present in the luminal membrane of the descending thin limbs of the loop of Henle



**Fig. 24.1** Functional units of the nephron based on presence of aquaporins AQP1 and AQP2. The solid line represents a nephron; AQP1 is represented as a small pink ovals and AQP2 as blue ovals. With regard to water excretion, we divide the nephron into three functional units (delineated by rectangles with dashed lines) on the basis of the presence of AQP1 or AQP2. AQP1 is always present in the proximal tubule cells—the first functional nephron unit (1). AQP1 is also present in the descending thin limb of the loop of Henle (DtL) of the juxtamedullary nephrons, which constitute about 15% of the total nephrons—the second functional nephron unit (2). Vasopressin causes the insertion of AQP2 into the luminal membrane of principal cells in the cortical collecting duct (CCD) and the medullary collecting duct (MCD); together, these nephron segments constitute the final or third functional unit (3). Even in the absence of vasopressin, there seems to be a small degree of water permeability in the inner MCD, which is called residual water permeability (RWP). (Modified from Halperin ML, Kamel KS, Goldstein MB. *Fluid, Electrolyte, and Acid-Base Physiology: A Problem-Based Approach*. ed 4. Philadelphia: Elsevier; 2012.)

(DtLs) of the juxtamedullary nephrons, which constitute about 15% of the total number of nephrons. Of note, AQP1 channels are not present in the luminal membrane of the DtLs of the superficial nephrons; thus, the entire loop of Henle of these nephrons (i.e., 85% of all the nephrons) is relatively impermeable to water.<sup>2</sup>

Vasopressin causes the insertion of AQP2 channels into the luminal membrane of principal cells in the cortical collecting duct (CCD), and the medullary collecting duct (MCD). Notwithstanding, even in the absence of vasopressin action, there seems to be a small degree of water permeability in the inner MCD, called basal or residual water permeability (RWP).

**Driving Force.** Water is drawn from a compartment with a lower effective osmolality to one with a higher effective osmolality. Effective osmoles are osmoles that have a difference in their concentrations between two compartments. The magnitude of this driving force could be very large, because a difference of 1 mOsm/kg H<sub>2</sub>O generates a pressure of 19.3 mm Hg. Vasopressin causes the insertion of AQP2 in the luminal membrane of principal cells in the CCD and MCD. Hence, as soon as water reaches a nephron segment that has AQP2 in the luminal membrane of its cells, water will be reabsorbed until the effective osmolality in the luminal

fluid is nearly equal to the effective osmolality in the surrounding interstitial fluid compartment.

Because vasopressin causes the insertion of urea transporters in the luminal membrane of cells in the inner MCD, the concentration of urea in the interstitial fluid in the inner medulla becomes nearly equal to its concentration in the lumen in the inner MCD. Hence, at the usual rate of urea excretion, which does not exceed the transport capacity of the urea transporters, urea is not an effective urine osmole (i.e., the excretion of urea does not oblige the excretion of H<sub>2</sub>O).

### Concept 2

The urine volume during a water diuresis is determined by the volume of distal delivery of filtrate and the volume of filtrate that is reabsorbed in the inner MCD via its RWP.<sup>3</sup>

**Distal Delivery of Filtrate.** The volume of distal delivery of filtrate is the volume of glomerular filtration minus the volume of filtrate that is reabsorbed in the nephron segments prior to the CCD. There are data to suggest that AQP1 channels are not present in the luminal membranes of the DtLs of the superficial nephrons, which constitute 85% of the total number of nephrons.<sup>2</sup> If that is the case, then the entire loop of Henle of most of the nephrons is impermeable to water. Hence, the volume of distal delivery of filtrate should be approximately equal to the volume of glomerular filtration minus the volume that is reabsorbed by PT.

It was thought that about 66% of the glomerular filtration rate (GFR) is reabsorbed along the entire PT. This was based on the measured ratio of the concentration of inulin in fluid samples obtained from the lumen of the PT (TF) and its concentration in simultaneous plasma (P) samples— $(TF/P)_{inulin}$ —in micropuncture studies in rats. Because inulin is freely filtered at the glomerulus and is not reabsorbed or secreted in the PT, a  $(TF/P)_{inulin}$  value of around 3 has suggested that approximately 66% of the filtrate is reabsorbed in the PT. However, these micropuncture measurements underestimate the volume of fluid that is actually reabsorbed in the PT because these measurements were made at the last accessible portion of the PT at the surface of the renal cortex, and hence did not take into account that additional volume may be reabsorbed in the deeper part of the PT, including its pars recta portion.

If the entire loop of Henle of most nephrons can be assumed to lack AQP1 and thereby be largely impermeable to water, the volume of filtrate that enters the loop of Henle can be deduced from the minimum value for the  $(TF/P)_{inulin}$  obtained using the micropuncture technique from the early distal convoluted tubule (DCT), which has been done in rats. Because this value is around 6, a reasonable estimate of the proportion of filtrate that is reabsorbed in the rat PTs is close to five-sixths (83%). This value is close to the estimate of fractional reabsorption in the PT obtained with measurement of lithium clearance, which is thought to be a marker for fractional reabsorption in PT in human subjects.

If these findings can be extrapolated to humans with GFR values of 180 L/day, only about 30 L of filtrate/day ( $180 \div 6$ ) would be delivered to the early DCT if all nephrons were superficial nephrons. This value of the volume of filtrate delivered to the DCT needs to be adjusted downward because juxtamedullary nephrons have AQP1 along their DtLs and,

hence, are permeable to water. If these nephrons constitute 15% of the total number of nephrons and therefore receive 27 L of glomerular filtrate/day (15% of 180 L/day), and if five-sixths of the glomerular filtrate of these nephrons is reabsorbed along their PTs, around 4.5 L/day reach their DtLs. Because the interstitial osmolality rises threefold (from 300 to 900 mOsmol/kg H<sub>2</sub>O) in the outer medulla, two-thirds, or 3 L of the 4.5 L/day, are reabsorbed in the DtLs of these nephrons. Therefore, the volume of filtrate delivered to DCTs is likely to be around 27 L/day (around 30 L/day exit the PTs minus around 3 L/day that are reabsorbed in DtLs of the juxtamedullary nephrons).

**Residual Water Permeability.** There are two pathways for transport of water in the inner MCD, a vasopressin-responsive system via AQP2 and a vasopressin-independent system, RWP. Two factors may affect the volume of water reabsorbed via RWP. First, the driving force is the enormous difference in osmotic pressure between the luminal and interstitial fluid compartments in the inner MCD during a water diuresis. Second, the contraction of the renal pelvis, because each time the renal pelvis contracts, some of the fluid in the renal pelvis travels in a retrograde direction up toward the inner MCD, and some of that fluid may be reabsorbed via RWP after it reenters the inner MCD for a second (or maybe a third) time.

As calculated above, around 27 L/day are delivered to the distal nephron in a normal subject. The observed urine flow rate during maximum water diuresis is around 10 to 15 mL/min (around 14–22 L/day). If this maximum water diuresis could be maintained for 24 hours, then somewhat more than 5 L of water would be reabsorbed per day in the inner MCD via its RWP during water diuresis.

### Concept 3

Another component of the physiology of water diuresis is the formation of electrolyte-free water. This process of desalination occurs in nephron segments that can reabsorb Na<sup>+</sup> but are impermeable to water because they lack AQPs (i.e., the cortical and medullary thick ascending limbs (TALs) of the loop of Henle and the DCT).

Regulation of the reabsorption of Na<sup>+</sup> and Cl<sup>−</sup> in the medullary TAL (mTAL) seems to occur via dilution of the concentration of an inhibitor of this process in the medullary interstitial compartment (possibly ionized calcium) by water reabsorption from the water-permeable nephron segments in the renal medulla (i.e., the MCD and DtLs of the juxtamedullary nephrons).<sup>4</sup> During water diuresis, reabsorption of H<sub>2</sub>O in inner MCD via its RWP may serve the purpose of diluting the concentration of this inhibitor of the reabsorption of Na<sup>+</sup> and Cl<sup>−</sup> in the mTAL, hence allowing desalination of the luminal fluid and the formation of electrolyte-free water to occur in this nephron segment.

### Tools for Assessing Water Diuresis

**Urine Flow Rate.** The observed urine flow rate during peak water diuresis in normal adult subjects is around 10 to 15 mL/min (extrapolated to around 14–22 L/day). Notwithstanding, this high urine flow rate will not be sustained because the volume of distal delivery of filtrate will ultimately fall.

The urine flow rate declines when desmopressin (DDAVP) is given to a patient with central diabetes insipidus (DI). The urine flow rate, however, will be higher than that observed

in response to DDAVP in a normal subject who consumes a typical Western diet. The reason is that the medullary interstitial osmolality is likely to be lower owing to a prior medullary washout during the water diuresis.

**Osmole Excretion Rate.** The osmole excretion rate is equal to the product of the urine osmolality (U<sub>osm</sub>) and the urine flow rate (see Eq. 24.1). In subjects eating a typical Western diet, the rate of excretion of osmoles is 600 to 900 mOsmol/day, with electrolytes and urea each accounting for close to half of the total urine osmoles. During a water diuresis due to the absence of vasopressin action, the rate of excretion of osmoles does not directly affect the urine volume because AQP2 are not present in the luminal membrane of principal cells in the CCD and MCD. Nevertheless, the rate of excretion of osmoles must be calculated in the patient with a water diuresis because, if high, polyuria due to an osmotic diuresis may be unmasked after there is a renal response to the administration of DDAVP.

$$\text{Osmole excretion rate} = U_{\text{osm}} \times \text{urine flow} \quad (24.1)$$

**Urine Osmolality.** The U<sub>osm</sub> is equal to the number of excreted osmoles divided by the urine volume. Therefore during a water diuresis, a change in the U<sub>osm</sub> could reflect a change in the osmole excretion rate and/or in the volume of filtrate delivered to the distal nephron, which is the major factor that determines the urine volume during a water diuresis. For example, if the rate of excretion of osmoles is 800 mOsmol/day, the U<sub>osm</sub> will be 50 mOsmol/kg H<sub>2</sub>O if the 24-hour urine volume is 16 L and 100 mOsmol/kg H<sub>2</sub>O if the 24-hour urine volume is 8 L. This higher urine osmolality does not reflect a better urine concentrating ability, but a lower volume of distal delivery of filtrate.

**Electrolyte-Free Water Balance.** Tonicity is a term used to describe the effective osmolality of a solution. For plasma water, this can be approximated (in the absence of hyperglycemia) by the total concentration of cations (Na<sup>+</sup> + K<sup>+</sup>) and their accompanying anions. The basis of the calculation of electrolyte-free water balance is the determination of how much water needs to be added to or subtracted from a solution with a given total amount of Na<sup>+</sup> + K<sup>+</sup> to make its tonicity equal to the normal plasma tonicity—that is, 150 mmol of cations plus an equal concentration of anions in 1 L of water. This calculation is used to determine the basis of a change in plasma Na<sup>+</sup> concentration (P<sub>Na</sub>), and the appropriate therapy to achieve normonatremia. To calculate the electrolyte-free water balance, one must know the volume and concentrations of Na<sup>+</sup> + K<sup>+</sup> in the input and the output (mainly the urine).

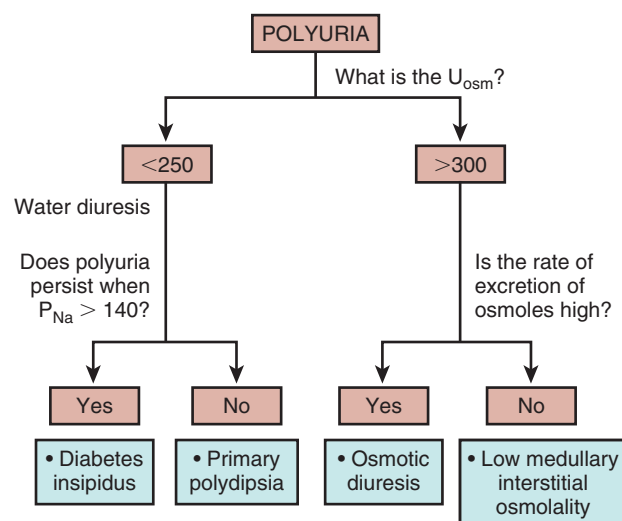
For example, consider a patient who receives 3 L of 0.9% saline (Na<sup>+</sup> concentration = 150 mmol/L) and excretes 3 L of urine with a concentration of Na<sup>+</sup> + K<sup>+</sup> of 50 mmol/L. There is no electrolyte-free water in the input because it has the same tonicity as the plasma water (150 mmol/L cations, plus an equal concentration of anions). With regard to the output, this patient excreted the equivalent of 1 L of isotonic salt solution and 2 L of electrolyte-free water. Hence, the patient has a negative electrolyte-free water balance of 2 L, and the P<sub>Na</sub> would be expected to rise. Another patient receives 3 L of 0.9% saline but excretes 3 L of urine with a concentration of Na<sup>+</sup> + K<sup>+</sup> of 200 mmol/L. As in the first example,

**Table 24.1** Comparison of a Tonicity Balance and Electrolyte-Free Water Balance in a Patient With Hypernatremia<sup>a</sup>

| Parameter                                 | Na <sup>+</sup> + K <sup>+</sup> (mmol) | Water (L) | Electrolyte-Free Water Balance (L) |
|-------------------------------------------|-----------------------------------------|-----------|------------------------------------|
| <b>Infusion of 3 L of Isotonic Saline</b> |                                         |           |                                    |
| • Input                                   | 450                                     | 3         | 0                                  |
| • Output                                  | 150                                     | 3         | 2                                  |
| • Balance                                 | +300                                    | 0         | -2                                 |
| <b>Infusion of 4 L of Isotonic Saline</b> |                                         |           |                                    |
| • Input                                   | 600                                     | 4         | 0                                  |
| • Output                                  | 150                                     | 3         | 2                                  |
| • Balance                                 | +450                                    | +1        | -2                                 |
| <b>No Intravenous Fluid Infusion</b>      |                                         |           |                                    |
| • Input                                   | 0                                       | 0         | 0                                  |
| • Output                                  | 150                                     | 3         | 2                                  |
| • Balance                                 | -150                                    | -3        | -2                                 |

<sup>a</sup>Three case examples are described, in each patient the  $P_{Na}$  rose from 140 to 150 mmol/L. In all three examples, the urine volume was 3 L with a concentration of  $Na^+ + K^+$  of 50 mmol/L. The only difference is the volume of isotonic saline infused in each case example. Note that although the balances for  $Na^+ + K^+$  and for water are very different in these three examples, the calculation of electrolyte-free water balance shows a negative balance of 2 L in all of them. Calculation of tonicity balance however, reveals that the patient in the first example has a positive balance of 300 mmol of  $Na^+ + K^+$ , while the patient in the second example has a positive balance of 450 mmol of  $Na^+ + K^+$  and a positive balance of 1 L of water, and the patient in the third example has a negative balance of 150 mmol of  $Na^+ + K^+$  and a negative balance of 3 L of water. The goals for therapy—to correct the hypernatremia and to return the volume and composition of the extracellular and intracellular fluid compartments to their normal values—are clear only after a tonicity balance is calculated.

there is no electrolyte-free water in the input. With regard to the output, because this patient would have needed to excrete 4 L (and not 3 L) of urine to make this total of 600 mmol of  $Na^+ + K^+$  (with 600 mmol of accompanying anions) into an isotonic solution, the patient has a positive electrolyte-free water balance of 1 L; hence, the  $P_{Na}$  will fall (i.e., 1 L of body water lost 150 mmol of  $Na^+ + K^+$  with 150 mmol of accompanying anions and hence, became 1 L electrolyte-free water). Calculation of the electrolyte-free water balance however, although it correctly predicts the change in  $P_{Na}$ , does not reveal whether its basis is a change in water balance or in fact a change in  $Na^+ + K^+$  balance. An example of this is shown in Table 24.1. Although the balances for  $Na^+ + K^+$  and for water are very different in the three examples used, calculation of the electrolyte-free water balance provided the same answer in all of them, a negative balance of 2 L of electrolyte-free water. Hence, calculation of the electrolyte-free water balance is not helpful to design the therapy needed to return the volume and composition of the extracellular fluid (ECF) and intracellular fluid (ICF) compartments to their normal values.



**Flow Chart 24.1** (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

**Tonicity Balance.** To determine the basis for a change in the  $P_{Na}$  and to define the proper therapy to return the volume and composition of the ECF and ICF compartments to their normal values, we calculate the tonicity balance. A tonicity balance refers to the balance of water and the balance of electrolytes that determine body tonicity (i.e.,  $Na^+ + K^+$ ).<sup>5</sup> To calculate a tonicity balance, one must examine the input and output volumes and the quantity of  $Na^+$  and  $K^+$  infused and the quantity of  $Na^+ + K^+$  excreted over the time period during which the  $P_{Na}$  changed (see Table 24.1). In practical terms, a tonicity balance can be calculated only in a hospital setting, where inputs and outputs are accurately recorded. In a febrile patient, balance calculations will not be as accurate because sweat losses are not measured. Nevertheless, restricting analysis of the output to the urine data would be sufficient in an acute setting if the changes in  $P_{Na}$  occurred over a relatively short period of time.

Even if measurements of the concentrations of  $Na^+$  and  $K^+$  in the urine are not available, if the  $P_{Na}$  at the beginning and end of a certain period, volume of the urine, and volume and amount of  $Na^+ + K^+$  in the fluid infused during that period are known, the clinician can use these data together with an estimation of total body water (TBW) to calculate the quantity of  $Na^+ + K^+$  excreted in the urine. Hence, the basis for the change in  $P_{Na}$  can be determined.

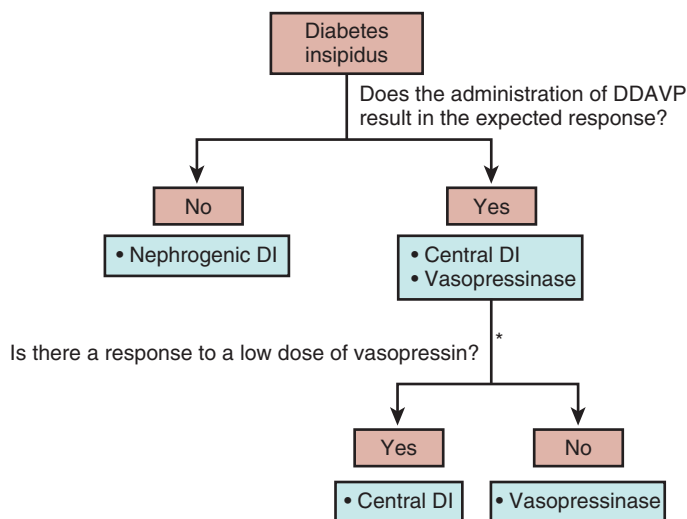
### CLINICAL APPROACH TO THE PATIENT WITH POLYURIA

The steps to take to determine the diagnosis in a patient with polyuria are outlined in Flow Charts 24.1 and 24.2.

#### Step 1: What Is the Urine Osmolality?

A value of the  $U_{osc}$  that is less than 250 mOsmol/kg  $H_2O$  indicates that the polyuria is due to a water diuresis. If the  $P_{Na}$  is less than 135 mmol/L, the polyuria is due to primary polydipsia. Once the  $P_{Na}$  has returned to the normal range, the urine flow rate should decrease, and the  $U_{osc}$  should rise to a value that is at least higher than the  $P_{osc}$ . Note that the prior water diuresis may have caused a lower





\*If not contraindicated due to pregnancy.

**Flow Chart 24.2** DI, Diabetes insipidus; DDAVP, desmopressin. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

medullary interstitial osmolality due to medullary washout, so the maximal urine-concentrating ability may not be achieved initially. In contrast, if the water diuresis persists when the  $P_{Na}$  is higher than 140 mmol/L, the cause of polyuria is DI.

If the  $U_{osm}$  is greater than 300 mOsmol/kg  $H_2O$ , calculate the osmole excretion rate (see Eq. 24.1). The usual value is close to 0.6 mOsmol/min (600–900 mOsmol/day) in subjects consuming a typical Western diet. If the osmole excretion rate is appreciably increased, then the polyuria is due to an osmotic diuresis. Conversely, if the osmole excretion rate is not increased, then the cause of the polyuria is a defect that leads to a lower medullary interstitial osmolality.

### Step 2: Examine the Renal Response to Vasopressin or Desmopressin

See Flow Chart 24.2.

If the cause of the polyuria is DI, this could be due to a lesion in the hypothalamic–posterior pituitary axis, which controls the production and release of vasopressin (central DI), the presence of a circulating vasopressinase that breaks down vasopressin, or a renal lesion that prevents the binding of vasopressin to its V2 receptor (V2R) or interferes with vasopressin effect to cause the insertion of AQP2 in the luminal membrane of principal cells in the CCD and MCD (nephrogenic DI).

In the diagnostic approach to the patient with a water diuresis, DDAVP should only be administered if the patient continues to have a water diuresis, despite  $P_{Na}$  greater than 140 mmol/L. In normal subjects consuming a typical Western diet, the expected urine flow rate in response to the administration of DDAVP is around 0.5 mL/min. This is because the inner medullary interstitial effective osmolality (nonurea osmolality) is around 450 mOsmol/kg  $H_2O$ , and the rate of excretion of effective osmoles (electrolytes) is around 450 mOsmol/day; hence, the expected urine flow rate is  $450/450 = 1$  L/day, or 0.7 mL/min. In a patient with central DI, the urine flow rate is expected to fall in response to the administration of DDAVP but, because of medullary

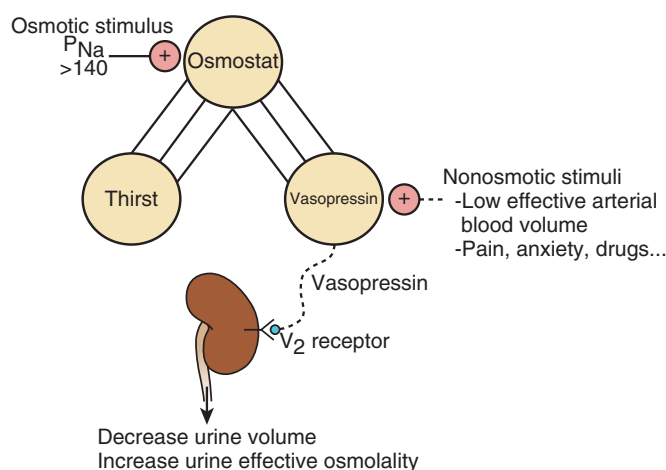
washout by the water diuresis, the urine flow rate is likely to be higher than 0.7 mL/min, perhaps around 1 mL/min if the effective inner medullary interstitial osmolality has fallen to say around 300 mOsmol/kg  $H_2O$ . The urine flow rate will be even higher if there is an appreciable increase in the rate of excretion of effective osmoles (e.g., because of prior expansion of the effective arterial blood volume with excessive administration of saline). In a patient with central DI, the  $U_{osm}$  should rise to a value that is at least higher than  $P_{osm}$  in response to the administration of DDAVP.

Similar responses in terms of a fall in urine flow rate and a rise in  $U_{osm}$  are also observed in patients with water diuresis caused by hydrolysis of vasopressin by circulating vasopressinase(s), because DDAVP is resistant to degradation by these enzymes. One could, in theory, examine the response to the administration of vasopressin after the effect of DDAVP has worn off, and the patient is having a water diuresis again. In contrast to the response to the administration of DDAVP, a patient in whom a vasopressinase has been released will not respond to the administration of a small dose of vasopressin. However, this should not be attempted in pregnant women because of the oxytocic effect of vasopressin.

If the urine osmolality and flow rate do not respond appropriately to DDAVP, then the diagnosis is nephrogenic diabetes insipidus.

### Step 3: Establish the Basis for Central Diabetes Insipidus

Central DI results from decreased release of vasopressin because of an inborn error (hereditary central DI) or because of a variety of infiltrative, neoplastic, vascular, or traumatic lesions that may involve the osmostat, the site where vasopressin is synthesized (the hypothalamic paraventricular and supraoptic nuclei), the tracts connecting these nuclei to the posterior pituitary, and/or a lesion involving the posterior pituitary (Fig. 24.2; see also Chapter 15). In a patient with hypernatremia, the absence of thirst suggests that the lesion involves the osmostat.



**Fig. 24.2** Water control system. The sensor is the osmostat or the tonicity stat (top circle), which detects a change in  $P_{Na}$  via an effect on the volume of its cells. The osmostat is linked to the thirst center (left circle) and to the vasopressin release center (right circle). Non-osmotic stimuli (e.g., certain drugs, nausea, pain, anxiety) also influence the release of vasopressin. Vasopressin release is also stimulated when there is a large decrease in the effective arterial blood volume. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

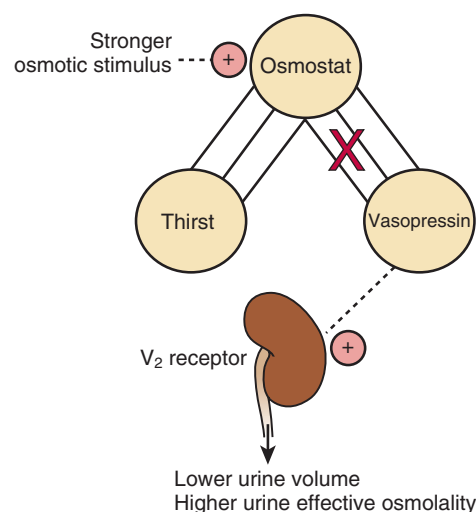
#### Step 4: Establish the Basis for Nephrogenic Diabetes Insipidus

If DDAVP fails to cause a rise in  $U_{osm}$  to a value that is equal to or higher than  $P_{osm}$ , the diagnosis is nephrogenic DI. Nephrogenic DI results from a lesion that prevents the binding of vasopressin to  $V_2R$  or one that interferes with its effect to cause the insertion of AQP2 into the luminal membrane of principal cells in the collecting ducts. Hereditary nephrogenic DI could be due to an X-linked recessive  $V_2R$  mutation (more common), or autosomal recessive or dominant AQP2 mutations. The most common cause of nonhereditary nephrogenic DI in an adult is the ingestion of lithium (see Chapter 15).

Patients with classic nephrogenic DI need to be distinguished from patients with a defect that leads to a lower medullary interstitial osmolality (e.g., due to a medullary interstitial disease). From a pathophysiologic perspective, these are two different disorders. Furthermore, in patients with classic nephrogenic DI, the urine flow rate is largely determined by the volume of distal delivery of filtrate and the volume reabsorbed via RWP in the inner MCD. In contrast, in patients with low medullary interstitial osmolality, the urine flow rate is determined by the rate of excretion of effective osmoles.

#### CLINICAL CASE 1: WHAT IS “PARTIAL” ABOUT PARTIAL CENTRAL DIABETES INSIPIDUS?

A 32-year-old previously healthy man had a recent basal skull fracture. Since his head injury, his urine output had been consistently about 4 L/day, and his  $U_{osm}$  around 200 mOsmol/kg  $H_2O$  in multiple 24-hour urine collections. Vasopressin was not detected in his plasma when his  $P_{Na}$  was around 140 mmol/L. During the daytime, his  $U_{osm}$  was consistently around 90 mOsmol/kg  $H_2O$ , and his  $P_{Na}$  was around 137 mmol/L. When he was given DDAVP, his urine flow rate decreased to 0.5 mL/min, and the  $U_{osm}$  rose to 900 mOsmol/kg  $H_2O$ . Interestingly, it was noted that if he stopped drinking water after supper, his sleep was not interrupted by a need



**Fig. 24.3** Lesion causing partial central diabetes insipidus. The top circle represents the osmostat or the tonicity stat, which is the sensor, the left circle represents the thirst center, and the right circle is the vasopressin release center. The X symbol represents a hypothetical lesion that causes the severing of some but not all the fibers connecting the osmostat to the vasopressin release center. A stronger osmotic stimulus can overcome the defect and stimulate sufficient vasopressin release to concentrate the urine. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

to void, and his  $U_{osm}$  was around 425 mOsmol/kg  $H_2O$  in his first voided urine samples in the morning. His urine flow rate fell to 0.5 mL/min, and his  $U_{osm}$  rose to 900 mOsmol/kg  $H_2O$ , after an infusion of hypertonic saline.

#### Questions and Discussion

Is this a water diuresis? Because the patient's urine volume was around 4 L/day and his  $U_{osm}$  was around 200 mOsmol/kg  $H_2O$ , the cause of his polyuria was a water diuresis (see [Flow Chart 24.1](#)). His osmole excretion rate was 800 mOsmol/day, which is within the usual rate of excretion of osmoles in a subject consuming a typical Western diet. In response to the administration of DDAVP, his urine flow rate decreased to 0.5 mL/min, and the  $U_{osm}$  rose to 900 mOsmol/kg  $H_2O$ . Hence, the diagnosis was central DI. Because his urine volume was only 4 L/day and not 10 to 15 L/day, he was thought to have partial central DI.

**Central diabetes insipidus.** The diagnosis of central DI was straightforward. Because he complained of thirst, his osmostat and his thirst center, as well as the fibers connecting them, appeared to be functionally intact (see [Fig. 24.2](#)). Interesting to note that the  $U_{osm}$  was greater than the  $P_{osm}$  in the first-voided morning urine ( $U_{osm} = 425$  mOsmol/kg  $H_2O$ ) when he stopped drinking water several hours before he went to sleep. This suggests that the vasopressin release center seems to function but only when there is a strong stimulus for the release of this hormone, presumably a higher  $P_{Na}$ . In keeping with this hypothesis is the fact that his urine flow rate fell to 0.5 mL/min and  $U_{osm}$  rose to 900 mOsmol/kg  $H_2O$  after an infusion of hypertonic saline. Therefore, a possible site for the lesion is destruction of some but not of all the fibers linking the osmostat to the vasopressin release center ([Fig. 24.3](#)).

Such a lesion might explain why polyuria was not present overnight if the patient stopped water intake several hours prior to going to sleep, because he might develop a negative water balance sufficient to cause his  $P_{Na}$  to rise to a level high enough to cause the release of vasopressin. This clinical case provides insights into the pathophysiology of partial central DI.<sup>6</sup>

**Primary polydipsia.** If the patient's  $P_{Na}$  was high enough early in the morning to stimulate the release of vasopressin, whereas during the daytime, when his  $P_{Na}$  was 137 mmol/L, his  $U_{osm}$  was consistently around 90 mOsmol/kg  $H_2O$ , this suggests that primary polydipsia was present while he was awake. Its basis probably was a learned behavior to avoid the uncomfortable feeling of thirst. This interpretation provides insights into understanding the pathophysiology of the polyuria in this patient and, importantly, to determine the options for therapy to decrease his urine volume.

What are the best options for therapy? The major point here is that a higher  $P_{Na}$  could stimulate the release of vasopressin in this patient. A rise in  $P_{Na}$  may be induced by a positive balance for  $Na^+$  or a negative balance for water. The patient chose to use oral NaCl tablets to raise his  $P_{Na}$  to control his daytime polyuria. This therapy would avoid the risk of acute hyponatremia, which might occur if he were given DDAVP and drank an excessive quantity of water by habit. In contrast, he selected water deprivation starting several hours before going to bed to raise his  $P_{Na}$  overnight, which seemed to permit him to have undisturbed sleep.

#### CLINICAL CASE 2: IN A PATIENT WITH CENTRAL DIABETES INSIPIDUS, WHY DID POLYURIA PERSIST AFTER THE ADMINISTRATION OF DESMOPRESSIN?

A 16-year-old male (weight, 50 kg; total body water, 30 L) underwent craniopharyngioma resection. We are asked to see the patient because of polyuria and hypernatremia. His urine output was 3 L over 5 hours (urine flow rate = 10 mL/min), and his  $P_{Na}$  rose from 140 to 150 mmol/L. During this time period, he received 3 L of isotonic saline. His  $U_{osm}$  was 120 mOsmol/kg  $H_2O$ , and the urine  $[Na^+ + K^+]$  concentration was 50 mmol/L. To confirm the diagnosis of central DI, he was given DDAVP, and the urine flow rate fell to 6 mL/min, the  $U_{osm}$  rose to 375 mOsmol/kg  $H_2O$ , and the urine  $[Na^+ + K^+]$  concentration rose to 175 mmol/L.

#### Questions and Discussion

Does the drop in the patient's urine flow rate to 6 mL/min represent a partial response to DDAVP? When DDAVP acts, the urine flow rate is directly proportional to the osmole excretion rate and inversely proportional to the medullary interstitial osmolality. The expected urine flow rate in response to DDAVP can be calculated by considering several factors. First, the usual rate of excretion of osmoles in an adult subject consuming a typical Western diet is about 900 mOsmol/day; approximately half are urea osmoles and half are electrolyte osmoles. As will be detailed later, at its usual rate of excretion, urea is not an effective osmole in the lumen of the inner MCD. Therefore, around 450 effective mOsmol are excreted each day. Interstitial osmolality in the inner medulla is around 900 mOsmol/kg  $H_2O$ . If half of this is due to urea osmoles and half is due to electrolyte osmoles, the effective osmolality of the interstitium in the inner medulla is therefore

around 450 mOsmol/kg  $H_2O$ . When DDAVP acts, the effective osmolality of the fluid exiting the inner medulla will be equal to the effective osmolality of the interstitium in the inner medulla—in other words, 450 mOsmol/kg  $H_2O$ . The 450 mOsmol of effective (electrolyte) osmoles will therefore bring 1 L of urine with them each day. This is a flow rate of 1000 mL/1440 min, or 0.7 mL/min. The osmole excretion rate in this patient before the administration of DDAVP was 1.2 mOsmol/min ( $U_{osm} = 120 \text{ mOsmol/kg } H_2O \times \text{urine flow rate} = 10 \text{ mL/min}$ ). This value is double the usual osmole excretion rate (900 mOsmol/1440 min = 0.6 mOsmol/min) in subjects consuming a typical Western diet. Another point merits emphasis with regard to the nature of the excreted osmoles. Prior to the administration of DDAVP, the concentration of  $Na^+ + K^+$  in the urine was 50 mmol/L. Hence, the vast majority of the osmoles excreted in the urine (100 osmoles— $50 \times 2$  to account for the accompanying anions—out of each 120 osmoles) were electrolyte osmoles, which are effective osmoles that obligate the excretion of  $H_2O$  when DDAVP acts. In this case, the likely source of this was the NaCl infusion that the patient was receiving concurrently.

It is also important to recognize that because of a prior water diuresis, there would be a degree of washout of the renal medulla. Hence, the maximum  $U_{osm}$  in response to the administration of DDAVP would be significantly lower than that observed in normal subjects. This may explain why this patient's  $U_{osm}$  rose to only 375 mOsmol/kg  $H_2O$  following the administration of DDAVP.

Therefore, the drop in the patient's urine flow rate to only about 6 mL/min does not represent a partial response to DDAVP but, rather, the expected response because of the high rate of excretion of effective osmoles (osmotic diuresis) and the low effective medullary interstitial osmolality (urinary concentrating defect due to medullary washout).

Why did the  $P_{Na}$  rise from 140 to 150 mmol/L during this large water diuresis? Although the tendency here is to assume that this patient's hypernatremia was due to a water deficit because he had a large water diuresis, the actual basis of the rise in his  $P_{Na}$  can be determined with a calculation of the tonicity balance.

**Water balance.** The patient received 3 L of isotonic saline, and hence had an input of 3 L of water. He excreted 3 L of urine, so there is a neutral balance of water.

**$Na^+ + K^+$  balance.** The patient received 450 mmol of  $Na^+$  (3 L  $\times$  150 mmol  $Na^+$ /L) and excreted only 150 mmol of  $Na^+ + K^+$  (3 L urine  $\times$  50 mmol  $Na^+ + K^+$ /L). Hence, he had a positive balance of 300 mmol of  $Na^+ + K^+$ . When one divides this surplus of  $Na^+ + K^+$  by the total body water (60% of body weight of 50 kg = 30 L), the rise in  $P_{Na}$  of 10 mmol/L is largely due to a positive balance of  $Na^+$  rather than a deficit of water. The proper treatment to restore body tonicity and the volume and composition of the ECF and ICF compartments in this patient was to induce a negative balance of 300 mmol of  $Na^+ + K^+$ .

#### OSMOTIC DIURESIS

##### Concept 4

When vasopressin acts, it causes the insertion of AQP2 into the luminal membrane of principal cells in the CCD and

**MCD.** Water is reabsorbed until the concentration of effective osmoles in the lumen of the CCD and MCD is nearly equal to their concentration in the surrounding cortical and medullary interstitial fluid, respectively. The volume of urine during an osmotic diuresis is directly proportional to the rate of excretion of effective osmoles and indirectly proportional to effective osmolality of the interstitial fluid in the inner medulla.

Because cells in the inner MCD have urea transporters in their luminal membranes when vasopressin acts, urea is usually an ineffective osmole—the concentration of urea is nearly equal in the interstitial fluid in the inner medulla and in the lumen of the inner MCD. Thus, urea does not obligate the excretion of water. The net result of excreting some extra urea is a higher  $U_{\text{osm}}$  but not a higher urine flow rate. Therefore, it is more correct to say that the urine flow rate is directly proportional to the number of nonurea or effective osmoles in the lumen of the inner MCD and is inversely proportional to the concentration of nonurea or effective osmoles in the interstitial fluid in the inner medulla.

Urea, however, may be an effective urine osmole when the rate of excretion of urea rises by a large amount because urea might not be absorbed fast enough to achieve equal concentrations in the lumen of the inner MCD and in the interstitial fluid in the inner medulla.

### Concept 5

The medullary interstitial osmolality falls during an osmotic diuresis because of medullary washout. During an osmotic diuresis, a larger number of liters of fluid is delivered to and is reabsorbed in the MCD during an osmotic diuresis than during antidiuresis. Hence, medullary washout occurs, and medullary interstitial osmolality falls. The  $U_{\text{osm}}$  is usually close to 600 mOsmol/kg  $H_2O$  during an osmotic diuresis. Lower values of the  $U_{\text{osm}}$  that are even close to  $P_{\text{osm}}$  may be observed when there is a large rate of excretion of osmoles.

### Tools for Evaluation of Osmotic Diuresis

**Urine Osmolality.** In a patient who has an osmotic diuresis, the  $U_{\text{osm}}$  should be higher than the  $P_{\text{osm}}$ .

**Osmole Excretion Rate.** In an adult during an osmotic diuresis, the rate of excretion of osmoles should be much greater than 1000 mOsmol/day ( $>0.7$  mOsmol/min).

**Nature of the Urine Osmoles.** The nature of the urine osmoles should be determined by measuring the rate of excretion of the individual osmoles in the urine. One can deduce which solute is likely to be responsible for the osmotic diuresis by measuring the concentration in plasma (e.g., glucose, urea). A large amount of mannitol is not commonly given; hence, it is unlikely to be the sole cause of a large and sustained osmotic diuresis. A saline-induced osmotic diuresis may occur if the EABV was overly expanded with the infusion of a large amount of saline, or if the patient has cerebral or renal salt wasting. For diagnosis of a state of salt wasting, there must be an appreciable excretion of  $Na^+$  when the EABV is definitely contracted.

**Sources of the Urine Osmoles.** In a patient with a glucose- or a urea-induced osmotic diuresis, it is important to determine whether these osmoles were derived from an exogenous source or from catabolism of endogenous proteins.

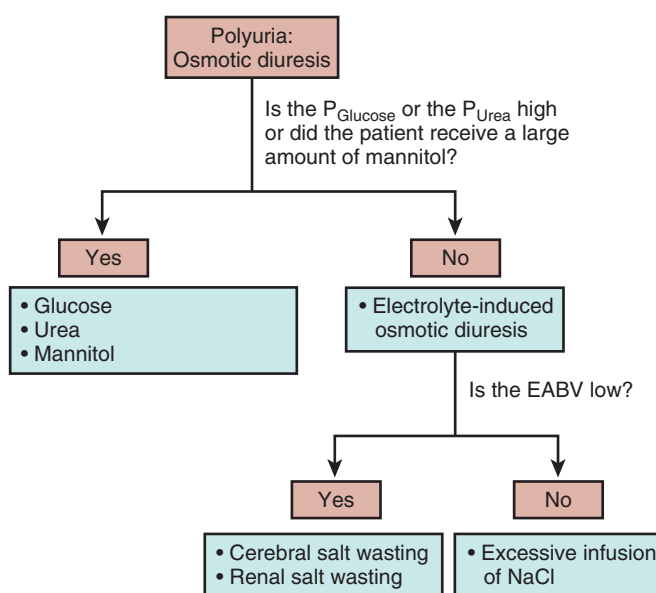
**Source of Urea.** Urea appearance rate. The rate of appearance of urea can be determined from the amount of urea that is retained in the body plus the amount that is excreted in the urine over a given period of time. The former can be calculated from the rise in the concentration of urea in plasma ( $P_{\text{urea}}$ ) and by assuming a volume of distribution of urea that is equal to total body water ( $\sim 60\%$  of body weight).

One can use the following calculation to determine whether the source of urea was the breakdown of exogenous or endogenous proteins if the intake of proteins is known. Close to 16% of the weight of protein is nitrogen. Therefore if 100 g of protein were oxidized, 16 g of nitrogen would be formed. The molecular weight of nitrogen is 14, so about 1140 mmol of nitrogen would be produced. Because each molecule of urea contains two atoms of nitrogen, about 570 mmol of urea would be produced from the oxidation of 100 g of protein (or 5.7 mmol of urea/1 g protein). Because each kilogram of lean body mass has around 180 g of protein, the breakdown of 1 kg of lean body mass will produce about 1026 mmol of urea (5.7 mmol of urea/1 g protein  $\times$  180 g of protein).

**Source of Glucose.** The production of glucose from endogenous sources is relatively small. In more detail, only 60% of the weight of protein can be converted to glucose. Hence, to produce enough glucose from protein to induce 1 L of osmotic diuresis, which requires the excretion of around 300 mmol of glucose, one would need the catabolism of 90 g of protein (equivalent to the catabolism of about 0.5 kg of lean body mass). Therefore, if there is a large glucose-induced osmotic diuresis, the glucose must be from an exogenous source (e.g., the ingestion of large volumes of fruit juice or sugar-containing soft drinks in a patient who lacks insulin actions).

### CLINICAL APPROACH TO THE PATIENT WITH AN OSMOTIC DIURESIS

The steps in the clinical approach to the patient with osmotic diuresis are illustrated in [Flow Chart 24.3](#).



**Flow Chart 24.3** EABV, Effective arterial blood volume. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)



**Step 1: Calculate the Osmole Excretion Rate**

If the  $U_{\text{osm}}$  is greater than the  $P_{\text{osm}}$ , and the osmole excretion rate significantly exceeds 1000 mOsmol/day (0.7 mOsmol/min), an osmotic diuresis is likely to be present.

**Step 2: Define the Nature of the Excreted Osmoles**

One can make a reasonable assessment of the likelihood that a solute will cause polyuria if its concentration in plasma is measured, the GFR is estimated, and the renal handling of that solute is known. One should also determine whether enough mannitol was administered to cause the observed degree of polyuria. Osmotic diuresis could be due to a saline diuresis if  $\text{Na}^+$  and  $\text{Cl}^-$  are excreted at high rates and represent the majority of the urine osmoles.

**Step 3: Identify the Source of the Osmoles in the Urine**

In a patient with a glucose- or urea-induced osmotic diuresis, it is important to decide whether these osmoles were derived from an exogenous source or from catabolism of endogenous proteins. The clinician should also be aware of the possibility of a large amount of glucose in the lumen of the gastrointestinal tract because when it is absorbed it may contribute to the osmotic diuresis.

In a patient with a saline-induced osmotic diuresis, one must determine why so much NaCl is being excreted. Some potential causes are prior excessive saline administration (a common situation in hospitalized patients), administration of a loop diuretic in a patient with marked degree of peripheral edema, cerebral salt wasting, or renal salt wasting.

**CLINICAL CASE 3: UNUSUALLY LARGE OSMOTIC DIURESIS IN A PATIENT WITH DIABETES MELLITUS**

A 50-kg, 14-year-old female had a long history of type 1 diabetes mellitus, which was poorly controlled because she did not take insulin regularly. In the past 48 hours, she was thirsty, drank large volumes of fruit juice, and noted that her urine volume was very high. On physical examination, her EABV was not appreciably contracted. The urine flow rate was 10 mL/min over a 100-min period while she was in the emergency room. Other laboratory data included the following: pH = 7.33, plasma bicarbonate ( $\text{HCO}_3^-$ ) concentration ( $\text{PHCO}_3$ ) = 24 mmol/L, plasma anion gap ( $P_{\text{anion gap}}$ ) = 16 mEq/L, plasma  $\text{K}^+$  concentration ( $P_{\text{K}}$ ) = 4.8 mmol/L, plasma creatinine concentration ( $P_{\text{Cr}}$ ) = was close to her usual values of 1.0 mg/dL (88  $\mu\text{mol/L}$ ), blood urea nitrogen (BUN) = 22 mg/dL ( $P_{\text{urea}}$  = 8 mmol/L), and hematocrit = 0.50. Of note, there was no decrease in her plasma glucose concentration ( $P_{\text{Glu}}$ ) over that time period, despite the excretion of a large amount of glucose in her urine. The following results were also reported:

| Parameter                                     | Admission    |               | After 100 min |               |
|-----------------------------------------------|--------------|---------------|---------------|---------------|
|                                               | Plasma       | Urine         | Plasma        | Urine         |
| Glucose, mg/dL<br>(mmol/L)                    | 1260<br>(70) | 5400<br>(300) | 1260<br>(70)  | 5400<br>(300) |
| $\text{Na}^+$ , mmol/L                        | 125          | 50            | 123           | 50            |
| Osmolality,<br>mOsmol/kg $\text{H}_2\text{O}$ | 320          | 450           | 316           | 450           |

**Questions and Discussion**

What is the basis of the polyuria?

$U_{\text{osm}}$ . The  $U_{\text{osm}}$  of 450 mOsmol/kg  $\text{H}_2\text{O}$  indicates that this polyuria was due to an osmotic diuresis. The  $U_{\text{osm}}$  was lower than the expected value during an osmotic diuresis, probably reflecting the very high osmole excretion rate, which caused a larger fall in the patient's medullary interstitial osmolality due to medullary washout as a result of the reabsorption of a large volume of water in her MCD.

**Osmole excretion rate.** The product of her  $U_{\text{osm}}$  (450 mOsmol/L) and urine flow rate (10 mL/min) yielded an osmole excretion rate of 4.5 mOsmol/min, a value that is sevenfold higher than the usual value in an adult subject consuming a typical Western diet ( $\approx 0.6$  mOsmol/min).

**Nature of the urine osmoles.** Because the patient's GFR was not low and her  $P_{\text{Glu}}$  was extremely high (1260 mg/dL, 70 mmol/L), the filtered load of glucose was markedly higher than the maximum tubular capacity for its reabsorption. Hence, this was a glucose-induced osmotic diuresis, confirmed by the finding of a  $U_{\text{Glu}}$  of about 300 mmol/L.

**Sources of the urine osmoles.** Of special emphasis, this patient's  $P_{\text{Glu}}$  did not decline despite such a high rate of excretion of glucose. In quantitative terms, the total content of glucose in her ECF compartment was 126 g—1260 mg/dL  $\times$  10 to convert to mg/L  $\times$  10 L ECFV  $\div$  1000. During this 100-minute period, she excreted 54 g of glucose—5400 mg/dL  $\times$  10 to convert to mg/L  $\times$  1 L  $\div$  1000. Therefore, although she excreted close to half of the content of glucose in her entire ECF compartment, there was no change in her  $P_{\text{Glu}}$ . Accordingly, to maintain this degree of hyperglycemia, she needed a large input of glucose over a short period of time. The only likely source of such a large amount of glucose was glucose that was retained in her stomach. As a reference, 1 L of apple juice contains about 135 g of glucose. Although the usual effect of hyperglycemia is to slow gastric emptying, this did not seem to be the case in this patient because she seemed to have had a rapid rate of exit of fluid from her stomach, with its subsequent absorption in her intestine.<sup>7</sup>

What dangers do you anticipate in this patient?

**Cerebral edema.** Brain cell swelling might occur if there were a significant fall in her effective  $P_{\text{osm}}$  (Eq. 24.2).<sup>8</sup> A fall in her effective  $P_{\text{osm}}$  may occur if the absorbed glucose and water from the gastrointestinal (GI) tract were retained in the body and glucose was subsequently metabolized; hence, there would be a gain of electrolyte-free water in the body.

$$\text{Effective } P_{\text{osm}} = 2(P_{\text{Na}} + P_{\text{Glu}}) \text{ (all values in mmol/L)} \quad (24.2)$$

**DECREASED EFFECTIVE ARTERIAL BLOOD VOLUME**

Principles governing the regulation of EABV are briefly discussed here and in more detail in Chapter 14.

**Concept 6**

Water crosses cell membranes rapidly through AQP channels to achieve osmotic equilibrium. Hence, the numbers of effective osmoles in the ECF and ICF compartments determine their respective volumes. The effective osmoles in the ECF

compartment are primarily  $\text{Na}^+$  and its attendant anions ( $\text{Cl}^-$  and  $\text{HCO}_3^-$ ); their content in the ECF compartment determines its volume.

### Concept 7

The hydrostatic pressure and the oncotic pressure across the capillary membrane are the major forces that determine the distribution of the ECF compartment volume between its intravascular and interstitial spaces.

The major driving force for outward movement of an ultrafiltrate of plasma across the capillary membrane is the hydrostatic pressure difference. The hydrostatic pressure at the venous end of the capillary is higher under conditions that lead to venous hypertension (e.g., venous obstruction, congestive heart failure).

The major driving force for inward fluid movement from the interstitial space to the intravascular space is the colloidal osmotic pressure difference. This is largely due to the higher concentration of albumin in the intravascular space (40 g/L) than in the interstitial space (10 g/L). Fluid accumulates in the interstitial space in patients with hypoalbuminemia (e.g., patients with nephrotic syndrome). Thus, the ECF volume (ECFV) in these patients may be increased but their EABV may be decreased.

### Concept 8

Control mechanisms for  $\text{Na}^+$  homeostasis are set to defend the EABV rather than the ECFV. EABV can be defined as the part of the ECFV that is located in the arterial blood system and that effectively perfuses tissues. Changes in the EABV are sensed by baroreceptors located in the large blood vessels (the carotid sinus and the aortic arch) and the afferent glomerular arterioles. These are stretch receptors that detect changes in the filling of these vessels.

## TOOLS FOR EVALUATING DECREASED EFFECTIVE ARTERIAL BLOOD VOLUME

### Quantitative Assessment of the Extracellular Fluid Volume

The physical examination, the plasma concentrations of  $\text{HCO}_3^-$ , creatinine, urea, and urate, as well as the fractional excretions of  $\text{Na}^+$ ,  $\text{Cl}^-$ , urea, and urate are useful at times to suggest that the EABV is contracted. Nevertheless, they do not provide a quantitative estimate of the ECFV. A quantitative assessment of the ECFV can be obtained using the hematocrit (Table 24.2) or the concentration of total proteins in plasma, assuming that their values were normal to begin with.<sup>9</sup> The hematocrit is the ratio of red blood cells (RBCs) volume to blood volume, as given in the following formula:

$$\text{Hematocrit} = \text{RBC volume} / (\text{RBC volume} + \text{plasma volume}) \quad (24.3)$$

Blood volume in an adult subject is around 70 mL/kg body weight (i.e., around 5 L in a 70-kg subject). With a blood volume of 5 L and a hematocrit of 0.40, the RBC volume is 2 L and the plasma volume is 3 L (Eq. 24.3). When the hematocrit is 0.50, and assuming no change in the RBC volume (2 L), the blood volume is 4 L. Hence, the plasma volume is 2 L (i.e., reduced by one-third from its normal value of 3 L). If one ignores changes in Starling

**Table 24.2 Use of the Hematocrit to Estimate the Extracellular Fluid Volume<sup>a</sup>**

| Hematocrit | % Change ECFV |
|------------|---------------|
| 0.40       | 0             |
| 0.50       | -33           |
| 0.60       | -60           |

<sup>a</sup>The assumptions made when using this calculation are that the patient does not have anemia or erythrocytosis, that the red blood cell (RBC) volume is 2 L, and that the plasma volume is 3 L (blood volume 5 L). The formula is:

$$\text{Hematocrit} = \text{RBC volume} / (\text{RBC volume} + \text{plasma volume})$$

Values between those listed can be deduced by interpolation. ECFV, Extracellular fluid volume.

**Table 24.3 Urine Electrolyte Values in a Patient With a Contracted Effective Arterial Blood Volume<sup>a</sup>**

| Condition                             | Urine $\text{Na}^+$ | Urine $\text{Cl}^-$ |
|---------------------------------------|---------------------|---------------------|
| <b>Vomiting</b>                       |                     |                     |
| Recent                                | High                | Low                 |
| Remote                                | Low                 | Low                 |
| <b>Diuretics</b>                      |                     |                     |
| Recent                                | High                | High                |
| Remote                                | Low                 | Low                 |
| Diarrhea or laxative abuse            | Low                 | High                |
| Bartter syndrome or Gitelman syndrome | High                | High                |

<sup>a</sup>Values of the urine electrolyte concentration must be adjusted when polyuria is present. Urine  $\text{Cl}^-$  is high in patients with diarrhea or laxative abuse if they have a high rate of excretion of  $\text{NH}_4^+$ . *High*, Urine concentration >15 mmol/L; *low*, urine concentration <15 mmol/L.

forces for simplicity, the ECFV should have declined by approximately one-third of its normal volume (see Table 24.2).

### Urinary Tests for Low Effective Arterial Blood Volume

The expected response to a low EABV is to excrete as little  $\text{Na}^+$  and  $\text{Cl}^-$  as possible in the urine. Because 24-hour urine collections to calculate rates of excretion of  $\text{Na}^+$  and  $\text{Cl}^-$  are usually not practical in clinical circumstances in which a quick assessment is required, clinicians use the urinary concentrations of  $\text{Na}^+$  ( $U_{\text{Na}}$ ) and  $\text{Cl}^-$  ( $U_{\text{Cl}}$ ) in a spot urine sample (Table 24.3) to assess the renal response to the presence of a low EABV. These, however, are concentration terms, which may not necessarily reflect low rates of excretion if the urine flow rate is high. To avoid this type of error, the  $U_{\text{Na}}$  and  $U_{\text{Cl}}$  should be related to the concentration of creatinine in the urine ( $U_{\text{Cr}}$ ) because the rate of excretion of creatinine is relatively constant throughout the day; its 24-hour excretion rate can be estimated based on assessment of the patient's muscle mass. There are some caveats in using the urine  $\text{Na}^+$  and  $\text{Cl}^-$  excretion rates to assess EABV.<sup>10</sup>

**Low Rate of Excretion of Sodium and Chloride.** A low rate of excretion of  $\text{Na}^+$  and  $\text{Cl}^-$  in a patient with a decreased EABV suggests either a loss of NaCl via nonrenal routes (e.g., sweat, GI tract) or that there was a prior renal loss of  $\text{Na}^+$  and  $\text{Cl}^-$  (e.g., remote use of diuretics). In the absence of a low EABV, a low rate of excretion of  $\text{Na}^+$  and  $\text{Cl}^-$  may reflect a low intake of NaCl.

**High Rate of Excretion of Sodium but Little Excretion of Chloride.** In a patient with a low EABV, a high rate of excretion of  $\text{Na}^+$  but not  $\text{Cl}^-$  suggests that there is an anion other than  $\text{Cl}^-$  being excreted with  $\text{Na}^+$ . If the anion is  $\text{HCO}_3^-$  (the urine pH is alkaline), suspect recent vomiting. The anion could also be one that was ingested or administered (e.g., penicillinate anion), in which case the urine pH would be close to 6.

**High Rate of Excretion of Chloride but Little Excretion of Sodium.** In a patient with a low EABV, a high rate of excretion of  $\text{Cl}^-$  but not  $\text{Na}^+$  suggests that a cation other than  $\text{Na}^+$  is being excreted with  $\text{Cl}^-$ . Most often the cation is  $\text{NH}_4^+$ , and the setting is diarrhea or laxative abuse, causing hyperchloremic metabolic acidosis.

**Excretions of Sodium and Chloride Are Not Low.** In a patient who has a low EABV, a high rate of excretion of both  $\text{Na}^+$  and  $\text{Cl}^-$  suggests the absence of a stimulator of the reabsorption of  $\text{Na}^+$  and  $\text{Cl}^-$  (e.g., aldosterone deficiency), the presence of an inhibitor of their reabsorption (e.g., a diuretic), or an intrinsic renal lesion that has effects similar to those of a diuretic (e.g., Bartter syndrome or Gitelman syndrome). The pattern of excretion of electrolytes in multiple spot urine samples throughout the day can provide a helpful clue to the cause of the loss of NaCl. For example, the use of diuretics is suspected if  $U_{\text{Na}}$  and  $U_{\text{Cl}}$  in spot urine samples are very low at times but high at others. By contrast, in a patient with Bartter syndrome or Gitelman syndrome, these rates of excretion are persistently high.

### Fractional Excretion of Sodium or Chloride

Calculation of the fractional excretion of  $\text{Na}^+$  ( $\text{FE}_{\text{Na}}$ ) or  $\text{Cl}^-$  ( $\text{FE}_{\text{Cl}}$ ) expresses the amount of  $\text{Na}^+$  or  $\text{Cl}^-$  that is excreted in the urine as a percentage of the amount that is filtered. For example, adults consuming a typical Western diet excrete about 150 mmol/day each of  $\text{Na}^+$  and  $\text{Cl}^-$ . Because a normal GFR is about 180 L/day, the kidney filters about 27,000 mmol of  $\text{Na}^+$  and about 20,000 mmol of  $\text{Cl}^-$  per day ( $P_{\text{Na}}$  or  $P_{\text{Cl}} \times \text{GFR}$ ). Hence,  $\text{FE}_{\text{Na}}$  is about 0.5%, and  $\text{FE}_{\text{Cl}}$  is about 0.75%. (The formula for this calculation is shown in Eq. 24.4.)

$$\text{FE}_{\text{Na}} = 100 \times ([U_{\text{Na}}/P_{\text{Na}}]/[U_{\text{Cr}}/P_{\text{Cr}}]) \quad (24.4)$$

Note that  $U_{\text{Na}}$  or  $U_{\text{Cl}}$  and  $P_{\text{Na}}$  or  $P_{\text{Cl}}$  must be in the same units (generally mmol/L), and  $U_{\text{Cr}}$  and  $P_{\text{Cr}}$  must be in the same units (e.g., either both in mg/dL or both in mmol/L).

One should bear in mind three practical points when using the  $\text{FE}_{\text{Na}}$  or  $\text{FE}_{\text{Cl}}$ :

1. The excretions of  $\text{Na}^+$  and  $\text{Cl}^-$  are directly related to the dietary intake of NaCl.

Hence, a low  $\text{FE}_{\text{Na}}$  or  $\text{FE}_{\text{Cl}}$  may represent a low intake of NaCl rather than a low EABV.

2. The numeric values for  $\text{FE}_{\text{Na}}$  and  $\text{FE}_{\text{Cl}}$  will be twice as high, at steady state, in a euvolemic subject who consumes 150 mmol of NaCl daily whose GFR is reduced by 50% as in another subject who consumes the same amount of NaCl and has a normal GFR.

Hence, the numeric values of  $\text{FE}_{\text{Na}}$  and  $\text{FE}_{\text{Cl}}$  must be interpreted in the context of the GFR at the time the measurements are made.

3. As is the case with the use of  $U_{\text{Na}}$  or  $U_{\text{Cl}}$ , the  $\text{FE}_{\text{Na}}$  may be high in a patient with a low EABV when there is an unusually large excretion of another anion (e.g.,  $\text{HCO}_3^-$ ) in the urine, and the  $\text{FE}_{\text{Cl}}$  may be high in a patient with a low EABV when there is an unusually large excretion of another cation (e.g.,  $\text{NH}_4^+$ ) in the urine.

Calculations of  $\text{FE}_{\text{Na}}$  and  $\text{FE}_{\text{Cl}}$  are commonly used in patients with acute kidney injury in the differential diagnosis of prerenal azotemia versus acute tubular necrosis (in the absence of diuretics).<sup>11</sup> The advantage of using fractional excretion in this setting over the urinary  $\text{Na}^+$  and  $\text{Cl}^-$  concentrations is that the use of  $\text{FE}_{\text{Na}}$  and  $\text{FE}_{\text{Cl}}$  adjusts these concentration terms for water reabsorption in the nephron.

### Determining the Nephron Site Where There Is a Defect in Reabsorption of Sodium

If another compound or an ion that should have been reabsorbed in a given nephron segment is being excreted, one has presumptive evidence for a defect in that nephron segment. For example, if the defect is in the PT, one might find glucosuria in the absence of hyperglycemia (e.g., Fanconi syndrome).

The presence of hypokalemia may suggest a defect in the TAL of the loop of Henle or in the DCT with enhanced rate of  $\text{K}^+$  secretion in the aldosterone-sensitive distal nephron (ASDN, which includes the second portion of the DCT [DCT2], the connecting segment, and the CCD) due to both increased distal delivery of  $\text{Na}^+$  and increased aldosterone levels because of decreased EABV. The presence of hypocalciuria would suggest that the site of the lesion is in the DCT (e.g., Gitelman syndrome, thiazide diuretics), whereas hypercalciuria suggests a lesion in the TAL (e.g., Bartter syndrome or use of loop diuretics). The presence of hyperkalemia suggests a lesion in the ASDN (e.g., Addison disease, use of potassium-sparing diuretics such as spironolactone).

### CLINICAL CASE 4: ASSESSMENT OF THE EFFECTIVE ARTERIAL BLOOD VOLUME

A 25-year-old woman was assessed by her family physician because she was feeling weak. Although she admitted to being concerned about her body image, she denied vomiting and the intake of diuretics. Her blood pressure was 90/60 mm Hg, her pulse rate was 110 beats/min, and her jugular venous pressure was low. Measurements in a venous blood sample showed  $P_{\text{HCO}_3} = 24$  mmol/L,  $P_{\text{anion gap}} = 17$  mEq/L,  $P_{\text{K}} = 2.9$  mmol/L, hematocrit = 0.50, and plasma albumin concentration ( $P_{\text{Alb}}$ ) = 5.0 g/dL (50 g/L). The urine electrolyte values were  $U_{\text{Na}}$  less than 5 mmol/L,  $U_{\text{Cl}} = 42$  mmol/L, urinary  $\text{K}^+$  concentration ( $U_{\text{K}}$ ) = 10 mmol/L, and  $U_{\text{Cr}} = 7$  mmol/L.

### Questions and Discussion

How severe is the degree of this patient's ECFV contraction? The elevated value for the hematocrit (0.50) provides

quantitative information about her ECFV (see Table 24.2), as does the elevated  $P_{\text{Alb}}$ . Based on a hematocrit of 0.50, her ECFV was reduced by around 33%. If she had prior anemia, the degree of her ECFV contraction might be even greater.

What is the cause of her low EABV? The low  $U_{\text{Na}}$  implies that her EABV was low. The high  $U_{\text{Cl}}$  (42 mmol/L), which exceeded the sum of her  $U_{\text{Na}} + U_{\text{K}}$ , indicates that there was another cation in that urine, most likely  $\text{NH}_4^+$ .

**Interpretation.** Calculating the content of  $\text{HCO}_3^-$  in this patient's ECF compartment ( $P_{\text{HCO}_3} \times \text{her current ECFV}$ ) revealed that she had a deficit of  $\text{NaHCO}_3$ . Loss of  $\text{NaCl}$  plus  $\text{NaHCO}_3$  via the GI tract was suspected as the cause of contracted EABV. The patient later admitted to the frequent use of a laxative. Hypokalemia could be explained by the loss of  $\text{K}^+$  via the GI tract because her low  $U_{\text{K}}/U_{\text{Cr}}$  ratio suggests that the loss of  $\text{K}^+$  was extrarenal (see section on potassium). Hypokalemia is associated with acidosis in PT cells, which stimulates ammoniogenesis. The increased rate of excretion of the cation  $\text{NH}_4^+$  obligates the excretion of  $\text{Cl}^-$ , despite the presence of a low EABV.

## HYPONATREMIA

*Hyponatremia* is defined as a  $P_{\text{Na}}$  less than 135 mmol/L.

### Concept 9

Because water moves across cell membranes to achieve osmotic equilibrium, acute hyponatremia is associated with the swelling of brain cells. Brain cells adapt by exporting effective osmoles, mainly  $\text{K}^+$  (and an anion from their ICF) and a number of organic solutes (e.g., taurine, myoinositol).

If hyponatremia is present for more than 48 hours, these adaptive changes have proceeded sufficiently to return brain cells back close to their normal volume. In this setting, a rapid increase in the  $P_{\text{Na}}$  shrinks cerebral vascular endothelial cells, leading to opening of the blood-brain barrier, allowing lymphocytes, complement, and cytokines to enter the brain, damage oligodendrocytes, and cause demyelination. Microglial activation also plays a role in this process. This has important implications for the management of patients with hyponatremia (see Chapter 15).

### Concept 10

Chronic hyponatremia is usually due to a defect in the renal excretion of water. There are two causes for diminished renal excretion of water, low distal delivery of filtrate and the actions of vasopressin.

Hyponatremia is not a specific disease; rather, it is a diagnostic category with many different causes. The traditional approach to the pathophysiology of hyponatremia centers on a reduced electrolyte-free water excretion due to the actions of vasopressin. In some clinical settings, release of vasopressin is thought to be due to a diminished EABV. Nevertheless, at least in some patients with hyponatremia, the degree of decrease in EABV does not seem to be large enough to cause the release of vasopressin. We suggest that hyponatremia may develop in some patients in the absence of vasopressin actions. Two factors are important in this context, the volume of filtrate that is delivered to the distal nephron and the volume of water that is reabsorbed in the inner MCD through its RWP.<sup>3</sup>

The volume of distal delivery of filtrate is reduced if the GFR is decreased or the fractional reabsorption of  $\text{NaCl}$  in the PT is increased.

The fractional reabsorption of  $\text{NaCl}$  in the PT is increased in response to a decreased EABV. Decreased EABV can be due to a total body deficit of  $\text{NaCl}$  (e.g., diuretic use in a patient who consumes little salt,  $\text{NaCl}$  loss in diarrhea fluid or sweat) or to a disorder that causes a low cardiac output. Because there is an obligatory minimal loss of  $\text{Na}^+$  in each liter of urine during a large water diuresis, a deficit of  $\text{Na}^+$  can be created during the polyuria induced by a large intake of water in a subject who consumes little  $\text{NaCl}$  (e.g., a patient with beer potomania).

The driving force for water reabsorption via RWP is the osmotic pressure gradient generated by the difference in osmolality between the luminal fluid in the inner MCD and that in the medullary interstitium. As discussed previously, we estimate that somewhat more than 5 L of water is reabsorbed per day in the inner MCD by RWP during water diuresis.

In some patients, hyponatremia is caused by reduced electrolyte-free water excretion due to actions of vasopressin, but the release of vasopressin is not caused by a decreased EABV. This category is called the syndrome of the inappropriate secretion of antidiuretic hormone (SIADH). SIADH, however, is a diagnosis of exclusion; it cannot be made if the patient has a condition that may lead to a low distal delivery of filtrate. A rare cause of SIADH is a mutation in the gene encoding V2R, causing V2R to be constitutively active. This disorder is called “nephrogenic syndrome of inappropriate antidiuresis,” and it is suspected in a patient with SIADH of undetermined cause in whom the vasopressin level is undetectable and who does not respond to the administration of a V2R antagonist (e.g., tolvaptan) with a water diuresis.

## TOOLS FOR EVALUATING HYPONATREMIA

### Measurement of Plasma Sodium Concentration

Pseudohyponatremia is present when the  $P_{\text{Na}}$  measured by the laboratory is lower than the actual ratio of  $\text{Na}^+$  to plasma water in the patient. This occurs when the method used requires dilution of the plasma sample. This is because 7% of the plasma volume is a nonaqueous volume (i.e., lipids and proteins), which is not taken into account when adjusting for the volume of the diluent. Hence, although the concentration of  $\text{Na}^+$  in plasma water is 150 mmol/L, the measured value by flame photometry is 140 mmol/L, and lower values will be obtained by this method if the nonaqueous volume in plasma is larger (e.g., in a patient with hypertriglyceridemia or hyperparaproteinemia) because the same volume of diluent is added to a smaller volume of plasma water. With the use of an ion-selective electrode, the activity of  $\text{Na}^+$  in the aqueous plasma volume is measured; nevertheless, because of the use of automatic aspirators and dilutors to prepare the plasma samples, pseudohyponatremia remains an issue. This error in the measurement of  $P_{\text{Na}}$  is detected by the finding of a normal  $P_{\text{osm}}$  value in the absence of high concentration of other osmoles, such as urea, glucose, or alcohol.<sup>12</sup>

**Hyponatremia Due to Hyperglycemia.** In the absence of actions of insulin, glucose is an effective osmole for skeletal muscle and, if hyperglycemia is associated with a rise in plasma effective osmolality, there will be a shift of water out of skeletal



muscle. Although a number of formulas have been proposed to predict the fall in  $P_{Na}$  based on a given rise in  $P_{Glu}$ , these formulas are based on theoretic calculations with the addition of glucose without water to the ECF compartment. Different correction factors have been proposed based on assumptions made about ECFV and the volume of distribution of glucose in the absence of insulin actions. Moreover, because patients with hyperglycemia have variable fluid intake and variable loss of water and  $Na^+$  in the urine, one cannot assume a fixed relationship between the rise in  $P_{Glu}$  and the fall in  $P_{Na}$ .

#### Detection of a Low Effective Arterial Blood Volume

A difficulty in making the diagnosis of SIADH is that patients may have a mild to modest degree of EABV contraction, causing a low distal delivery of filtrate that cannot always be detected by the physical examination. The following laboratory tests may provide helpful clues to detect the presence of EABV contraction. At times, however, EABV expansion with the infusion of saline may be required to rule out low distal delivery of filtrate as the cause of hyponatremia. Absence of water diuresis despite EABV expansion confirms the diagnosis of SIADH.

**Concentrations of Sodium and Chloride in Urine.** The assessment of  $U_{Na}$  and  $U_{Cl}$  are helpful to detect the presence of a low EABV, and they may also provide clues about its cause. In the absence of the caveats noted above, a  $U_{Na}$  and  $U_{Cl}$  higher than 30 mmol/L is thought to be in keeping with euvolemia and the diagnostic category of SIADH.

**Concentrations of Urea and Urate in Plasma.** Expansion of the EABV diminishes the rate of reabsorption of urea and urate in PT, resulting in higher rates of their fractional excretions and lower levels of their plasma concentrations. Because patients with SIADH are likely to have an expanded EABV, a  $P_{Urea}$  less than 3.6 mmol/L ( $BUN < 21.6$  mg/dL), a plasma urate level less than 0.24 mmol/L ( $< 4$  mg/dL), a fractional excretion of urea greater than 55 %, and a fractional excretion of urate greater than 12% are thought to be more in keeping with the diagnostic category of SIADH. The fractional excretions of urea and urate, however, are not reliable markers in patients who are thought to have cerebral salt wasting and decreased EABV because the defect in  $Na^+$  reabsorption in these patients seems to involve the PT. Therefore, these patients will have a high fractional excretion of urea and urate despite decreased EABV.

Because the reabsorption of urea in PT is strongly influenced by the EABV, the relative rise in  $P_{Urea}$  is usually larger than the rise in  $P_{Creatinine}$  in patients with low EABV. A ratio of  $P_{Urea}/P_{Creatinine}$  (where both are in mmol/L units) greater than 100, or  $BUN/P_{Creatinine}$  (where both are in mg/dL units) greater than 20, suggests a low EABV. This, however, may not be the case if protein intake is low.

**Other Tests.** A rise in the  $P_{Cr}$  and a low or a high  $P_{HCO_3}$  level may suggest that the EABV is low.

The clinical approach to the diagnosis of the cause of chronic hyponatremia is illustrated in [Flow Chart 24.4](#).

#### CLINICAL CASE 5: HYPONATREMIA WITH BROWN SPOTS

A 22-year-old woman had myasthenia gravis. In the previous 6 months, she noted a marked decline in her energy and

weight loss (down from 50 to 47 kg). She often felt faint when standing up quickly. On physical examination, her blood pressure was 80/50 mm Hg, pulse rate was 126 beats/min, jugular venous pressure was below the level of the sternal angle, and there was no peripheral edema. Brown pigmented spots were evident in her buccal mucosa. The biochemistry data are as follows.

| Findings                      | Plasma                         | Urine               |
|-------------------------------|--------------------------------|---------------------|
| $Na^+$ , mmol/L               | 112                            | 130                 |
| $K^+$ , mmol/L                | 5.5                            | 24                  |
| Urea                          | BUN: 28 mg/dL<br>(10 mmol/L)   | Urea:<br>130 mmol/L |
| Creatinine                    | 1.7 mg/dL<br>(150 $\mu$ mol/L) | 6.0 mmol/L          |
| Osmolality,<br>mOsm/kg $H_2O$ | 240                            | 438                 |

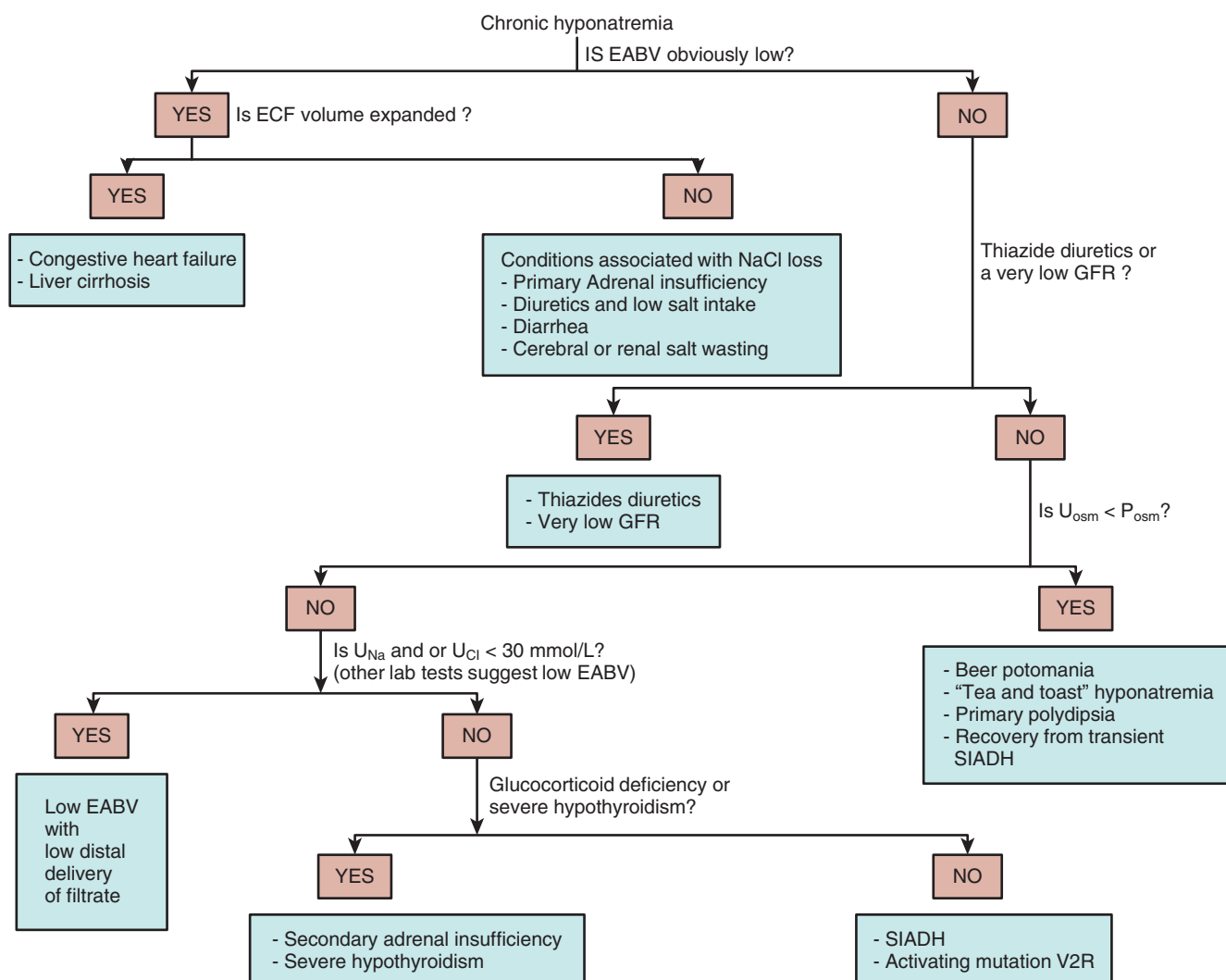
#### Questions and Discussion

What is the most likely basis for the very low EABV? The very contracted EABV (manifested by the low blood pressure and tachycardia), the low  $P_{Na}$ , the high  $P_K$  of 5.5 mmol/L, and the renal salt wasting strongly suggest that the most likely diagnosis is adrenal insufficiency. This is likely due to autoimmune adrenalitis because the patient also had myasthenia gravis. The low EABV was due to renal salt wasting because of lack of aldosterone and was also caused in part by a lower degree of contraction of venous capacitance vessels because of glucocorticoid deficiency.

What dangers to the patient are present on presentation? Two potential emergencies dominate the initial management, a very contracted EABV and the lack of cortisol. To deal with the former and restore hemodynamic stability, the initial intravenous infusion can be given as 0.9% saline. Once the patient is hemodynamically stable, to expand her EABV further without changing her  $P_{Na}$ , the intravenous fluid should be changed to one that is isotonic to the patient. The second potential danger is related to the lack of cortisol and is dealt with by administering cortisol.

It did not seem that the patient had an acute component of hyponatremia requiring the induction of a rapid initial rise in her  $P_{Na}$  because she did not have significant symptoms that could be related to increased intracranial pressure and she also denied a recent large water intake.

What dangers should be anticipated during therapy, and how can they be avoided? Re-expansion of the patient's EABV could lead to an increased excretion of water due to an increased distal delivery of filtrate and suppression of the release of vasopressin. In addition, the administration of cortisol would improve her hemodynamic state and also inhibit the release of corticotropin-releasing hormone and, hence, of vasopressin. The net result of this therapy would be to cause water diuresis and thereby a dangerous rise in the  $P_{Na}$ . Because the patient had a small muscle mass (and therefore a small total body water volume), the excretion of a relatively small volume of electrolyte-free water could lead to too rapid a rise in  $P_{Na}$ . In addition, because of her poor nutritional state (which becomes even more evident if one interprets her weight loss in conjunction with a large gain of water in her cells, hence her loss of muscle mass was larger than revealed by her weight loss), she was at high risk for osmotic demyelination if her  $P_{Na}$  were to rise rapidly. The



**Flow Chart 24.4** EABV, Effective arterial blood volume; ECF, extracellular fluid; GFR, glomerular filtration rate; SIADH, syndrome of inappropriate diuretic hormone secretion. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

rise in her  $P_{Na}$  should not exceed a maximum of 4 to 6 mmol/L/24 hours. Accordingly, we would administer DDAVP early in therapy to prevent water diuresis.

#### CLINICAL CASE 6: HYPONATREMIA IN A PATIENT ON THIAZIDE DIURETIC THERAPY

A 71-year-old woman was prescribed a thiazide diuretic for the treatment of hypertension. She had chronic kidney disease (CKD) due to ischemic nephropathy, with an estimated GFR of 28 mL/min/1.73 m<sup>2</sup> (around 40 L/day). She consumed a low-salt, low-protein diet and drank several large cups of water and tea a day following the advice to remain hydrated. A month later, she presented to her family doctor feeling unwell. Her blood pressure was 130/80 mm Hg, her heart rate was 80 beats/min, there were no postural changes in her blood pressure or heart rate, and her jugular venous pressure was about 1 cm below the level of the sternal angle. Her  $P_{Na}$  was 112 mmol/L. Other laboratory results are as follows.

| Findings                               | Plasma    | Urine     |
|----------------------------------------|-----------|-----------|
| Na <sup>+</sup> , mmol/L               | 112       | 22        |
| K <sup>+</sup> , mmol/L                | 3.6       | 10        |
| HCO <sub>3</sub> <sup>-</sup> , mmol/L | 28        | 0         |
| Urea, mmol/L                           | 8         | 241       |
| Creatinine, μmol/L (mg/dL)             | 145 (1.3) |           |
| Creatinine mmol/L (g/L)                |           | 6.1 (0.7) |
| Osmolality mOsmol/kg H <sub>2</sub> O  | 240       | 325       |

#### Questions and Discussion

What is the most likely basis for the chronic hyponatremia in this patient? Although the patient was taking a thiazide diuretic, the degree of decrease in her EABV did not seem to be large enough to cause the release of vasopressin. The patient had a low baseline estimated GFR of 28 mL/min/1.73 m<sup>2</sup> (around 40 L/day). The use of diuretics and the low-salt diet led to a deficit of Na<sup>+</sup> and a mild reduction in the EABV. Even a relatively small decrease in EABV leads to increased release

of catecholamines, and activation of the renin angiotensin aldosterone system by  $\beta$ -adrenergic stimulation, both of which increase the reabsorption of sodium and water in the PT. If the patient were to reabsorb 90% of her GFR of 40 L/day (which may be even lower because of the mild reduction in her EABV) in the PT, instead of 83% (as discussed previously, this is the percentage of the filtrate that is usually reabsorbed in the PT in the absence of low EABV), less than 4 L/day of filtrate would be delivered distally. This is the maximum volume of urine she could excrete. This volume exceeds the usual daily intake of water, but hyponatremia might still develop in such a patient because there is water reabsorption by RWP along the inner MCD, even in the absence of vasopressin action.

The ability to generate electrolyte-free water is also diminished by the effect of thiazide diuretics to inhibit the reabsorption of NaCl (without water) in DCT. With regard to the volume of water that may be reabsorbed in her inner MCD, the driving force is the difference in osmolality between the luminal fluid and the interstitial fluid in the inner MCD. Because of the low rate of excretion of osmoles, and because the volume of filtrate delivered to the inner MCD in the absence of vasopressin is larger than during antidiuresis, the osmolality of fluid in the lumen of the inner MCD will be low. If the osmole excretion rate in this patient is 300 mOsmol/day, and if 4 L are delivered to the inner MCD, the osmolality of the luminal fluid in the inner MCD would be 75 mOsmol/kg H<sub>2</sub>O. Even if the medullary interstitial osmolality is substantially lower than normal—for example, 375 mOsmol/kg H<sub>2</sub>O—there is still an enormous osmotic driving force for water reabsorption along the inner MCD because a difference of 1 mOsmol/kg H<sub>2</sub>O generates a pressure of about 19.3 mm Hg.

A recent study suggested another mechanism for the decreased ability to excrete electrolyte-free water in patients with thiazide-induced hyponatremia, that is also independent of vasopressin. This study showed that nearly half the patients with thiazide-induced hyponatremia carry a single-nucleotide polymorphism in the prostaglandin transporter (PGT) encoded by *SLCO2A1*. Under normal conditions, the effect of vasopressin to cause the insertion of AQP2 into the luminal membrane of principal cells is counterbalanced by increased renal prostaglandin E<sub>2</sub> (PGE<sub>2</sub>) production. PGE<sub>2</sub> is directed to the basolateral side of the collecting duct by this prostaglandin transporter, where binding of PGE<sub>2</sub> to its basolateral EP1 and EP3 receptors sends out signals for the retrieval of AQP2 from the apical membrane. Administration of a thiazide diuretic leads to increased PGE<sub>2</sub> production, although the mechanism is not clear. In some patients with the *SLCO2A1* variant, because of the decreased transport capacity of PGT, the concentration of PGE<sub>2</sub> in the tubular lumen increases. Binding of PGE<sub>2</sub> to its luminal EP4 receptors has the opposite effect and signals for the insertion of AQP2 into the apical membrane of principal cells. This results in increased water reabsorption, independently of vasopressin.<sup>13</sup>

What dangers should be anticipated during therapy, and how can they be avoided? Understanding this pathophysiology has clinical implications for the management of the patient with hyponatremia. Initially, this patient's hyponatremia was thought to be caused by stimulation of vasopressin release due to decreased EABV owing to her intake of a thiazide diuretic; hence, she was given isotonic saline to re-expand

her EABV. Even a relatively small volume of saline (especially if it were given as a bolus) might be sufficient to reduce the fractional reabsorption of filtrate in the PT and increase its distal delivery. If the fractional reabsorption in PT were decreased to 83% of the GFR of 40 L/day, distal delivery of filtrate would increase to about 7 L/day. This may exceed the capacity for water reabsorption by RWP (or via the PGE<sub>2</sub>/EP4-mediated insertion of AQP2 into the luminal membrane of principal cells), so water diuresis would ensue. Because of the patient's small muscle mass, even a modest water diuresis might be large enough to cause a rapid rise in  $P_{Na}$  and increase the risk of osmotic demyelination syndrome, especially if she were malnourished or potassium-depleted.

## POTASSIUM

Hypokalemia and hyperkalemia are common electrolyte disorders in clinical practice that may cause life-threatening cardiac arrhythmias. The analysis of the urine composition provides essential information to establish the underlying pathophysiology and plan therapy.

The regulation of K<sup>+</sup> homeostasis has two main aspects:

1. Control of the transcellular distribution of K<sup>+</sup>, which is vital for survival because it limits acute changes in the  $P_K$ .
2. Regulation of K<sup>+</sup> excretion by the kidney, which maintains whole-body K<sup>+</sup> balance; this is, however, a much slower process.

## TRANSCELLULAR DISTRIBUTION OF POTASSIUM

### Concept 11

Three factors regulate the movement of K<sup>+</sup> across cell membranes—the concentration difference for K<sup>+</sup>, the electrical voltage across cell membranes, and the presence of open K<sup>+</sup> channels in cell membranes.

*Approximately 98% of K<sup>+</sup> in the body is in cells.* Although the concentration difference would favor the movement of K<sup>+</sup> out of cells via K<sup>+</sup> channels in the cell membrane, K<sup>+</sup> ions are retained inside the cells by an electrical force because the cell interior has a negative voltage caused by the negatively charged intracellular organic phosphates. These chemical and electrical forces eventually come into balance, and the equilibrium potential for K<sup>+</sup> ( $E_K$ ) is achieved. Because cell membranes have much higher permeability to K<sup>+</sup> ions than to Na<sup>+</sup> ions, the resting membrane potential (RMP) of cells is close to  $E_K$ .

*The shift of K<sup>+</sup> into cells requires an increase in cell interior negative voltage.* This can be achieved by increasing flux via the Na<sup>+</sup>-K<sup>+</sup> adenosine triphosphatase pump (Na<sup>+</sup>-K<sup>+</sup>-ATPase). This is because Na<sup>+</sup>-K<sup>+</sup>-ATPase is an electrogenic pump that exports three Na<sup>+</sup> ions out of the cell while importing only two K<sup>+</sup> ions into the cell. Hence, activation of Na<sup>+</sup>-K<sup>+</sup>-ATPase leads to the net export of positive charges out of the cell. There are three ways to increase ion pumping acutely by the Na<sup>+</sup>-K<sup>+</sup>-ATPase:

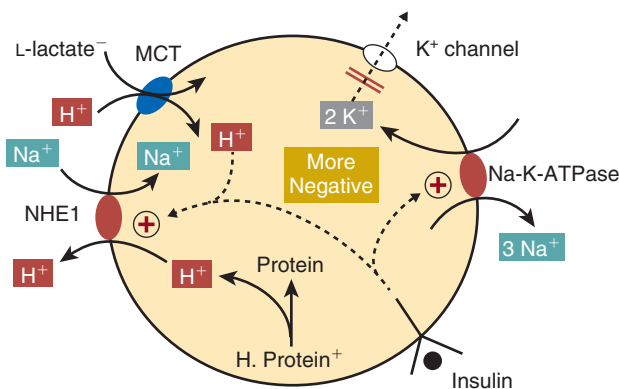
1. A rise in the concentration of its limiting substrate—intracellular Na<sup>+</sup>
2. An increase in the affinity (lower  $K_m$  [concentration for half-maximal activation]) for Na<sup>+</sup> and K<sup>+</sup> or an increase

in the  $V_{\max}$  (maximal pump turnover rate) of the  $\text{Na}^+\text{-K}^+$ -ATPase units in cell membranes

3. An increase in the number of active  $\text{Na}^+\text{-K}^+$ -ATPase pump units in the cell membrane through the recruitment of new units

A long-term increase in  $\text{Na}^+\text{-K}^+$ -ATPase pump activity requires the synthesis of new pump units, as occurs with exercise training, excess thyroid hormones, or high dietary  $\text{K}^+$  intake.

Insulin causes a shift of  $\text{K}^+$  into cells as it promotes the translocation of  $\text{Na}^+\text{-K}^+$ -ATPase units from an intracellular pool to the cell membrane. Insulin causes the phosphorylation of FXYP1 (phospholemann) via atypical protein kinase C, which increases the  $V_{\max}$  of  $\text{Na}^+\text{-K}^+$ -ATPase. Insulin also activates the  $\text{Na}^+\text{-H}^+$  exchanger isoform 1 (NHE1) and hence increases the electroneutral entry of  $\text{Na}^+$  into cells.  $\beta_2$ -Adrenergic agonists induce the phosphorylation of FXYP1 via cyclic adenosine monophosphate (cAMP)-mediated activation of protein kinase A. NHE1 is also activated by a rise in the intracellular  $\text{H}^+$  concentration. Monocarboxylic acids (e.g., L-lactic acid, ketoacids) do not cause hyperkalemia. This is because they enter cells on the monocarboxylic acid cotransporter (MCT). An increase in intracellular  $\text{H}^+$  concentration in the submembrane area where NHE1 is located may activate NHE1, which leads to an increase in the electroneutral entry of  $\text{Na}^+$  into cells and, in the presence of insulin, a shift of  $\text{K}^+$  into cells (Fig. 24.4).<sup>14</sup> A shift of  $\text{K}^+$  out of cells may occur in patients with metabolic acidosis due to the gain of inorganic acids, which are not transported by the MCT (e.g., in patients with diarrhea causing the loss of  $\text{NaHCO}_3$  [gain of  $\text{HCl}$ ]).



**Fig. 24.4** Possible mechanism for how monocarboxylic acids may cause a shift of  $\text{K}^+$  into cells. The circle represents a cell. The  $\text{Na}^+\text{-H}^+$  exchanger isoform 1 (NHE1) in cell membranes is activated by insulin and by a high concentration of  $\text{H}^+$  in the cell interior because  $\text{H}^+$  binds to the modifier site of NHE1. The concentration of  $\text{H}^+$  near NHE1 could rise when monocarboxylic acids (lactic acid in this example) enter cells on the monocarboxylic acid cotransporter (MCT) and their  $\text{H}^+$  ions are released close to NHE1. In the presence of insulin, which activates the  $\text{Na}^+\text{-K}^+$ -ATPase and NHE1 in cell membranes, electroneutral entry of  $\text{Na}^+$  into cells is increased. The subsequent transport of  $\text{Na}^+$  out of cells via the electrogenic  $\text{Na}^+\text{-K}^+$ -ATPase increases the cell interior negative voltage and causes the retention of  $\text{K}^+$  in cells. The source of bulk of  $\text{H}^+$  ions transported out of the cell is  $\text{H}^+$  ions that were bound to intracellular proteins (shown as H. Proteins<sup>+</sup>) (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

## RENAL EXCRETION OF POTASSIUM

Renal  $\text{K}^+$  excretion. Control of  $\text{K}^+$  secretion occurs primarily in the ASDN, which includes the DCT2, the connecting segment, and the CCD (see Chapter 6 for a detailed discussion). Two factors influence the rate of excretion of  $\text{K}^+$ —the net secretion of  $\text{K}^+$  by principal cells in the ASDN, and the flow rate in the ASDN.

### $\text{K}^+$ Secretion in the Aldosterone-Sensitive Distal Nephron

The process for secretion of  $\text{K}^+$  by principal cells in the ASDN requires two elements.<sup>15</sup> First, a lumen-negative transepithelial voltage must be generated by the electrogenic reabsorption of  $\text{Na}^+$  (i.e., reabsorption of  $\text{Na}^+$  via the amiloride-sensitive sodium channel, ENaC, without an accompanying anion, which is usually  $\text{Cl}^-$ ). Second, a sufficient number of open renal outer medullary  $\text{K}^+$  (ROMK) channels in the luminal membranes of principal cells must be present.

Aldosterone actions lead to an increase in the number of open ENaC units in the luminal membranes of principal cells in the ASDN.

It was thought that the paracellular pathway plays an important role in the reabsorption of  $\text{Cl}^-$  in the ASDN, but the large peritubular-luminal concentration difference for  $\text{Cl}^-$  and the relatively small luminal-peritubular electrical driving force make this mechanism unlikely. An electroneutral, thiazide-sensitive, and amiloride-resistant  $\text{NaCl}$  transport process has been identified in the luminal membrane of  $\beta$ -intercalated cells of the CCD in mice. This seems to be mediated by the parallel activity of the  $\text{Na}^+$ -independent  $\text{Cl}^-/\text{HCO}_3^-$  exchanger (pendrin) and the  $\text{Na}^+$ -dependent  $\text{Cl}^-/\text{HCO}_3^-$  exchanger (NDCBE), resulting in electroneutral  $\text{NaCl}$  reabsorption.<sup>16</sup>

An increase in luminal fluid concentration of  $\text{HCO}_3^-$  and/or an alkaline luminal fluid pH seem to increase the amount of  $\text{K}^+$  secreted in the ASDN.<sup>17</sup> It has been suggested that this effect might be due to a decrease in the paracellular permeability of  $\text{Cl}^-$ . A different mechanism for the effect of luminal  $\text{HCO}_3^-$  may be that because a gradient for  $\text{HCO}_3^-$  is needed to increase flux through the  $\text{Cl}^-/\text{HCO}_3^-$  exchanger; pendrin, an increase in luminal  $\text{HCO}_3^-$  concentration may inhibit flux through pendrin, and hence NDCBE, thereby decreasing the rate of electroneutral  $\text{NaCl}$  reabsorption, and in turn increasing the rate of electrogenic reabsorption of  $\text{Na}^+$  and hence the secretion of  $\text{K}^+$ .

ROMK channels are the most important  $\text{K}^+$  channels for the secretion of  $\text{K}^+$  in the ASDN. Big  $\text{K}^+$  conductance—BK or maxi- $\text{K}^+$ —channels seem to play an important role in flow-dependent  $\text{K}^+$  secretion, but their role in the physiologic regulation of renal excretion of  $\text{K}^+$  and its disorders is not clear.<sup>18</sup>

A complex network of with-no-lysine (WNK) kinases, WNK4 and WNK1, via effects on the thiazide-sensitive  $\text{NaCl}$  cotransporter (NCC) in the DCT1 and ROMK in the ASDN, may function as a switch to change the aldosterone response of the kidney to conserve  $\text{Na}^+$  or excrete  $\text{K}^+$ .<sup>19</sup> WNK4, and a full-length, kinase-active form of WNK1 (L-WNK1), can increase the activity of NCC, thereby diminishing the delivery of  $\text{NaCl}$  to the ASDN and hence the rate of electrogenic reabsorption of  $\text{Na}^+$  and the ability to generate a large negative luminal voltage. Increasing delivery of  $\text{Na}^+$  and hence the flow rate in the ASDN may also increase  $\text{K}^+$  secretion via the



flow-activated BK channels. Both WNK4 and L-WNK1 also induce endocytosis of ROMK.<sup>20</sup> See Chapter 6 for a detailed discussion of this complex and rapidly evolving field.

### Concept 12

When vasopressin acts, the flow rate in the ASDN is determined by the number of effective osmoles present in luminal fluid. Vasopressin causes the insertion of AQP2 channels into the luminal membranes of principal cells, the osmolality of fluid in the terminal CCD becomes equal to the  $P_{\text{osm}}$  and hence is relatively fixed. Therefore the number of osmoles present in the luminal fluid in the terminal CCD determines the number of liters of fluid that exit the terminal CCD. These osmoles are largely urea,  $\text{Na}^+$ ,  $\text{Cl}^-$ , and  $\text{K}^+$ , with an accompanying anion. Owing to the process of intrarenal urea recycling, the largest fraction of the osmoles delivered to the ASDN is urea osmoles. In subjects eating a typical Western diet, the amount of urea that recycles would be approximately 600 mmol/day. This process of urea recycling adds an extra 2 L to the flow rate in the terminal CCD (600 mOsmol divided by a luminal fluid osmolality that is equal to plasma osmolality—that is,  $\approx 300$  mOsmol/kg  $\text{H}_2\text{O}$ ).

In a quantitative analysis, Kamel and Halperin have illustrated that even in patients with a large defect in the ability to generate a lumen-negative voltage in the ASDN, a significant degree of hyperkalemia is not likely to develop with a usual  $\text{K}^+$  intake unless there is decreased flow rate in the ASDN. Restricting protein intake may decrease the amount of urea that recycles and hence the rate of flow in the ASDN.<sup>21</sup>

### Concept 13

There is no normal rate of  $\text{K}^+$  excretion in the urine because normal subjects in a steady state excrete all the  $\text{K}^+$  they eat and absorb from the GI tract. To assess the renal response to hypokalemia, we use the observed rate of excretion of  $\text{K}^+$  in patients who developed hypokalemia due to a nonrenal cause. In subjects who became  $\text{K}^+$ -depleted because of low dietary  $\text{K}^+$  intake, the rate of  $\text{K}^+$  excretion fell to 10 to 15 mmol/day. In normal subjects given a large load of  $\text{K}^+$  ( $>200$  mmol/day) on a chronic basis, the rate of renal excretion of  $\text{K}^+$  rose to match their intake, with only a modest rise in their  $P_{\text{K}}$ . Hence, the development of chronic hyperkalemia requires a defect in renal  $\text{K}^+$  excretion.

## TOOLS FOR EVALUATING A PATIENT WITH A DYSKALEMIA

### Assess the Rate of Excretion of Potassium in the Urine

A 24-hour urine collection is not necessary to assess the daily rate of excretion of  $\text{K}^+$ . Taking advantage of the fact that creatinine is excreted at a near-constant rate throughout the day, we use the ratio of the concentration of  $\text{K}^+$  in the urine ( $U_{\text{K}}$ ) to the concentration of creatinine in the urine ( $U_{\text{Cr}}$ )— $U_{\text{K}}/U_{\text{Cr}}$ . This approach has the following advantages. The data can be available in a short time period, and more relevant information is gathered if one knows the stimulus, the  $P_{\text{K}}$ , that influences the rate of excretion of  $\text{K}^+$  during that time frame. The limitation is that there is a diurnal variation in  $\text{K}^+$  excretion, but this does not negate the

advantages. The expected  $U_{\text{K}}/U_{\text{Cr}}$  in patients with hypokalemia due to an intracellular shift of  $\text{K}^+$  and in those with chronic hypokalemia due to extrarenal loss of  $\text{K}^+$  is less than 18 mmol  $\text{K}^+$ /g creatinine or less than 2 mmol  $\text{K}^+$ /mmol creatinine. The expected ratio in a patient with hyperkalemia and a normal renal response is more than 200 mmol  $\text{K}^+$ /g creatinine or more than 20 mmol  $\text{K}^+$ /mmol creatinine. To develop *chronic* hyperkalemia, however, one must have a defect in the renal excretion of  $\text{K}^+$ ; hence,  $U_{\text{K}}/U_{\text{Cr}}$  does not provide clinically useful information in that situation. Because of diurnal variation in  $\text{K}^+$  excretion, a 24-hour urine collection, rather than a random  $U_{\text{K}}/U_{\text{Cr}}$ , may be needed to assess the contribution of dietary  $\text{K}^+$  intake to the degree of hyperkalemia.

### Transtubular Potassium Concentration Gradient

The transtubular potassium concentration gradient (TTKG) was developed to provide a semiquantitative reflection of the driving force for  $\text{K}^+$  secretion in the ASDN. The rationale behind this calculation was to adjust  $U_{\text{K}}$  for the amount of water that was reabsorbed in downstream nephron segments (i.e., the MCD) to estimate the concentration of  $\text{K}^+$  in the luminal fluid in the terminal CCD ( $K_{\text{CCD}}$ ). To calculate the  $[K_{\text{CCD}}]$ , we suggested dividing the  $U_{\text{K}}$  by the ratio of  $U_{\text{osm}}$  to the  $P_{\text{osm}}$  ( $U_{\text{osm}}/P_{\text{osm}}$ ) because the osmolality in the luminal fluid in the terminal CCD should be equal to  $P_{\text{osm}}$  when vasopressin causes the insertion of AQP2 into the luminal membranes of principal cells in the CCD.

The assumption made with use of the  $U_{\text{osm}}/P_{\text{osm}}$  ratio is that most of the osmoles delivered to the MCD are not reabsorbed in this nephron segment, and therefore the rise in osmolality from its value in the terminal CCD and its value in the urine reflects water reabsorption in the MCD. Although the amount of electrolytes reabsorbed in the MCD should not pose a problem, this is not true for urea because of intrarenal urea recycling. We estimated that in subjects eating a typical Western diet, almost 600 mmol of urea is reabsorbed downstream from the CCD/day. It follows that the calculated  $K_{\text{CCD}}$  obtained from  $(U_{\text{K}} \div [U_{\text{osm}}/P_{\text{osm}}])$  is likely to be appreciably higher than the actual value in vivo.<sup>21</sup> Therefore we no longer use the TTKG in the clinical assessment of patients with dyskalemia. Rather, we use the  $U_{\text{K}}/U_{\text{Cr}}$  ratio to provide the information needed to assess the renal response in these patients.

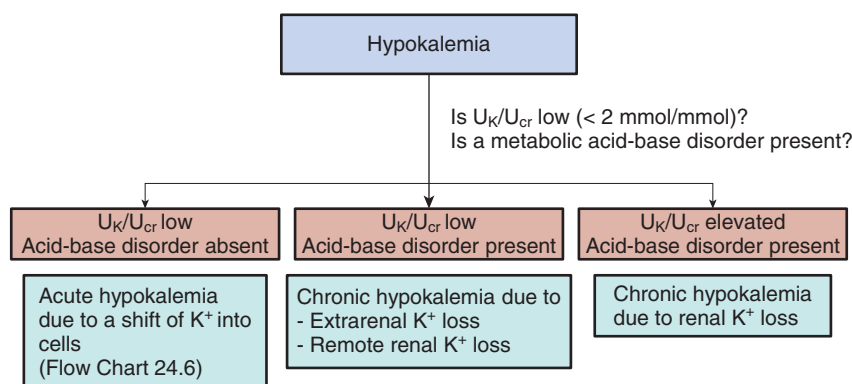
### Establishing the Basis for the Abnormal Rate of Excretion of Potassium

In a patient with hypokalemia, a higher than expected rate of excretion of  $\text{K}^+$  implies that there is a higher lumen-negative voltage in the ASDN, and that open ROMK channels are present in the luminal membranes of principal cells. The higher lumen-negative voltage is due to a higher rate of electrogenic  $\text{Na}^+$  reabsorption in the ASDN. The converse is true in a patient with hyperkalemia, in whom there is a lower than expected rate of excretion of  $\text{K}^+$ .

The clinical indices that help in the differential diagnosis of the pathophysiology of the abnormal rate of electrogenic reabsorption of  $\text{Na}^+$  in the ASDN are an assessment of the EABV and the presence or absence of hypertension. The measurement of the plasma renin concentration or activity ( $P_{\text{Renin}}$ ) and the plasma aldosterone concentration ( $P_{\text{Aldo}}$ ) are helpful in this differential diagnosis (Table 24.4).

**Table 24.4** Use of Plasma Renin and Plasma Aldosterone Values to Assess the Basis of Hypokalemia or Hyperkalemia

| Parameter                                                        | Renin Value | Aldosterone Value |
|------------------------------------------------------------------|-------------|-------------------|
| <b>Lesions That Cause Hypokalemia</b>                            |             |                   |
| <b>Adrenal Gland</b>                                             |             |                   |
| • Primary hyperaldosteronism                                     | Low         | High              |
| • Glucocorticoid-remediable hyperaldosteronism                   | Low         | High              |
| <b>Kidney</b>                                                    |             |                   |
| • Renal artery stenosis                                          | High        | High              |
| • Malignant hypertension                                         | High        | High              |
| • Renin-secreting tumor                                          | High        | High              |
| • Liddle syndrome                                                | Low         | Low               |
| • Disorders involving 11 $\beta$ -hydroxysteroid dehydrogenase-2 | Low         | Low               |
| <b>Lesions That Cause Hyperkalemia</b>                           |             |                   |
| <b>Adrenal Gland</b>                                             |             |                   |
| Addison disease                                                  | High        | Low               |
| <b>Kidney</b>                                                    |             |                   |
| Pseudohypoaldosteronism type 1                                   | High        | High              |
| Hyporeninemic hypoaldosteronism                                  | Low         | Low               |

**Flow Chart 24.5** (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

## CLINICAL APPROACH TO THE PATIENT WITH HYPOKALEMIA

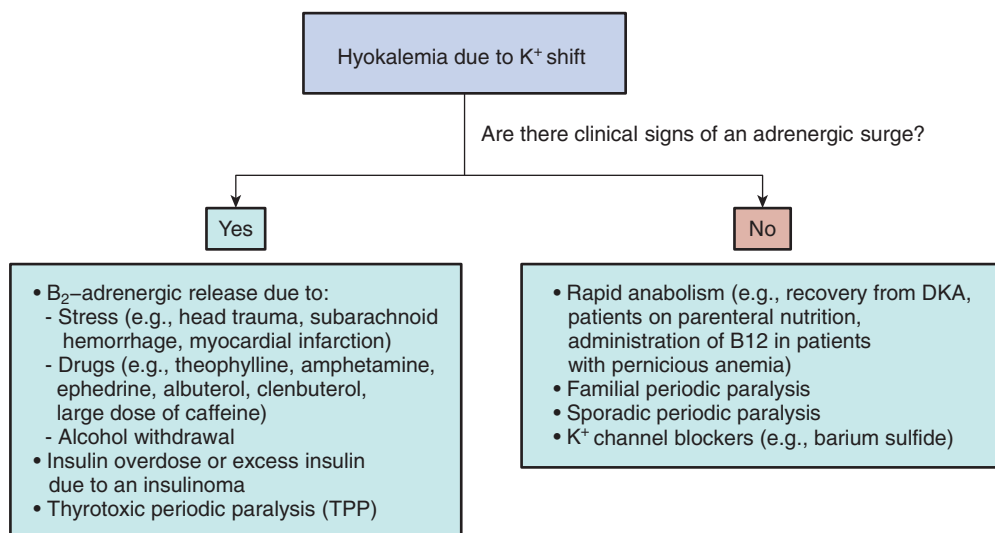
### Step 1. Deal With Medical Emergencies That May Be Present on Presentation, and Anticipate and Prevent Risks That May Arise During Therapy

The major emergencies related to hypokalemia are cardiac arrhythmias and respiratory muscle weakness leading to respiratory failure. Patients with chronic hyponatremia and hypokalemia are also at a high risk for the development of osmotic demyelination, with a rapid rise in  $P_{Na}$ . Administration of KCl may lead to a rapid rise in  $P_{Na}$ <sup>22</sup> because  $K^+$  ions enter and  $Na^+$  ions leave muscle cells. This movement of  $Na^+$  ions into the ECF compartment may also expand the EABV, leading to an increase in distal delivery of filtrate, resulting in water diuresis.

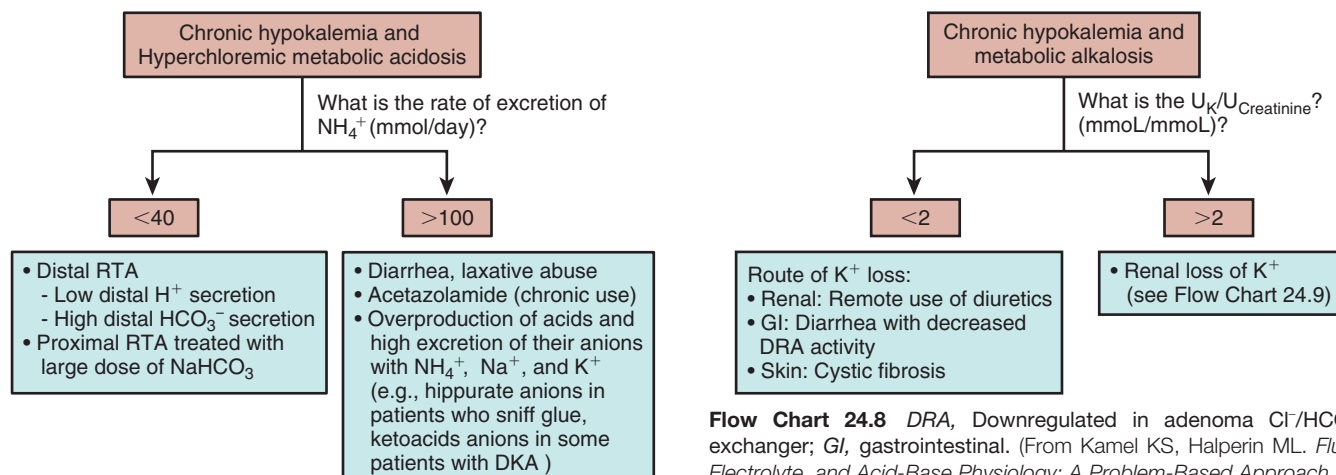
### Step 2. Determine Whether the Basis for Hypokalemia Is an Acute Shift of Potassium Into Cells

A low rate of excretion of  $K^+$  ( $U_K/U_{Cr} < 18$  mmol of  $K^+$ /g of creatinine or  $< 2$  mmol of  $K^+$ /mmol of creatinine) and the absence of a metabolic acid-base disorder suggest that the major basis of hypokalemia is an acute shift of  $K^+$  into cells (Flow Chart 24.5).<sup>23</sup>

Having established that the major pathophysiology of the hypokalemia is an acute shift of  $K^+$  into cells, the next step is to determine whether an adrenergic surge may have caused this shift of  $K^+$  (Flow Chart 24.6). In these settings, tachycardia, a wide pulse pressure, and systolic hypertension are often present. It is very important to recognize patients with these features because the administration of nonspecific beta-blockers can lead to a very rapid recovery without the



**Flow Chart 24.6** DKA, Diabetic ketoacidosis. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)



**Flow Chart 24.7** DKA, Diabetic ketoacidosis; RTA, renal tubular acidosis. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

administration of a large amount of KCl and hence avoid the risk of the development of rebound hyperkalemia when the stimulus for shift of K<sup>+</sup> ions abates.<sup>24</sup>

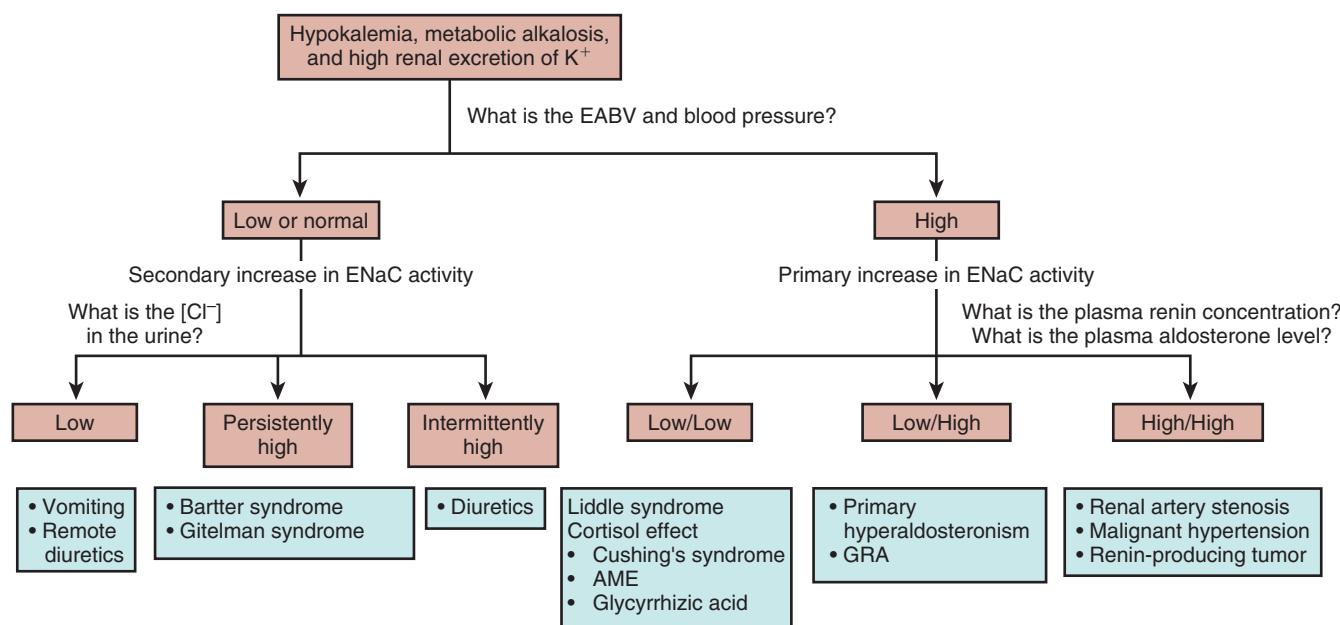
### Step 3. Examine the Acid-Basis Status in the Patient With Chronic Hypokalemia

In the patient with chronic hypokalemia, the first step is to examine the acid-base status in plasma.

**Subgroup With Metabolic Acidosis.** The group of patients with metabolic acidosis (usually hyperchloremic metabolic acidosis) can be divided into two categories according to the rate of excretion of NH<sub>4</sub><sup>+</sup> in the urine (Flow Chart 24.7). The rate of excretion of NH<sub>4</sub><sup>+</sup> can be estimated with use of the urine osmolal gap (see discussion of metabolic acidosis).

**Subgroup With Metabolic Alkalosis.** The first step in the patient with metabolic alkalosis is to determine whether the site of loss of K<sup>+</sup> is renal or extrarenal on the basis of the assessment of the rate of renal excretion of K<sup>+</sup> using the U<sub>K</sub>/U<sub>Cr</sub> (Flow Chart 24.8). Patients with a low value for this ratio (i.e., <18 mmol of K<sup>+</sup>/g of creatinine or <2 mmol of K<sup>+</sup>/mmol of creatinine) have conditions with loss of K<sup>+</sup> via nonrenal routes, such as in sweat (e.g., patients with cystic fibrosis) or via the GI tract (e.g., patients with diarrhea associated with decreased activity of the colonic luminal Cl<sup>-</sup>/HCO<sub>3</sub><sup>-</sup> exchanger, DRA, such as congenital chloridorrhea, villous adenoma, and some patients with laxative abuse). On the other hand, patients in whom the U<sub>K</sub>/U<sub>Cr</sub> is higher than these values have a condition associated with a renal loss of K<sup>+</sup>.<sup>24a</sup> The steps to take to determine the underlying pathophysiology in this latter group of patients are outlined in Flow Chart 24.9.

In essence, we are trying to determine the cause of a higher rate of electrogenic reabsorption of Na<sup>+</sup> in the ASDN. The



**Flow Chart 24.9** AME, Apparent mineralocorticoid excess syndrome; EABV, effective arterial blood volume; GRA, glucocorticoid remediable aldosteronism. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

primary reason is an increased number of open ENaC units in the luminal membranes of principal cells in the ASDN. This increase could be due to two groups of disorders.

The first group involves a secondary increase in ENaC activity due to the release of aldosterone in response to a low EABV. Patients with these disorders are not likely to have high blood pressure. The most common causes are vomiting and the use of diuretic agents. In some patients, the diuretic effect may be due to an inherited disorder affecting NaCl reabsorption in the medullary TAL (i.e., Bartter syndrome) or in the DCT (i.e., Gitelman syndrome). Ligands that occupy the calcium-sensing receptor in the medullary TAL (e.g., ionized calcium in a patient with hypercalcemia), drugs (e.g., gentamicin, cisplatin), and possibly cationic proteins (e.g., cationic monoclonal immunoglobulins in a patient with multiple myeloma) could potentially mimic Bartter syndrome. The use of urine electrolyte values in the differential diagnosis of hypokalemia in a patient with a contracted EABV is summarized in Table 24.3. It is currently recommended that hypertensive patients who develop hypokalemia while taking diuretics be screened for primary hyperaldosteronism.

The second group of disorders involves conditions that are associated with a primary increase in ENaC activity (e.g., primary hyperreninemic hyperaldosteronism, primary hyperaldosteronism, disorders in which cortisol acts as a mineralocorticoid in the ASDN, or a constitutively active ENaC in the luminal membrane of principal cells in the ASDN). The EABV in these patients is not low, and the blood pressure is commonly high.

In some patients a decreased rate of electroneutral Na<sup>+</sup> reabsorption may contribute to the increased rate of electrogenic Na<sup>+</sup> reabsorption and enhanced kaliuresis. This may be the case when Na<sup>+</sup> is delivered to the ASDN

with little Cl<sup>-</sup> (e.g., delivery of Na<sup>+</sup> with HCO<sub>3</sub><sup>-</sup> in a patient with recent vomiting or with an anion of a drug such as penicillin).

Magnesium (Mg<sup>2+</sup>) deficiency is frequently associated with hypokalemia. This relationship is largely due to the underlying disorders that cause losses of both Mg<sup>2+</sup> and K<sup>+</sup> (e.g., diarrhea, diuretic therapy, Gitelman syndrome). K<sup>+</sup> secretion in the ASDN is mediated by ROMK, a process that is inhibited by intracellular Mg<sup>2+</sup>. A decrease in intracellular Mg<sup>2+</sup>, caused by Mg<sup>2+</sup> deficiency, releases the Mg<sup>2+</sup>-mediated inhibition of ROMK. Mg<sup>2+</sup> deficiency alone, however, does not necessarily cause hypokalemia, because an increase in the rate of electrogenic reabsorption of Na<sup>+</sup> is required to enhance the rate of secretion of K<sup>+</sup>.

#### CLINICAL CASE 7: HYPOKALEMIA AND A LOW RATE OF POTASSIUM EXCRETION

A 28-year-old Asian woman presented with sudden onset of generalized muscle weakness and inability to ambulate on awakening one morning. She had lost 7 kg of body weight in the last 2 months but denied nausea, vomiting, diarrhea, or the use of diuretics, laxatives, exogenous thyroid hormone, herbal medications, or illicit drugs. The attack was not preceded by strenuous exercise or the consumption of a carbohydrate-rich meal. She had no family history of hypokalemia, periodic paralysis, or hyperthyroidism. On physical examination, she was alert and oriented; blood pressure was 150/70 mm Hg, heart rate was 116 beats/min, and respiratory rate was 18 breaths/min. The thyroid gland was not obviously enlarged, and there was no exophthalmos. Symmetric flaccid paralysis with areflexia was present in all four limbs. The remainder of the physical examination was unremarkable. The pH and PCO<sub>2</sub> values shown in the following table of laboratory findings were from an arterial blood sample,



whereas all other blood data were from a venous blood sample. The electrocardiogram (ECG) showed sinus tachycardia and prominent U waves.

| Findings                               | Blood                    | Urine                    |
|----------------------------------------|--------------------------|--------------------------|
| K <sup>+</sup> , mmol/L                | 1.8                      | 12                       |
| Creatinine                             | 0.7 mg/dL<br>(62 μmol/L) | 1.9 g/L<br>(16.8 mmol/L) |
| Na <sup>+</sup> , mmol/L               | 140                      | 179                      |
| Cl <sup>-</sup> , mmol/L               | 108                      | 184                      |
| pH                                     | 7.41                     | —                        |
| PCO <sub>2</sub> , mm Hg               | 36                       | —                        |
| HCO <sub>3</sub> <sup>-</sup> , mmol/L | 23                       | —                        |
| Glucose, mg/dL                         | 112                      | 0                        |

### Questions and Discussion

Is there a medical emergency? Because the ECG did not show changes due to hypokalemia other than U waves, and because respiratory muscle weakness causing hypoventilation was not present, as assessed from the arterial PCO<sub>2</sub>, there were no emergencies related to hypokalemia at this time.

What is the basis of the hypokalemia? A U<sub>K</sub>/U<sub>Cr</sub> ratio less than 1 (mmol/mmol) and the absence of a metabolic acid-base disorder on presentation suggest that the basis of the severe hypokalemia in this patient was an acute shift of K<sup>+</sup> into cells.

Possible reasons for potassium shift into cells: The presence of tachycardia, systolic hypertension, and wide pulse pressure suggest that an adrenergic surge was the cause of the acute shift of K<sup>+</sup> into cells. On further laboratory testing, she was found to have hyperthyroidism, so the diagnosis was thyrotoxic periodic paralysis (TPP). Patients with TPP often have only subtle signs and symptoms of thyrotoxicosis. In contrast to standard teaching, most patients with TPP do not have clear precipitating factors such as strenuous exercise or the consumption of a carbohydrate-rich meal. Increased Na<sup>+</sup>-K<sup>+</sup>-ATPase activity has traditionally been implicated in the pathogenesis of TPP. Studies have shown that susceptibility to TPP can be conferred by mutations in the skeletal muscle-specific, inward-rectifying K<sup>+</sup> (Kir) channel, Kir2.6.

What are the options for therapy? The patient received intravenous KCl at a rate of 10 mmol/hour by the administration of a solution of 0.9% saline with 40 mEq/L KCl at a rate of 250 mL/hour. In a patient with severe hypokalemia, one should not use a dextrose-containing solution because this may cause the release of insulin, which could induce a further shift of K<sup>+</sup> into cells and aggravate the severity of hypokalemia. Although she received only 80 mmol of KCl, rebound hyperkalemia (P<sub>K</sub>, 5.7 mmol/L) developed; the P<sub>K</sub> returned to normal 6 hours later. Studies have suggested that hypokalemia in patients with TPP can be rapidly corrected without the risk of rebound hyperkalemia by the administration of a nonselective beta-blocker and only a small dose of KCl.

### CLINICAL CASE 8: HYPOKALEMIA AND HIGH RATE OF POTASSIUM EXCRETION

Progressive muscle weakness developed over the last 6 hours in a 76-year-old Asian man that became so severe that he was unable to move. He reported that he exercised that morning and ate a large carbohydrate meal for breakfast, which he usually does after he has a good exercise session. He had no other neurologic symptoms. He denied vomiting,

or diarrhea, or the use of diuretics or laxatives. Hypokalemia (P<sub>K</sub>, 3.3 mmol/L) and hypertension had been noted 1 year ago but had not been investigated further. On this admission, his blood pressure was 160/96 mm Hg and his heart rate was 70 beats/min. A neurologic examination revealed symmetric flaccid paralysis with areflexia but no other findings. The laboratory data prior to therapy are shown in the following table. The pH and PCO<sub>2</sub> values are from an arterial blood sample. Results of measurement of the P<sub>Renin</sub> and P<sub>Aldo</sub>, which became available later, showed that both levels were low. The plasma cortisol value was in the normal range.

| Findings                                | Blood                    | Urine                   |
|-----------------------------------------|--------------------------|-------------------------|
| K <sup>+</sup> , mmol/L                 | 1.8                      | 26                      |
| Na <sup>+</sup> , mmol/L                | 147                      | 132                     |
| Cl <sup>-</sup> , mmol/L                | 90                       | 138                     |
| Creatinine                              | 0.8 mg/dL<br>(70 μmol/L) | 0.6 g/L<br>(5.3 mmol/L) |
| pH                                      | 7.55                     | —                       |
| PCO <sub>2</sub> , mm Hg                | 40                       | —                       |
| HCO <sub>3</sub> <sup>-</sup> , mmol/L  | 45                       | 0                       |
| Osmolality, mOsm/kg<br>H <sub>2</sub> O | 302                      | 482                     |

### Questions and Discussion

What is the cause of hypokalemia in this patient? In the presence of hypokalemia, the U<sub>K</sub>/U<sub>Cr</sub> was 5 (mmol/mmol), and metabolic alkalosis was also present. Hence, the hypokalemia was largely due to a disorder that caused excessive loss of K<sup>+</sup> in the urine. The acute presentation with extreme weakness was likely due to an acute shift of K<sup>+</sup> into cells in conjunction with a chronic disorder that caused the loss of K<sup>+</sup>. This component, of an acute shift of K<sup>+</sup> into cells, was attributed to the predominant β<sub>2</sub>-adrenergic effect during rest after vigorous exercise, and the release of insulin due to the large carbohydrate intake during breakfast prior to the onset of symptoms.

On clinical assessment, the patient's EABV volume was thought not to be contracted, and he had hypertension. Therefore, the increased electrogenic reabsorption of Na<sup>+</sup> in the ASDN was due to a primary increase in ENaC activity. The differential diagnosis was guided by measurements of the P<sub>Renin</sub> and P<sub>Aldo</sub> (see Table 24.4). Because both the P<sub>Aldo</sub> and P<sub>Renin</sub> were suppressed, the differential diagnosis was between disorders in which cortisol acts as mineralocorticoid and those with constitutively active ENaC in the luminal membranes of principal cells in the ASDN. Inherited disorders in which ENaC is constitutively active (Liddle syndrome) seemed unlikely, considering the patient's age. Plasma cortisol values were not elevated. Computed tomography of the chest did not show a lung mass. Although the patient denied consuming licorice or chewing tobacco, it turned out that he used an herbal preparation containing large amounts of glycyrrhizic acid (the active ingredient in licorice) to sweeten his tea. The patient was treated initially with intravenous KCl; the weakness improved when the P<sub>K</sub> reached 2.5 mmol/L. Oral KCl supplementation was continued. Two weeks later, P<sub>K</sub> and blood pressure values returned to normal levels, and his body weight had decreased from 78 to 74 kg as a result of natriuresis after stopping the glycyrrhizic acid-containing herbal preparation.

## CLINICAL APPROACH TO THE PATIENT WITH HYPERKALEMIA

### Step 1. Address Emergencies

Hyperkalemia constitutes a medical emergency, primarily because of its effect on the heart, which may lead to cardiac conduction abnormalities, arrhythmias and, ultimately, asystole.

### Step 2. Determine Whether the Cause of the Hyperkalemia Is an Acute Shift of Potassium Out of Cells *in vivo* or Pseudohyperkalemia

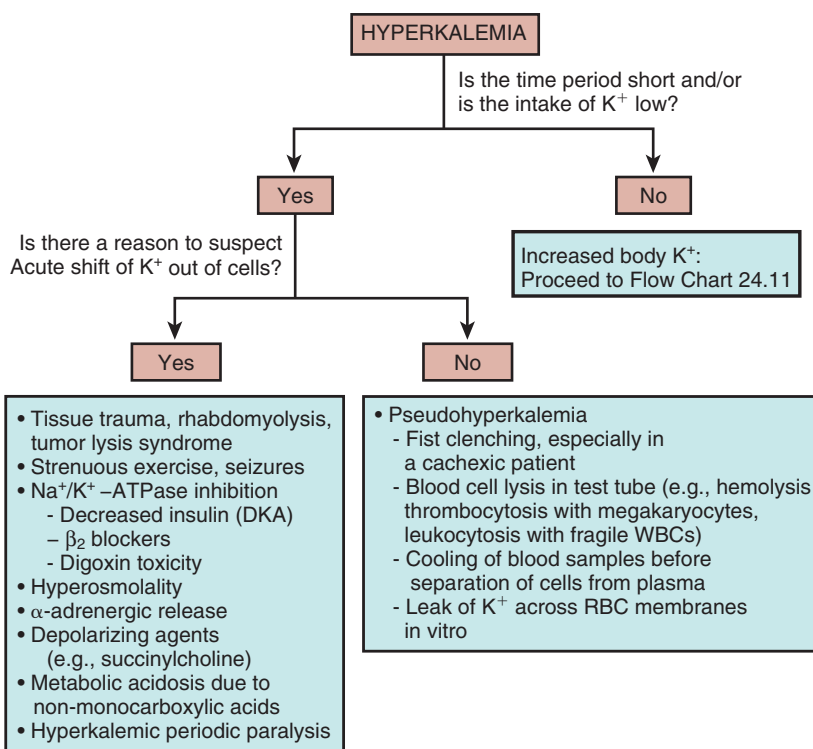
Flow Chart 24.10 illustrates the initial approach for determining the cause of the hyperkalemia. If the time course for the development of hyperkalemia was short and/or the hyperkalemia developed while the intake of  $K^+$  was low, then the following three categories should be considered.

**Destruction of Cells in the Body.** Cell destruction could be due, for example, to rhabdomyolysis or tumor lysis syndrome.

**Shift of  $K^+$  Out of Cells in the Body.** Shift of  $K^+$  out of cells in the body may occur in conditions in which there is a less negative voltage in cells. Such conditions include lack of a stimulus for  $Na^+K^+$ -ATPase (e.g., lack of insulin in patients with diabetic ketoacidosis [DKA],  $\beta_2$ -adrenergic blockade), and conditions in which there is an  $\alpha$ -adrenergic surge causing inhibition of the release of insulin or directly causing a shift of  $K^+$  out of hepatic cells. Digoxin inhibits  $Na^+K^+$ -ATPase, so digoxin overdose can result in hyperkalemia. A shift of  $K^+$  may also occur in conditions with metabolic acidosis due to nonmonocarboxylic acids—that is, acids that cannot be

transported on the MCT (e.g., metabolic acidosis due to a gain of HCl from loss of  $NaHCO_3$  in a patient with diarrhea, ingestion of citric acid). Acute hyperkalemia may occur during exhaustive exercise or in patients with status epilepticus. Severe hyperkalemia has been described as a complication of the administration of mannitol for the treatment or prevention of cerebral edema. This is because a rise in effective osmolality in the interstitial fluid causes the movement of water out of cells via AQP channels in cell membranes, which raises the concentration of  $K^+$  in the ICF and provides a chemical driving force for the movement of  $K^+$  out of cells. Succinylcholine depolarizes muscle cells, resulting in the efflux of  $K^+$  through acetylcholine receptors in conditions that may lead to the upregulation of acetylcholine receptors (e.g., burns, neuromuscular injury [upper or lower motor neuron], disuse atrophy, prolonged immobilization). Fluoride can open the  $Ca^{2+}$ -sensitive  $K^+$  channels; as a result, fluoride intoxication can lead to fatal hyperkalemia. A positive family history for acute hyperkalemia suggests that there may be a molecular basis for this disorder (e.g., hyperkalemic periodic paralysis).

**Pseudohyperkalemia May Be Present.** The presence of changes in the ECG related to hyperkalemia rules out pseudohyperkalemia as the sole cause of the hyperkalemia. Pseudohyperkalemia is caused by the release of  $K^+$  during or after venipuncture. Excessive fist clenching during blood sampling may increase  $K^+$  release from local muscle and thus may raise the measured  $P_K$  by as much as 1 mmol/L. Pseudohyperkalemia can be present in cachectic patients in whom the normal T tubule architecture in skeletal muscle may be disturbed. Mechanical trauma to red blood cells



**Flow Chart 24.10** DKA, Diabetic ketoacidosis; RBC, red blood cells; WBC, white blood cells. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology: A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

during venipuncture, due for example to the use of a very small-gauge needle, can result in their hemolysis and the release of  $K^+$  in the test tube.  $K^+$  ions are normally released from platelets during blood clotting, so pseudohyperkalemia may be noted in patients with thrombocytosis (especially with megakaryocytosis). Pseudohyperkalemia may also be present in patients with severe leukocytosis, especially due to fragile leukemia cells, because of the breakdown of cells during venipuncture, when shaken by pneumatic transport, or during centrifugation. Cooling of blood prior to the separation of cells from plasma is another cause of pseudohyperkalemia. There are several hereditary subtypes of pseudohyperkalemia caused by an increase in passive  $K^+$  permeability of erythrocytes. The  $P_K$  increases in blood samples from patients with this disorder when left at room temperature.

### Step 3. What Is the Rate of Potassium Excretion?

In a patient with chronic hyperkalemia, pseudohyperkalemia should be ruled out first. In normal subjects, a  $K^+$  load can augment the rate of excretion of  $K^+$  to more than 200 mmol/day, with only a modest increase in  $P_K$ . Thus, patients with chronic hyperkalemia generally have some defect in renal  $K^+$  excretion. In a steady state, they excrete what they eat

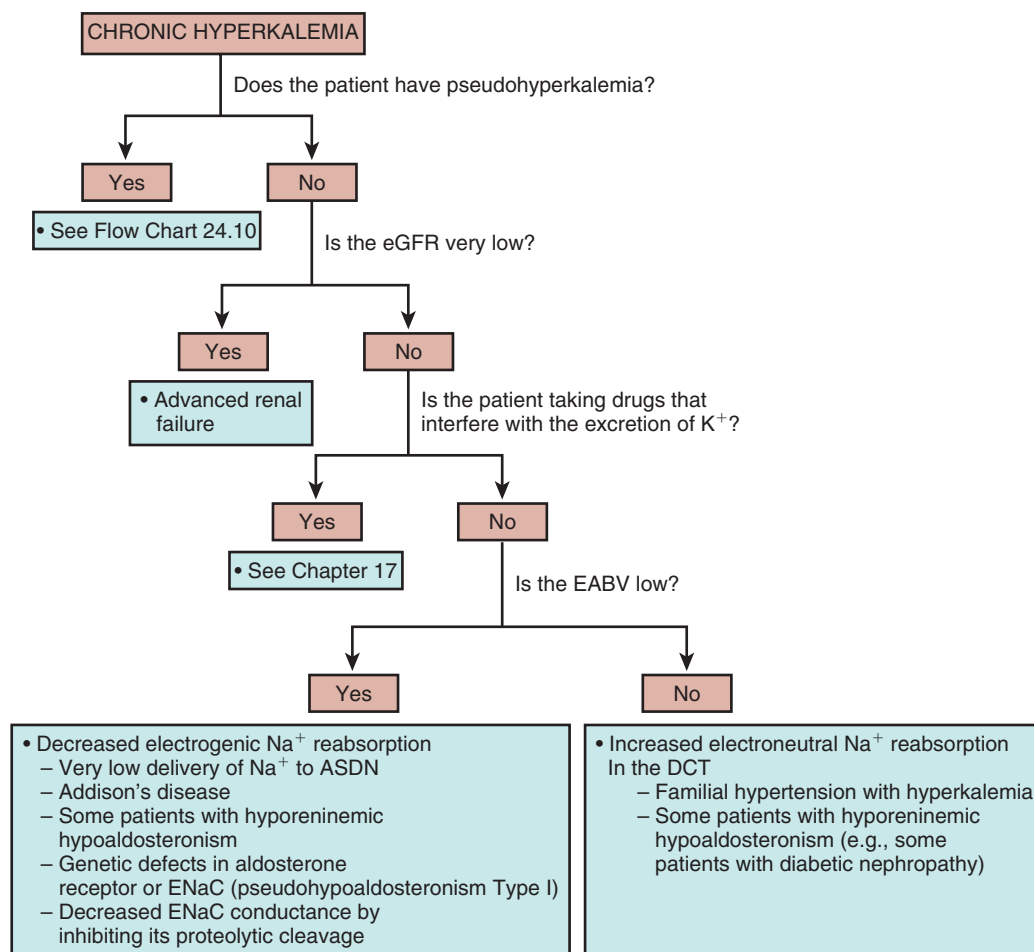
(minus the amount of  $K^+$  lost in stool), but at the expense of maintaining hyperkalemia. Therefore, the value of assessing the rate of  $K^+$  excretion is primarily to determine the contribution of  $K^+$  intake to the degree of hyperkalemia. A 24-hour urine collection is necessary for this purpose, rather than determination of  $U_K/U_{Cr}$  in a spot urine sample, because of the diurnal variation in  $K^+$  excretion.

### Step 4. What Is the Basis for the Defect in Renal Potassium Excretion?

The clinical approach to determine the basis for the renal defect in  $K^+$  excretion is illustrated in [Flow Chart 24.11](#). More than one cause may be present in a single patient (e.g., intake of a drug that blocks the renin-angiotensin-aldosterone system [RAAS] in a patient with chronic renal insufficiency).

**Does the Patient Have Advanced Chronic Renal Failure?** The estimated GFR is generally less than 15 ml/minute if this is the sole cause of hyperkalemia.

**Is the patient taking drugs that interfere with the renal excretion of potassium?** Drugs that may interfere with the renal excretion of  $K^+$  include inhibitors of the RAAS axis, inhibitors of ENaC (including amiloride, trimethoprim, and



**Flow Chart 24.11** ASDN, Aldosterone sensitive distal nephron; DCT, distal convoluted tubule; EABV, effective arterial blood volume; eGFR, estimated glomerular filtration rate; ENaC, epithelial sodium channel. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

pentamidine), and serine protease inhibitors that interfere with activation of ENaC via proteolytic cleavage (nafamostat). See Chapter 17 for further discussion.

**Does the Patient Have a Disorder That Leads to Diminished Reabsorption of Sodium via ENaC in the ASDN?** One subgroup of patients may have a very low delivery of  $\text{Na}^+$  to the ASDN due to marked decrease in the EABV. A second subgroup consists of patients who have lesions that lead to a diminished number of open ENaC units in the luminal membranes of principal cells in the ASDN. This includes patients who have hypoaldosteronism (e.g., adrenal insufficiency) and those with molecular defects that involve the aldosterone receptor or ENaC. Patients in this subgroup have a low EABV, higher than expected rates of excretion of  $\text{Na}^+$  and  $\text{Cl}^-$  in a setting of low EABV and a high  $P_{\text{Renin}}$ . The  $P_{\text{Aldo}}$  is helpful to determine the reason for this diminished  $\text{Na}^+$  reabsorption via ENaC in the ASDN.

A subset of patients with hypoaldosteronism has low  $P_{\text{Renin}}$  (hyporeninemic hypoaldosteronism) and a low EABV. Their lesions may be destruction of or a biosynthetic defect in the juxtaglomerular apparatus, leading to a low  $P_{\text{Renin}}$  and thereby a low  $P_{\text{Aldo}}$ . Patients with such disorders are expected to have a significant rise in  $U_{\text{K}}/U_{\text{Cr}}$  with the administration of exogenous mineralocorticoids for several days.

**Does the Patient Have a Disorder That Increases Electroneutral Sodium Reabsorption in the Distal Convoluted Tubule?** In this group of patients, the site of the lesion is the DCT1, where there is increased reabsorption of  $\text{Na}^+$  and  $\text{Cl}^-$  via NCC, due to its activation by WNK4 or L-WNK1. Suppression of release of aldosterone by an expanded EABV leads to a diminished number of open ENaC units in the luminal membranes of principal cells in the ASDN. This, together with decreased delivery of  $\text{Na}^+$  from the DCT lumen to the ASDN, and also a decreased flow rate, abrogates electrogenic  $\text{Na}^+$  reabsorption and flow-induced K secretion (see Chapter 6). These kinases also cause endocytosis of ROMK from the luminal membranes of principal cells in the ASDN. Such patients tend to have an expanded EABV, hypertension, and suppressed  $P_{\text{Renin}}$  and  $P_{\text{Aldo}}$  (hyporeninemic hypoaldosteronism). Patients with this pathophysiology are expected to show a good response to the administration of thiazide diuretics in terms of lowering of blood pressure and correction of hyperkalemia.

The clinical picture in patients with the syndrome of familial hyperkalemia with hypertension (also known as pseudohypoaldosteronism type II or Gordon syndrome) resembles that of a gain-of-function in the thiazide-sensitive NCC. Major deletions in the gene encoding for WNK1 and missense mutations in the gene encoding for WNK4 have been reported in these patients.<sup>25</sup> A set of clinical findings similar to those in patients with familial hyperkalemia with hypertension may occur in other patients, most commonly those with diabetic nephropathy.<sup>20</sup> Support for the hypothesis that suppression of renin release in these patients is the result of EABV expansion are the findings that circulating atrial natriuretic peptide blood levels are elevated in these patients and many show response to NaCl restriction or furosemide therapy with an increased  $P_{\text{Renin}}$ . Another example of this pathophysiology is the hyperkalemia in patients treated with calcineurin inhibitors.<sup>26</sup>

**Does the Patient Have a Disorder That Increases Electroneutral Reabsorption of Sodium in the Cortical Collecting Duct?** We hypothesize that the pathophysiology of the hyperkalemia in some patients may be due to increased electroneutral  $\text{Na}^+$  reabsorption in the CCD via parallel increases in the transport activity of pendrin and the NDCBE. This may be the pathophysiology for what used to be thought of as chloride shunt disorder.<sup>27</sup> These patients will also have an expanded EABV and suppressed  $P_{\text{Renin}}$  and  $P_{\text{Aldo}}$  (hyporeninemic hypoaldosteronism). Increasing the concentration of  $\text{HCO}_3^-$  in the luminal fluid in the CCD may diminish flux through pendrin and thereby NDCBE. Hence we speculate that this subset of patients may be more responsive, in terms of increasing  $\text{K}^+$  excretion, to the induction of bicarbonaturia by the administration of the carbonic anhydrase inhibitor acetazolamide than to the administration of thiazide diuretics. This speculation, however, needs to be examined in a clinical study.

### Step 5. Is a Low Flow Rate in the ASDN Contributing to Hyperkalemia?

Because of the process of intrarenal urea recycling, a large fraction of the osmoles delivered to the ASDN is urea osmoles. A low protein intake may decrease the amount of urea that recycles and hence the rate of flow in the ASDN. The usual rate of excretion of urea in subjects consuming a typical Western diet is about 400 mmol/day. If the rate of excretion of urea is appreciably lower than that, a low flow rate in the ASDN may be a contributing factor to the degree of hyperkalemia.

### CLINICAL CASE 9: HYPERKALEMIA IN A PATIENT TAKING TRIMETHOPRIM

*Pneumocystis jiroveci* pneumonia (PJP) developed in a 35-year-old cachectic man with human immunodeficiency virus (HIV) infection. On admission, he was febrile, there were no physical findings indicating contraction of his EABV volume, and all plasma electrolyte values were in a normal range. He was treated with cotrimoxazole (sulfamethoxazole and trimethoprim). Three days later, he was noted to have low blood pressure, his EABV was low, and his  $P_{\text{K}}$  rose to 6.8 mmol/L. An ECG showed tall, peaked, narrow-based T waves. The urine volume was 0.8 L/day, and the  $U_{\text{osm}}$  value was 350 mOsm/kg  $\text{H}_2\text{O}$ ; other laboratory findings were as follows.

| Findings                  | Blood            | Urine                             |
|---------------------------|------------------|-----------------------------------|
| $\text{K}^+$ , mmol/L     | 6.8              | 14                                |
| $\text{Na}^+$ , mmol/L    | 130              | 60                                |
| $\text{Cl}^-$ , mmol/L    | 105              | 43                                |
| Creatinine                | 0.9 mg/dL        | 0.8 g/L                           |
| pH                        | 7.30             | —                                 |
| $\text{PCO}_2$ , mm Hg    | 30               | —                                 |
| $\text{HCO}_3^-$ , mmol/L | 15               | 0                                 |
| Urea                      | BUN:<br>14 mg/dL | Urea concentration:<br>280 mmol/L |

### Questions and Discussion

What is the cause of hyperkalemia in this patient? The steps to follow are provided in [Flow Charts 24.10 and 24.11](#). Although an element of pseudohyperkalemia could have been present because of repeated fist clenching during



venipuncture in this cachectic patient, the ECG changes indicated that he had true hyperkalemia.

Is the time course for the development of hyperkalemia short and/or did hyperkalemia develop while the intake of  $K^+$  was low? The  $U_K$  was 14 mmol/L, and his rate of excretion of  $K^+$  was extremely low in the presence of hyperkalemia ( $U_K/U_{Cr}$  of 17.5 [mmol  $K^+$ /g creatinine]), so one might conclude that the major basis for the hyperkalemia is the low rate of  $K^+$  excretion. This severe degree of hyperkalemia developed over a relatively short period of time, and while the patient was consuming very little  $K^+$ . Therefore, a shift of  $K^+$  from cells rather than a large positive external balance for  $K^+$  is likely the major cause of hyperkalemia. The cause of this exit of  $K^+$  from cells could be inhibition of the release of insulin by the  $\alpha$ -adrenergic effect of adrenaline released in response to the low EABV. Nevertheless, he also had a large defect in renal  $K^+$  excretion. Because the EABV was low and  $U_{Na}$  and  $U_{Cl}$  were inappropriately high in the presence of a contracted EABV, the low  $U_K/U_{Cr}$  was due to diminished reabsorption of  $Na^+$  in the ASDN (see [Flow Chart 24.11](#)). The initial diagnosis was adrenal insufficiency due to an infection in a patient with HIV. However, plasma cortisol was measured and was appropriately high. In addition, there was no increase in  $U_K/U_{Cr}$  in response to the administration of an exogenous mineralocorticoid. It was concluded that the diminished  $Na^+$  reabsorption in the ASDN was due to inhibition of ENaC by the trimethoprim that was used to treat the PJP. Both the  $P_{Renin}$  and  $P_{Aldo}$  were high (results became available later), as expected in this setting (see [Table 24.4](#)).

**Interpretation.** Renal salt wasting due to blockade of ENaC by trimethoprim led to the development of a contracted EABV. The major cause of hyperkalemia was a shift of  $K^+$  out of cells, probably because of inhibition of insulin release by the binding of catecholamines, released in response to the low EABV, to pancreatic islet cell  $\alpha$ -adrenergic receptors. Blockade of ENaC by trimethoprim diminished the rate of electrogenic  $Na^+$  reabsorption in the ASDN and the generation of lumen-negative voltage for  $K^+$  secretion. Because of the low EABV and the low intake of proteins, there was a low rate of flow in the ASDN. This, in addition to diminishing the rate of  $K^+$  excretion, caused the concentration of trimethoprim to be higher in the lumen of the ASDN; hence, trimethoprim became a more effective blocker of ENaC.

What are the implications of the pathophysiology of hyperkalemia for the choice of treatment in this patient? Because of the presence of changes in the ECG related to hyperkalemia, insulin (with glucose to prevent hypoglycemia) was given to induce a shift of  $K^+$  into cells. The patient was given saline to expand his EABV, which would also suppress the release of catecholamines and remove the  $\alpha$ -adrenergic inhibition of insulin release. Because the basis of hyperkalemia in this patient is a shift of  $K^+$  out of cells, inducing a large loss of  $K^+$  is not needed because there is no total body  $K^+$  surplus. The question arose, thereafter, as to whether trimethoprim should be discontinued. Because the drug was needed to treat the PJP, a means to remove its renal ENaC-blocking effect was sought. The concentration of trimethoprim would fall in the lumen of the ASDN if the flow rate in the ASDN were to rise through an increase in the number of osmoles delivered to this nephron segment. To achieve this aim, one could increase the delivery of NaCl to the ASDN by inhibiting the reabsorption of  $Na^+$  and  $Cl^-$  in the TAL of the loop of Henle

using a loop diuretic, plus the infusion of enough NaCl to prevent EABV contraction. Because it is the cationic form of trimethoprim that blocks ENaC, increasing the delivery of  $HCO_3^-$  to the ASDN by inhibiting its reabsorption in the PT with administration of the carbonic anhydrase IV inhibitor acetazolamide could also be considered to lower the concentration of  $H^+$  in the luminal fluid in the ASDN and thereby the concentration of the cationic form of the drug. Enough  $NaHCO_3$  should be given to prevent the development of metabolic acidosis.

#### CLINICAL CASE 10: CHRONIC HYPERKALEMIA IN A PATIENT WITH TYPE 2 DIABETES MELLITUS

A 50-year-old man with a history of type 2 diabetes mellitus was referred for investigation of hyperkalemia; his  $P_K$  ranged from 5.5 to 6 mmol/L in a number of measurements that were done over the last several weeks. He was previously taking an angiotensin-converting enzyme (ACE) inhibitor for the treatment of hypertension, this medication was discontinued but hyperkalemia persisted. He is currently taking amlodipine, 10 mg, once a day. He was noted to have microalbuminuria but no other history of macrovascular or microvascular disease related to diabetes mellitus. On physical examination, his blood pressure was 160/90 mm Hg, his jugular venous pressure was about 2 cm above the level of the sternal angle, and he had pitting edema of the ankles bilaterally. Results of laboratory investigations were as follows:

| Parameter                      | Findings                |
|--------------------------------|-------------------------|
| $P_{Na}$ , mmol/L              | 140                     |
| $P_K$ , mmol/L                 | 5.7                     |
| $P_{Cl}$ , mmol/L              | 108                     |
| $P_{Renin}$ , ng/L             | 4.50 (range, 9.30–43.4) |
| $P_{Aldo}$ , pmol/L            | 321 (range 111–860)     |
| $PHCO_3$ , mmol/L              | 19                      |
| $P_{Alb}$ , g/L (mg/dL)        | 40 (4.0)                |
| $P_{Cr}$ , $\mu$ mol/L (mg/dL) | 100 (1.2)               |

#### Questions and Discussion

What is the cause of hyperkalemia in this patient? The first step is to rule out pseudohyperkalemia. The presence of hyperchloremic metabolic acidosis (HCMA) would suggest true hyperkalemia. Hyperkalemia is associated with an alkaline PT cell pH, which leads to the inhibition of ammoniogenesis.  $K^+$  ions compete with  $NH_4^+$  for transport on the  $Na^+-K^+-2Cl^-$  cotransporter in the TAL of the loop of Henle, which leads to decreased medullary interstitial availability of  $NH_3$ . Studies have suggested that hyperkalemia is also associated with decreased expression of the ammonia transporter Rh type C glycoprotein in the collecting duct.<sup>28</sup>

The patient did not have advanced renal dysfunction and was not currently taking drugs that might interfere with renal excretion of  $K^+$ . The  $P_{Renin}$  was decreased and  $P_{Aldo}$  was suppressed compared with what is expected in a patient with hyperkalemia. He was then thought to have hyporeninemic hypoaldosteronism, commonly labeled as type IV renal tubular acidosis (RTA). This disorder is traditionally thought to be the result of destruction of, or a biosynthetic defect in, the juxtaglomerular apparatus (JGA), leading to low  $P_{Renin}$  and thereby to low  $P_{Aldo}$ . If this were the pathophysiology, one would expect the patient to have renal salt wasting with a decreased EABV and the absence of hypertension. These

features are not found in many patients with this disorder. Another hypothesis is that suppression of renin release in patients with this disorder is the result of EABV expansion because levels of atrial natriuretic peptide in the blood are elevated, and many cases respond to NaCl restriction or the administration of furosemide with an increased  $P_{\text{Renin}}$ . The basis of the disorder remains to be established. It is possible that the reabsorption of  $\text{Na}^+$  and  $\text{Cl}^-$  may be augmented in the DCT as in patients with familial hyperkalemia with hypertension. Of interest in regard to patients with type 2 diabetes mellitus, who may have hyperinsulinemia and the metabolic syndrome, is the finding that long-term insulin infusion in rats is associated with the retention of NaCl owing to its enhanced reabsorption in different nephron segments, including the DCT.<sup>20</sup> Studies in hyperinsulinemic db/db mice have suggested that the phosphatidylinositol-3-kinase (PI3K)/Akt signaling pathway activates the WNK-NCC phosphorylation cascade.<sup>29</sup>

Differentiation between these two groups of patients with hyporeninemic hypoaldosteronism—those with a JGA defect or destruction and those with excessive  $\text{Na}^+$  and  $\text{Cl}^-$  reabsorption in the DCT—has implications for therapy. The use of exogenous mineralocorticoids (e.g.,  $9\alpha$ -fludrocortisone) is of benefit for the first group because it results in both a kaliuresis and re-expansion of the EABV. Diuretic therapy would pose a threat to these patients because it would cause a more severe degree of EABV contraction. In contrast, mineralocorticoids may aggravate the hypertension in patients with excessive reabsorption of  $\text{Na}^+$  and  $\text{Cl}^-$  in the DCT. In this group, the administration of thiazide diuretics to inhibit NCC should lead to both kaliuresis and lowering of the blood pressure.

## METABOLIC ALKALOSIS

Metabolic alkalosis is an electrolyte disorder that is accompanied by an elevated  $\text{PHCO}_3$  and a high plasma pH. Most patients with metabolic alkalosis have a deficit of NaCl, KCl, and/or HCl, any of which may lead to a higher  $\text{PHCO}_3$ .

### Concept 14

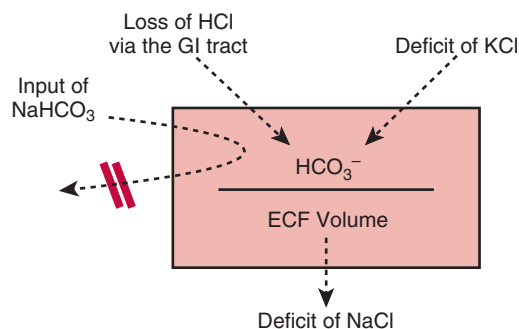
The concentration of  $\text{HCO}_3^-$  in the ECF compartment (assessed by the measurement of the concentration of  $\text{HCO}_3^-$  in plasma), is the ratio of the content of  $\text{HCO}_3^-$  in the ECF compartment (numerator) and the ECFV (denominator), as shown in Eq. 24.5.

$$[\text{HCO}_3^-] \text{ in ECF} = \text{quantity of } \text{HCO}_3^- \text{ in ECF} / \text{ECFV} \quad (24.5)$$

A rise in the concentration of  $\text{HCO}_3^-$  in the ECF compartment might be due to an increase in its numerator (positive balance of  $\text{HCO}_3^-$ ) and/or a decrease in its denominator (low ECFV) (Fig. 24.5; see Eq. 24.5). A quantitative assessment of the ECFV is critical to estimate the quantity of  $\text{HCO}_3^-$  in the ECF compartment and thereby to determine the basis of the metabolic alkalosis.

### Concept 15

Electroneutrality must be present in every body compartment and in the urine. We do not use the term  $\text{Cl}^-$  depletion alkalosis because it does not provide an adequate description of the pathophysiology of metabolic alkalosis. The deficit of

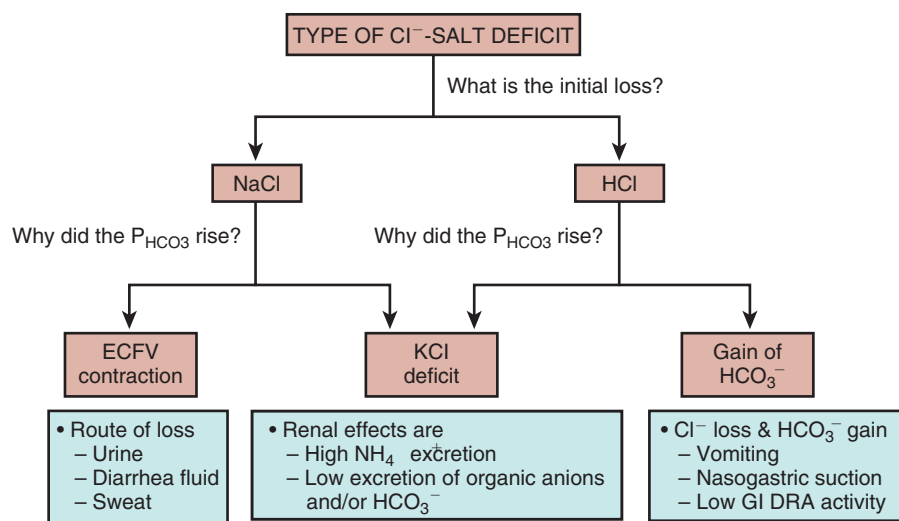


**Fig. 24.5** Basis for a high concentration of  $\text{HCO}_3^-$  in the extracellular fluid (ECF) compartment. The rectangle represents the ECF compartment. The concentration of  $\text{HCO}_3^-$  is the ratio of the content of  $\text{HCO}_3^-$  in the ECF compartment (numerator) and the ECF volume (ECFV) (denominator). The major causes for a rise in the content of  $\text{HCO}_3^-$  in the ECF compartment are a deficit of HCl and a deficit of KCl (the latter leads increased excretion of  $\text{NH}_4\text{Cl}$  and diminished excretion of organic anions in the urine). The major cause for a fall in the ECFV is a deficit of NaCl. An intake of  $\text{NaHCO}_3$  is not sufficient on its own to cause a sustained increase in the content of  $\text{HCO}_3^-$  in the ECF compartment unless there is also reduced renal output of  $\text{NaHCO}_3$  (indicated by the double red lines on the arrow on the left side of the figure) due to marked reduction in the glomerular filtration rate or there is another lesion that leads to maintaining the stimuli for the reabsorption of  $\text{NaHCO}_3$  in the proximal convoluted tubule. Double red lines on the arrow on the left portion of the figure indicate the reduced renal output of  $\text{NaHCO}_3$ . GI, Gastrointestinal. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

$\text{Cl}^-$  must be defined as being due to a deficit of HCl, KCl, and/or NaCl to determine why the  $\text{PHCO}_3$  level has gone up, what changes have occurred in the composition of the ECF and ICF compartments, and what is the appropriate therapy.<sup>30</sup> Although balance data are not available in clinical settings, with a quantitative assessment of the ECFV, tentative conclusions about the contribution of deficits of each of the different  $\text{Cl}^-$ -containing compounds to the development of metabolic alkalosis can be deduced (Flow Chart 24.12; see also discussion of Clinical Case 11).

### Concept 16

There is no tubular maximum for renal  $\text{HCO}_3^-$  reabsorption. Central to understanding the pathophysiology of metabolic alkalosis is that contrary to the widely held view, there is no tubular maximum for the reabsorption of  $\text{HCO}_3^-$ . If there were a tubular maximum for the reabsorption of  $\text{HCO}_3^-$ , then if the filtered load of  $\text{HCO}_3^-$  were to exceed this maximum, the excess  $\text{HCO}_3^-$  would be spilled out in the urine. Therefore, maintaining a high concentration of  $\text{HCO}_3^-$  in the ECF compartments requires a marked reduction in the GFR or a derangement in the tubular handling of  $\text{HCO}_3^-$  that increases the reabsorption of  $\text{HCO}_3^-$  above this tubular maximum. The vast majority of  $\text{HCO}_3^-$  reabsorption occurs in the PT via  $\text{H}^+$  secretion, mediated by the  $\text{Na}^+/\text{H}^+$  exchanger 3 (NHE3). The usual circulating levels of angiotensin II (Ang II), the usual concentration of  $\text{H}^+$  in PT cells, provide sufficient stimuli to NHE3 for the reabsorption of most of the filtered  $\text{HCO}_3^-$ . Experiments in which a tubular maximum for the reabsorption of  $\text{HCO}_3^-$  was demonstrated were carried out with the



**Flow Chart 24.12** DRA, Downregulated in adenoma Cl<sup>−</sup>/HCO<sub>3</sub><sup>−</sup> exchanger; ECFV, extracellular fluid volume; GI, gastrointestinal. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

administration of a load of NaHCO<sub>3</sub>. This, however, may have resulted in removal of the two stimuli for HCO<sub>3</sub><sup>−</sup> reabsorption in PT because the Na<sup>+</sup> load expands the EABV, causing suppression of the release of Ang II, and the administered HCO<sub>3</sub><sup>−</sup> load would increase the HCO<sub>3</sub><sup>−</sup> concentration in the peritubular capillaries in the PT, causing inhibition of the reabsorption of HCO<sub>3</sub><sup>−</sup>. In other experimental studies, in which a large increase in PHCO<sub>3</sub> was achieved without expanding the EABV, no bicarbonaturia was observed, in keeping with the notion that there is no tubular maximum for HCO<sub>3</sub><sup>−</sup> reabsorption.<sup>31,32</sup> Filtered HCO<sub>3</sub><sup>−</sup> ions are reabsorbed and retained in the ECF compartment as long as the above-mentioned stimuli for their reabsorption are maintained.

A deficit of NaCl or HCl may cause a higher PHCO<sub>3</sub> and may also lead to a secondary deficit of K<sup>+</sup> and hypokalemia. A deficit of K<sup>+</sup> may be associated with an acidified PT cell pH, which can then both initiate and sustain a high PHCO<sub>3</sub> as a result of new renal HCO<sub>3</sub><sup>−</sup> generation (higher rate of excretion of NH<sub>4</sub><sup>+</sup>) and enhanced reabsorption of HCO<sub>3</sub><sup>−</sup> and organic anions (which are metabolized to produce HCO<sub>3</sub><sup>−</sup>) in the PT (see [Flow Chart 24.12](#)).

## TOOLS FOR ASSESSMENT OF METABOLIC ALKALOSIS

### QUANTITATIVE ESTIMATE OF THE EXTRACELLULAR FLUID VOLUME

It is critical to have a quantitative estimate of the ECFV to determine its content of HCO<sub>3</sub><sup>−</sup> and thereby why the PHCO<sub>3</sub> rose. As discussed earlier in this chapter, we use the hematocrit for this purpose, provided that anemia or polycythemia are not present (see [Table 24.2](#)).

### BALANCE DATA FOR SODIUM, POTASSIUM, AND CHLORIDE

Balance data for Na<sup>+</sup>, K<sup>+</sup>, and Cl<sup>−</sup> are rarely available in clinical medicine. Nevertheless, they can be deduced if one has a quantitative estimate of the ECFV and measurements of the P<sub>Na</sub>, P<sub>Cl</sub>, and PHCO<sub>3</sub>. One cannot know the balances for K<sup>+</sup>

from these calculations, but its rough magnitude can be deduced by comparing the differences in the content of Na<sup>+</sup> versus that of Cl<sup>−</sup> and HCO<sub>3</sub><sup>−</sup> in the ECF compartment (see [Clinical Case 11](#)).

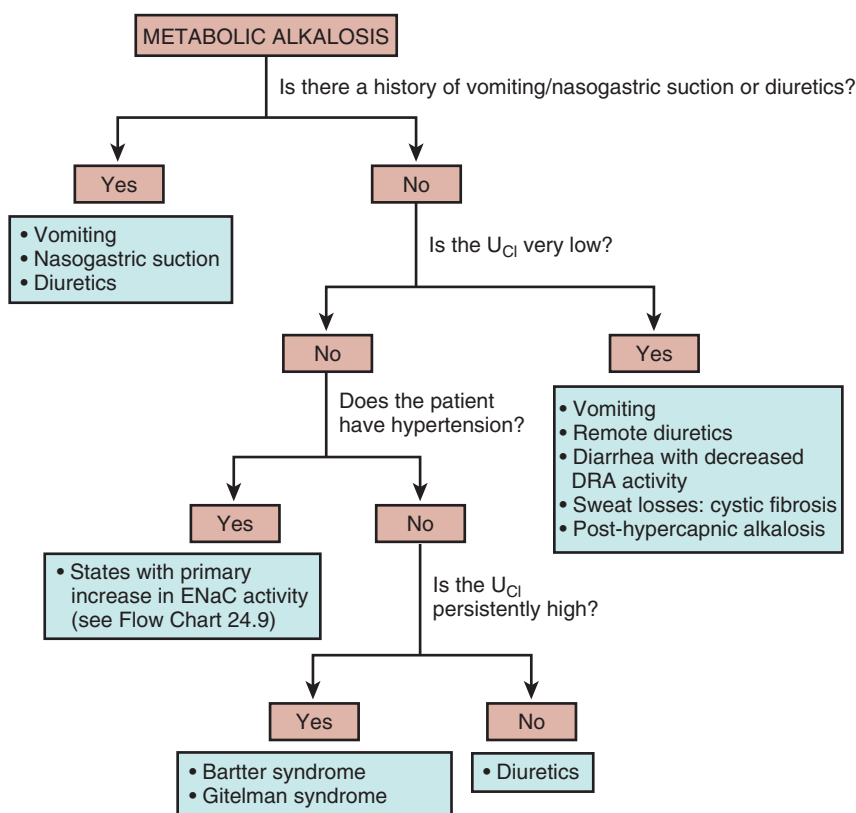
## CLINICAL APPROACH TO THE PATIENT WITH METABOLIC ALKALOSIS

Our clinical approach to a patient with metabolic alkalosis is outlined in [Flow Chart 24.13](#). The first step is to rule out the common causes of metabolic alkalosis—namely, vomiting and the use of diuretics. Some patients may deny vomiting or the use of diuretics; measuring urine electrolyte levels may provide helpful clues if one of these diagnoses is suspected (see [Table 24.3](#)).

A very useful initial test to detect the cause of metabolic alkalosis is to examine the U<sub>Cl</sub>. A very low U<sub>Cl</sub> (<20 mmol/L) is expected when there is a deficit of HCl and/or NaCl. The U<sub>Cl</sub> may not be low if there is recent intake of diuretics (see [Table 24.3](#)). If the U<sub>Cl</sub> is not low, assessment of the EABV and blood pressure helps identify patients with disorders of primary high ENaC activity in the ASDN (see [Flow Chart 24.9](#); the EABV is not low, hypertension is present) and distinguish them from patients with recent use of diuretics and those with Bartter syndrome or Gitelman syndrome (EABV is low, absence of hypertension). Measurements of U<sub>Cl</sub> in multiple spot urine samples are helpful to separate patients with Bartter or Gitelman syndrome (persistent high U<sub>Cl</sub>) from those with diuretic abuse (U<sub>Cl</sub> is high only while the diuretic is acting).

### CLINICAL CASE 11: METABOLIC ALKALOSIS WITHOUT VOMITING OR USE OF DIURETICS

After a forced 6-hour intense training exercise in the desert in the heat of the day, an elite corps soldier was the only one in his squad who collapsed. He perspired profusely during the training exercise and drank a large volume of water and sugar-containing fluids. He denied vomiting or the intake of any medications. Physical examination revealed a markedly contracted EABV. Initial laboratory data are shown in the



**Flow Chart 24.13** DRA, Downregulated in adenoma  $\text{Cl}^-/\text{HCO}_3^-$  exchanger; ENaC, epithelial sodium channel. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

table. The pH and  $\text{PCO}_2$  values are from an arterial blood sample; all other findings are from a venous blood sample.

| Parameter                          | Findings |
|------------------------------------|----------|
| $\text{P}_{\text{Na}}$ , mmol/L    | 125      |
| $\text{P}_{\text{K}}$ , mmol/L     | 2.7      |
| $\text{P}_{\text{Cl}}$ , mmol/L    | 70       |
| Hematocrit                         | 0.50     |
| pH                                 | 7.50     |
| $\text{P}_{\text{HCO}_3}$ , mmol/L | 38       |
| $\text{PCO}_2$ , mm Hg             | 47       |

### Questions and Discussion

What are the major threats to the patient, and how should they dictate therapy?

**Acute hyponatremia.** The danger is brain herniation due to increased intracranial pressure from swelling of brain cells. **Basis of hyponatremia.** The patient weighed 80 kg and had a muscular build, so his initial total body water (TBW) was about 50 L (ECFV  $\approx$  15 L; ICF volume  $\approx$  35 L). Because he had hyponatremia, his ICF volume was expanded because of a water gain. The percentage expansion of ICF volume is close to the percentage fall in  $\text{P}_{\text{Na}}$  (i.e.,  $\approx$  11%). Hence, he had a water gain in his ICF compartment of about 4 L. With a hematocrit of 0.50, his ECFV was decreased by one-third from its normal value of about 15 L to about 10 L; accordingly,

he lost 5 L of ECFV. This represents the loss of 5 L of water and 700 mmol of  $\text{Na}^+$  ( $5 \text{ L} \times \text{P}_{\text{Na}} 140 \text{ mmol/L}$ ). In addition, as the  $\text{P}_{\text{Na}}$  decreased from 140 to 125 mmol/L, each of the remaining 10 L of ECFV had a loss of 15 mmol of  $\text{Na}^+$ . Hence, his total  $\text{Na}^+$  loss was 850 mmol. In total body balance terms, he had a loss of 850 mmol of  $\text{Na}^+$  and 1 L of water—a loss of 5 L of water from the ECF compartment and a gain of 4 L of water in the ICF compartment. Therefore, the primary basis of his hyponatremia was  $\text{Na}^+$  loss.

**Hemodynamic instability.** An infusion of isotonic saline was started in the field, and the patient was hemodynamically stable on arrival at the emergency department. After it was recognized that his  $\text{P}_{\text{Na}}$  was 125 mmol/L, intravenous therapy was changed from isotonic saline to 3% hypertonic saline. The goal of therapy was to raise the  $\text{P}_{\text{Na}}$  to 130 mmol/L. Because the hyponatremia was acute, there was little if any risk of osmotic demyelination from rapidly raising the  $\text{P}_{\text{Na}}$  to 130 mmol/L.

**Hypokalemia.** The hypokalemia did not represent an emergency because he did not have a cardiac arrhythmia or respiratory muscle weakness. An intravenous infusion of isotonic saline supplemented with 40 mmol/L of KCl was started. The  $\text{P}_{\text{K}}$  was followed closely.

What is the basis for metabolic alkalosis? To distinguish between HCl, KCl, and NaCl deficits, a quantitative analysis of the degree of contraction of the ECFV is needed. As mentioned previously, with a hematocrit of 0.50, this patient's



ECFV was decreased by one-third, from its normal value of 15 L to about 10 L, so he had lost 5 L of ECFV.

*Deficit of HCl.* There was no history of vomiting, so an HCl deficit is a very unlikely basis for the metabolic alkalosis.

*Deficit of NaCl.* The decrease in his ECFV was about 5 L. One can now calculate how much this degree of ECFV contraction would raise the  $\text{PHCO}_3$  (divide the normal content of  $\text{HCO}_3^-$  in the ECF compartment ( $15 \text{ L} \times 25 \text{ mmol/L}$ , or 375 mmol) by the new ECFV (10 L), and the result is 37.5 mmol/L. This value is remarkably close to the measured  $\text{PHCO}_3$ , 38 mmol/L, suggesting that the major reason for the rise in the  $\text{PHCO}_3$  is the fall in his ECFV.

$\text{Na}^+$  deficit. As calculated above, the deficit of  $\text{Na}^+$  in the ECF compartment was about 850 mmol.

$\text{Cl}^-$  deficit. Multiplying the  $\text{P}_{\text{Cl}}$  before the training exercise (103 mmol/L) by the normal ECFV (15 L) yields a  $\text{Cl}^-$  content of about 1545 mmol. After the training exercise, the  $\text{P}_{\text{Cl}}$  was 70 mmol/L, and the ECF volume was 10 L, so the ECF  $\text{Cl}^-$  content was 700 mmol. Accordingly, the deficit of  $\text{Cl}^-$  was about 840 mmol, a value close to the deficit of  $\text{Na}^+$ .

*Deficit of KCl.* There was little difference between the deficits of  $\text{Na}^+$  and  $\text{Cl}^-$ , so there was no appreciable deficit of KCl to account for a drop in the  $\text{P}_{\text{K}}$  to 2.7 mmol/L. Especially in this muscular elite soldier, the loss of  $\text{K}^+$  would have to be very large to account for this degree of hypokalemia. Accordingly, the major mechanism for the hypokalemia is likely to be a shift of  $\text{K}^+$  into cells due to a  $\beta_2$ -adrenergic surge and possibly the alkalemia.

Routes for NaCl loss. The next issue is to examine possible routes for a large loss of NaCl in such a short time. Because diarrhea and polyuria were not present, the only route for a large NaCl loss was via sweat. To have a high electrolyte concentration in sweat and a large sweat volume, the likely underlying lesion would be cystic fibrosis. The diagnosis of cystic fibrosis was confirmed later by molecular studies.

What is the therapy for metabolic alkalosis in this patient? Because the basis for the metabolic alkalosis was largely an acute deficit of NaCl, the patient required a positive balance of about 850 mmol of NaCl to replace the deficit. He was initially given hypertonic saline to raise his  $\text{P}_{\text{Na}}$  to 130 mmol/L. If all of this deficit of NaCl were replaced with 3% hypertonic saline, which would also give him about 1.5 L of  $\text{H}_2\text{O}$ , his new total body  $\text{Na}^+$  will be around 6975 mmol (initial TBW 49 L  $\times$  initial  $\text{P}_{\text{Na}}$  125 mmol/L = 6975 mmol; to that add 850 mmol = 7825 mmol), his new TBW will be around 50.5 L, and the  $\text{P}_{\text{Na}}$  would rise to about 138 mmol/L. He might, however, still have a large volume of water in the GI tract that could be absorbed later and cause the  $\text{P}_{\text{Na}}$  to drop. Hence, the  $\text{P}_{\text{Na}}$  should be followed closely. After the administration of only 40 mmol of KCl, the patient's  $\text{P}_{\text{K}}$  rose to 3.8 mmol/L, adding support to our conclusion that the major cause of the hypokalemia was an acute shift of  $\text{K}^+$  into cells.

## METABOLIC ACIDOSIS

Metabolic acidosis is a process that causes a drop in the  $\text{PHCO}_3$  and a rise in the concentration of  $\text{H}^+$  in plasma. Metabolic acidosis represents a diagnostic category with many different

causes (see Chapter 16). The risks for the patient depend on the underlying disorder that caused the metabolic acidosis, the ill effects of the binding of  $\text{H}^+$  to intracellular proteins in vital organs (e.g., the brain and the heart), and possible dangers associated with the anions that accompanied the  $\text{H}^+$  load (e.g., chelation of ionized calcium by citrate in a patient with metabolic acidosis due to ingestion of citric acid).<sup>33</sup>

As previously articulated in Concept 14, the concentration of  $\text{HCO}_3^-$  in the ECF compartment is equal to the ratio of the content of  $\text{HCO}_3^-$  in the ECF compartment to the ECFV. It is important to distinguish between acidemia and acidosis. The term acidemia simply describes an elevation in the concentration of  $\text{H}^+$  in plasma. Acidemia may not be present in a patient who has metabolic acidosis if there is, for example, a large decrease in the ECFV, which may sufficiently raise the  $\text{PHCO}_3$ , even though there is a decreased content of  $\text{HCO}_3^-$  in the ECF compartment (e.g., in the patient with severe diarrhea<sup>34</sup> and in some patients with DKA). For a diagnosis of metabolic acidosis to be made in this setting, a quantitative estimate of the ECFV is needed to assess its content of  $\text{HCO}_3^-$  in the ECF compartment.

### Concept 17

$\text{H}^+$  ions must be removed by the bicarbonate buffer system (BBS) to avoid their binding to intracellular proteins. Binding of  $\text{H}^+$  to proteins could change their charge, shape, and possibly their functions. The bulk of BBS is in the ICF compartment and the interstitial spaces of skeletal muscles.<sup>35</sup>

## TOOLS FOR ASSESSMENT OF METABOLIC ACIDOSIS

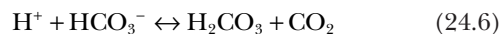
The tools for assessing metabolic acidosis are listed in Table 24.5.

### QUANTITATIVE ASSESSMENT OF THE EXTRACELLULAR FLUID VOLUME

As discussed earlier in this chapter, we use the hematocrit to obtain a quantitative assessment of the ECFV, provided that anemia or polycythemia were not present (see Table 24.2).

#### Tools to Assess the Removal of Hydrogen by the Bicarbonate Buffer System

As shown in Eq. 24.6, the BBS is driven by pull—that is, by a lower  $\text{PCO}_2$ . Effective buffering of  $\text{H}^+$  by the BBS requires a low  $\text{PCO}_2$  primarily in the interstitial space and ICF of skeletal muscles.



Acidemia stimulates the respiratory center, leading to a fall in arterial  $\text{PCO}_2$ . Although the arterial  $\text{PCO}_2$  sets the lower limit on the  $\text{PCO}_2$  in capillaries, it does not guarantee that the capillary  $\text{PCO}_2$  in skeletal muscles will be low enough to ensure effective buffering of  $\text{H}^+$  by the BBS. Because the free-flowing brachial venous  $\text{PCO}_2$  reflects the  $\text{PCO}_2$  in capillary blood of skeletal muscles in its drainage pool, it should provide a means of assessing the effectiveness of the BBS in patients with metabolic acidosis. The capillary  $\text{PCO}_2$  in skeletal muscles will be higher if the rate of blood flow to muscles is low—for example, as a result of a decreased EABV. If muscle oxygen consumption in this setting remains unchanged, more oxygen

**Table 24.5 Laboratory Tests for Diagnosis of Metabolic Acidosis**

| Question                                                                             | Parameter(s) Assessed                                                                                                                                                                   | Tool(s) to Use                                                                                         |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Is the total amount of $\text{HCO}_3^-$ in the ECF compartment low?                  | ECFV                                                                                                                                                                                    | Hematocrit or total plasma proteins                                                                    |
| Is metabolic acidosis due to overproduction of organic acids?                        | Appearance of new anions in the body or the urine                                                                                                                                       | Plasma anion gap<br>Urine anion gap                                                                    |
| Is metabolic acidosis due to ingestion of alcohol?                                   | Presence of alcohols as unmeasured osmoles                                                                                                                                              | Plasma osmolal gap                                                                                     |
| Is effective bicarbonate buffering in skeletal muscle available?                     | Adequacy of tissue perfusion and $\text{CO}_2$ clearance from muscle, enabling buffering of $\text{H}^+$ by $\text{HCO}_3^-$ in interstitial fluid and intracellular fluid compartments | Brachial venous $\text{PCO}_2$                                                                         |
| Is the renal response to chronic acidemia adequate?                                  | Examine the rate of excretion of $\text{NH}_4^+$                                                                                                                                        | Urine osmolal gap                                                                                      |
| If $\text{NH}_4^+$ excretion is high, which anion is excreted with $\text{NH}_4^+$ ? | Gastrointestinal loss of $\text{NaHCO}_3$<br>Acid added, but the anion is excreted in the urine                                                                                         | Urine $\text{Cl}^-$<br>Urine anion gap                                                                 |
| What is the basis for a low excretion of $\text{NH}_4^+$ ?                           | Low distal $\text{H}^+$ secretion<br>Low $\text{NH}_3$ availability<br>Both defects                                                                                                     | Urine pH >7.0<br>Urine pH $\approx$ 5.0<br>Urine pH $\approx$ 6.0                                      |
| Where is the defect in $\text{H}^+$ secretion?                                       | Distal $\text{H}^+$ secretion<br>Proximal $\text{H}^+$ secretion                                                                                                                        | $\text{PCO}_2$ in alkaline urine<br>Fractional excretion of $\text{HCO}_3^-$ , urine citrate excretion |

ECF, Extracellular fluid; ECFV, extracellular fluid volume.

From Kamel KS, Halperin ML. Fluid, electrolytes and acid-base physiology. A problem-based approach, ed 5. Philadelphia: Elsevier; 2017.

will be extracted from, and more  $\text{CO}_2$  will be added to, each liter of blood. The higher  $\text{PCO}_2$  in muscle capillaries will diminish the effectiveness of the BBS to remove extra  $\text{H}^+$ . Hence, the circulating  $\text{H}^+$  concentration rises, and a larger burden of  $\text{H}^+$  will be titrated by intracellular proteins in muscle, as well as by other organs, including the brain. Notwithstanding this, because of autoregulation of cerebral blood flow, it is likely that the  $\text{PCO}_2$  in brain capillary blood will change minimally unless there is a severe degree of contraction of the EABV and failure of autoregulation of cerebral blood flow. Hence, the BBS in the brain will continue to titrate much of this large  $\text{H}^+$  load. Considering, however, the limited amount of  $\text{HCO}_3^-$  in the brain, and that the brain receives a relatively larger proportion of the cardiac output (and of the  $\text{H}^+$  load), there is a risk of more  $\text{H}^+$  binding to intracellular proteins in the brain, further compromising their function (Fig. 24.6).<sup>36</sup> At usual rates of blood flow and metabolic work at rest, the brachial venous  $\text{PCO}_2$  is about 6 mm Hg greater than arterial  $\text{PCO}_2$ . If the blood flow rate to muscles is low, their venous  $\text{PCO}_2$  will be more than 6 mm Hg greater than the arterial  $\text{PCO}_2$ . Enough saline should be administered to increase this blood flow rate to muscle to achieve a brachial venous  $\text{PCO}_2$  that is no more than 6 mm Hg higher than the arterial  $\text{PCO}_2$ .

#### CLINICAL APPROACH: INITIAL STEPS

The initial steps in the clinical approach to a patient with metabolic acidosis are summarized in Flow Chart 24.14.

1. Identify threats for that patient and anticipate and prevent dangers that may arise during therapy.
2. Determine if effective buffering of  $\text{H}^+$  by the BBS in skeletal muscles is available.

## METABOLIC ACIDOSIS DUE TO ADDED ACIDS

### Concept 18

Addition of acids can be detected by the appearance of new anions. These new anions may remain in the body or may be excreted (e.g., in the urine or diarrhea fluid).

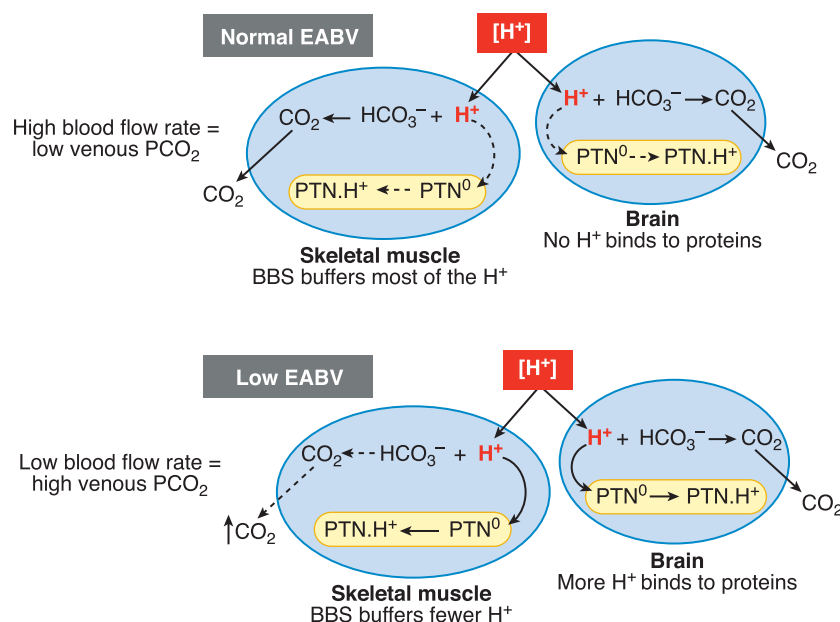
### TOOLS FOR ASSESSING METABOLIC ACIDOSIS DUE TO ADDED ACIDS

Tools for assessment of metabolic acidosis due to added acids are summarized in Table 24.5.

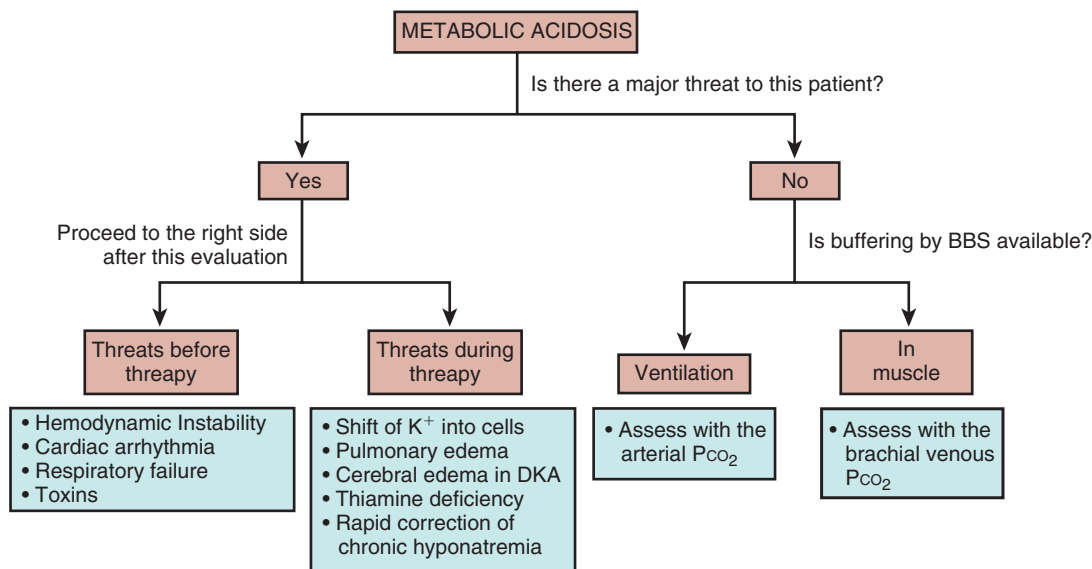
#### Detect New Anions in Plasma

The accumulation of new anions in plasma can be detected from a calculation of the plasma anion gap ( $P_{\text{anion gap}}$ ).<sup>37,38</sup> The major cation in plasma is  $\text{Na}^+$ , and the major anions are  $\text{Cl}^-$  and  $\text{HCO}_3^-$ . The difference between  $P_{\text{Na}}$  and ( $P_{\text{Cl}} + P_{\text{HCO}_3}$ ) reflects the other negative charges in plasma, which are predominantly due to the negative charge on albumin. Although it is commonly said that the normal value of the  $P_{\text{anion gap}}$  is  $12 \pm 2$  mEq/L, the mean value of a normal  $P_{\text{anion gap}}$  varies greatly among clinical laboratories because of different laboratory methods, and the range of normal values is wide. When using the calculation of the  $P_{\text{anion gap}}$  to detect the presence of new anions in plasma, one must adjust the baseline value of the  $P_{\text{anion gap}}$  for the  $P_{\text{Alb}}$ . As a rough estimate, the baseline value for the  $P_{\text{anion gap}}$  falls (or rises) by 2.5 mEq/L for every 10-g/L or 1-g/dL fall (or rise) in the  $P_{\text{Alb}}$ .<sup>39</sup> Even with this adjustment, it seems that net negative valence on albumin is increased if there is an appreciable decrease in the EABV.<sup>40</sup>

Stewart has recommended another approach to detect new anions in plasma, the strong ion difference (SID).<sup>41</sup> This



**Fig. 24.6** Buffering of  $H^+$  in the brain in a patient with metabolic acidemia and a contracted effective arterial blood volume (EABV). *Top*, Buffering of  $H^+$  in a patient with a normal EABV and thereby a low muscle venous  $PCO_2$ . The vast majority of  $H^+$  removal occurs by the bicarbonate buffer system (BBS) in the interstitial space and in cells of skeletal muscles. *Bottom*, Buffering of an  $H^+$  load in a patient with a contracted EABV and thereby a high muscle venous  $PCO_2$  values. A high muscle venous  $PCO_2$  prevents  $H^+$  removal by muscle BBS. As a result, the degree of acidemia may become more pronounced, and more  $H^+$  may bind to proteins (PTN- $H^+$ ) in other organs, including the brain. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)



**Flow Chart 24.14** BBS, Bicarbonate buffer system; DKA, diabetic ketoacidosis. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

approach is rather complex and offers only a minor advantage over the  $P_{anion\ gap}$  in that it includes a correction for the net negative charge on  $P_{Alb}$ .<sup>42</sup>

**Use of the Delta Plasma Anion Gap/Delta Plasma  $HCO_3^-$ .** The relationship between the rise in the  $P_{anion\ gap}$  and the fall in the  $PHCO_3$  (or the delta  $P_{anion\ gap}$ /delta  $PHCO_3$ ) is used to detect the presence of coexisting metabolic alkalosis (the rise in  $P_{anion\ gap}$  is larger than the fall in  $PHCO_3$ ) or the presence

of both an “acid overproduction” type and an “ $NaHCO_3$  loss” type of metabolic acidosis (the rise in the  $P_{anion\ gap}$  is smaller than the fall in the  $PHCO_3$ ).

There are several pitfalls in using this relationship. One is failure to adjust for changes in the ECFV.<sup>43</sup> Consider, for example, a patient with DKA who has a fall in  $PHCO_3$  by 15 mmol/L, from 25 to 10 mmol/L, and the “expected” 1:1 ratio of a rise in  $P_{anion\ gap}$  from a normal value of 12 mEq/L to 27 mEq/L. The patient had a normal ECFV of 10 L before

DKA developed but, as a result of the glucose-induced osmotic diuresis and natriuresis, the current ECFV is only 8 L. Although the fall in the  $\text{P}_{\text{HCO}_3^-}$  and the rise in the concentration of ketoacid anions (as judged from the rise in the  $\text{P}_{\text{anion gap}}$ ) are equal, the deficit of  $\text{HCO}_3^-$  and the amount of ketoacids added to the ECF compartment are not. The sum of the content of  $\text{HCO}_3^-$  and ketoacid anions in the ECF compartment prior to the development of DKA is 250 mmol ( $[25 + 0 \text{ mmol/L}] \times 10 \text{ L}$ ). Their sum in the ECF compartment after DKA developed, however, is only 200 mmol ( $[10 + 15] \text{ mmol/L} \times 8 \text{ L}$ ). The deficit of  $\text{HCO}_3^-$  in this example is 170 mmol, but the quantity of new anions in the ECF is only 120 mmol. This is because there is another component of the loss of  $\text{HCO}_3^-$  that occurred when ketoacids were added that is not detected by an increase in the  $\text{P}_{\text{anion gap}}$ . Some of the ketoacid anions were excreted in the urine with  $\text{Na}^+$  and/or  $\text{K}^+$ , an indirect form of  $\text{NaHCO}_3$  loss. Hence the rise in the  $\text{P}_{\text{anion gap}}$  underestimated the actual quantity of net production of ketoacids, and the fall in  $\text{P}_{\text{HCO}_3^-}$  underestimated the actual magnitude of the deficit of  $\text{HCO}_3^-$ . When the ECFV is re-expanded with saline, the degree of deficit of  $\text{HCO}_3^-$  will become evident. In addition, the fall in the  $\text{P}_{\text{anion gap}}$  will not be matched by a rise in the  $\text{P}_{\text{HCO}_3^-}$  because some ketoacid anions will be lost in urine when their filtered load is increased with the rise in GFR.

Another pitfall in the use of the delta  $\text{P}_{\text{anion gap}}/\text{delta } \text{P}_{\text{HCO}_3^-}$  is the failure to correct for the net negative valence attributable to  $\text{P}_{\text{Alb}}$ . When calculating the  $\text{P}_{\text{anion gap}}$ , one must adjust the base value for changes in the charge on the most abundant unmeasured anion in plasma, which is albumin. We emphasize that adjustments should be made for a decrease or an increase in  $\text{P}_{\text{Alb}}$ .

### Detect New Anions in the Urine

New anions can be detected with the calculation of the urine anion gap ( $\text{U}_{\text{anion gap}}$ ; Eq. 24.7).

$$\text{U}_{\text{anion gap}} = (\text{U}_{\text{Na}} + \text{U}_{\text{K}} + \text{U}_{\text{NH}_4}) - \text{U}_{\text{Cl}} \quad (24.7)$$

The concentration of  $\text{NH}_4^+$  in the urine ( $\text{U}_{\text{NH}_4}$ ) is estimated from the urine osmolal gap ( $\text{U}_{\text{osm gap}}$ ), as discussed in the next section. The nature of these new anions may sometimes be deduced by comparing their filtered load with their excretion rate. For example, when there is a very large quantity of the new anion in the urine in comparison with the rise in  $\text{P}_{\text{anion gap}}$ , one should suspect that this anion is secreted in the PT (e.g., hippurate anion from the metabolism of toluene) or is freely filtered and poorly reabsorbed by the PT (e.g., reabsorption of ketoacid anions may be inhibited by salicylate anions, D-lactate anions). On the other hand, a very low rate of excretion of new anions suggests that they are avidly reabsorbed in the PT (e.g., L-lactate anions).

### Detect Toxic Alcohols

The presence of alcohols in plasma can be detected by calculating the plasma osmolal gap ( $\text{P}_{\text{osm gap}}$ ; Eq. 24.8). This occurs because alcohols are uncharged compounds, have a low molecular weight, and usually large quantities are ingested.

$$\text{P}_{\text{osm gap}} = \text{measured } \text{P}_{\text{osm}} - (2 \times \text{P}_{\text{Na}} + \text{P}_{\text{Glu}} + \text{P}_{\text{Urea}}) \quad (24.8)$$

In Eq. 24.8, all terms are in mmol/L. If  $\text{P}_{\text{Glu}}$  is in mg/dL, divide by 180, and if  $\text{P}_{\text{Urea}}$  is expressed as mg/dL of urea nitrogen, divide by 2.8 to obtain units in mmol/L.

### CLINICAL APPROACH TO THE PATIENT WITH METABOLIC ACIDOSIS DUE TO ADDED ACIDS

The steps in the clinical approach to the patient with metabolic acidosis due to added acids are shown in [Flow Chart 24.15](#). If metabolic acidosis develops over a short period, the likely causes are overproduction of L-lactic acid (e.g., hypoxic L-lactic acidosis, ingestion of alcohol in a patient with thiamine deficiency) or ingestion of acids (e.g., metabolic acidosis due to ingestion of citric acid). Measurement of plasma levels of L-lactate and  $\beta$ -hydroxybutyrate confirms the diagnosis of L-lactic acidosis or ketoacidosis respectively. If suspected, measurements of blood levels of methanol, ethylene glycol, acetaminophen or acetylsalicylic acid should be obtained. Acetaminophen overdose may be associated with L-lactic acidosis or pyroglutamic acidosis. A detailed discussion of the various causes of metabolic acidosis due to added acids is provided in Chapter 16.

### CLINICAL CASE 12: SEVERE METABOLIC ACIDOSIS IN A PATIENT WITH CHRONIC ALCOHOLISM

A 52-year-old man presented to the emergency department with abdominal pain, visual disturbances, and shortness of breath.<sup>44</sup> He had a history of drinking excessive amounts of alcohol on a regular basis. He admitted to drinking approximately 1 L of vodka the day before hospital admission but denied ingesting any other substances. During the 24 hours before admission, he had not eaten at all. In the 5 hours before his admission, he had had several bouts of vomiting and did not drink any alcohol. His dietary intake had been generally very poor over the last several months because he had diminished appetite.

On physical examination, he was fully conscious and oriented. His respiration rate was rapid (40 breaths/min). His pulse rate was also rapid (150 beats/min), and his blood pressure was 120/58 mm Hg. Neurologic examination was unremarkable.

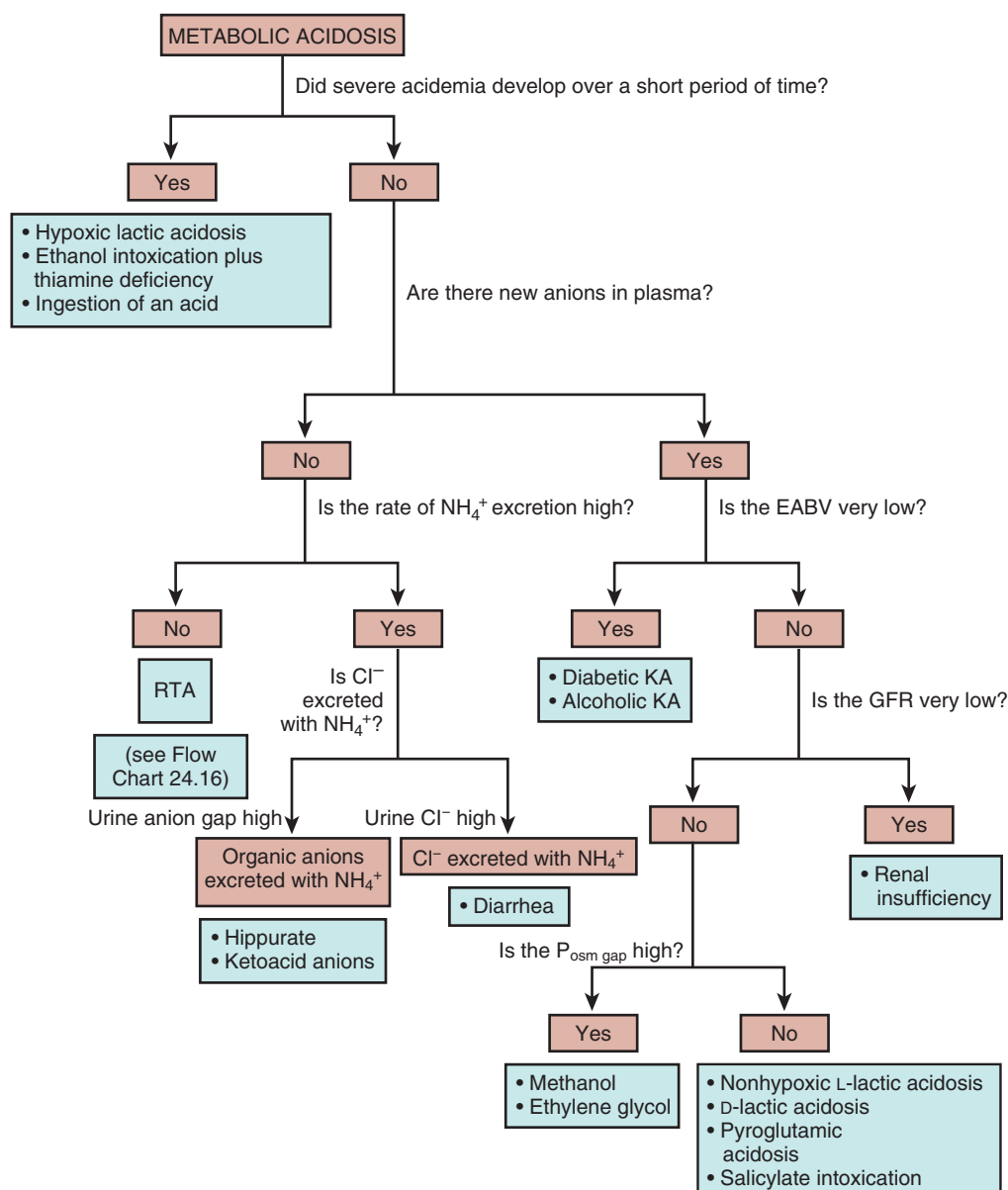
The patient's urine tested strongly positive for ketones. His initial laboratory results on admission to the emergency department are shown in the following table. The pH and  $\text{P}_{\text{CO}_2}$  are from an arterial blood sample, whereas all other findings are from a venous blood sample.

| Parameter                                                    | Findings |
|--------------------------------------------------------------|----------|
| $\text{P}_{\text{Na}}$ , mmol/L                              | 132      |
| $\text{P}_{\text{K}}$ , mmol/L                               | 5.4      |
| $\text{P}_{\text{Cl}}$ , mmol/L                              | 85       |
| $\text{P}_{\text{HCO}_3^-}$ , mmol/L                         | 3.3      |
| $\text{P}_{\text{anion gap}}$ , mEq/L                        | 44       |
| $\text{P}_{\text{osm}}$ , mOsmol/kg $\text{H}_2\text{O}$     | 325      |
| Hematocrit                                                   | 0.46     |
| pH                                                           | 6.78     |
| $\text{P}_{\text{CO}_2}$ , mm Hg                             | 23       |
| $\text{P}_{\text{Glu}}$ , mmol/L                             | 3.0      |
| $\text{P}_{\text{Alb}}$ , g/L                                | 36       |
| $\text{P}_{\text{osm gap}}$ , mOsmol/kg $\text{H}_2\text{O}$ | 42       |

### Questions and Discussion

What dangers may be present on admission or may arise during therapy?





**Flow Chart 24.15** RTA, Renal tubular acidosis; KA, ketoacidosis; GFR, glomerular filtration rate; *P<sub>osc</sub> gap*, plasma osmolal gap. (From Kamel KS, Halperin ML. Fluid, electrolytes, and acid-base physiology. A problem based approach, ed 5. Philadelphia: Elsevier, 2017.)

**Severe acidemia.** The patient had a severe degree of acidemia with a large increase in the  $P_{\text{anion gap}}$ , indicating overproduction of acids. For the time being, he is hemodynamically stable, but a quantitatively small additional  $\text{H}^+$  load would produce a disproportionately large fall in the  $\text{PHCO}_3$  and plasma pH. For example, halving the  $\text{PHCO}_3$  would cause the arterial pH to drop by 0.30 unit if the arterial  $\text{PCO}_2$  has not changed. By the same token, doubling of the  $\text{PHCO}_3$  would raise the plasma pH by 0.30 unit. A large dose of  $\text{NaHCO}_3$  would be needed to achieve this because the administered  $\text{NaHCO}_3$  might lead to back titration of some of the large  $\text{H}^+$  load that is bound to the patient's intracellular proteins and, in addition, he might still have ongoing production of acids.

**Toxic alcohol ingestion.** Because the patient had a severe degree of metabolic acidemia with a large  $P_{\text{osc gap}}$ , ingestion of

methanol or ethylene glycol was suspected. Aldehydes produced from the metabolism of these alcohols by the enzyme alcohol dehydrogenase in the liver are the major cause of toxicity because they rapidly bind to tissue proteins. Although the patient ingested a large amount of ethanol, which could have caused the large  $P_{\text{osc gap}}$ , and his urine was strongly positive for ketones, such a severe degree of acidemia is not usual in patients with alcoholic ketoacidosis. Because of the strong clinical suspicion of toxic alcohol ingestion, the patient was started on fomepizole (an inhibitor of alcohol dehydrogenase) while waiting for the determination of the level of these toxic alcohols in his blood.

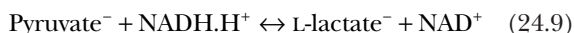
**Thiamine deficiency.** Malnourished patients who present with alcoholic ketoacidosis are at risk for the development of encephalopathy due to thiamine deficiency. Ketoacids,

when available, are the preferred brain fuel because they are derived from storage fat and, hence, proteins from lean body mass are spared as a source of glucose for the brain during prolonged starvation. After successful treatment of alcoholic ketoacidosis, ketoacids are no longer available as a brain fuel, so the brain must regenerate most of its ATP from the oxidation of glucose. Thiamine (vitamin B<sub>1</sub>) is a key cofactor for pyruvate dehydrogenase (PDH). The activity of PDH is diminished by the lack of thiamine, so the rate of regeneration of ATP in brain cells will not be sufficient for their biologic work. Therefore, glycolysis will be stimulated in the brain to make ATP. As a result, there will be a sudden rise in the production of H<sup>+</sup> and L-lactate anions in areas of the brain where the metabolic rate is the most rapid and/or areas that have the lowest reserve of thiamine. Thiamine must be administered early in therapy in such patients.

**More information.** Two hours after presentation, additional laboratory results became available. The patient's plasma L-lactate level was 23 mmol/L, and the assays for methanol and ethylene glycol were negative.

What is the cause of the severe degree of L-lactic acidosis in this patient? A rise in the concentration of L-lactate anions and H<sup>+</sup> can be caused by an increased rate of production and/or a decreased rate of removal of L-lactic acid. The rapid development and severity of L-lactic acidosis in this patient suggest that the L-lactic acidosis is largely due to overproduction of L-lactic acid.

The degree of L-lactic acidosis in patients presenting with alcohol intoxication is usually mild (plasma L-lactate level is usually <5 mmol/L) because it is due to the increased nicotinamide adenine dinucleotide, reduced form (NADH.H<sup>+</sup>)/nicotinamide adenine dinucleotide (NAD<sup>+</sup>) ratio owing to the ongoing production of NADH.H<sup>+</sup> through ethanol metabolism, which is largely restricted to the liver where the enzymes alcohol dehydrogenase and aldehyde dehydrogenase are expressed. Other organs in the body are capable of oxidizing the L-lactate produced by the liver and, hence, the degree of L-lactic acidosis is usually mild.



However, a severe degree of L-lactic acidosis may develop rapidly if there is a large intake of alcohol in a patient who is thiamine-deficient. The site of L-lactic acid production is likely to be the liver because there will be both accumulation of pyruvate (owing to diminished activity of PDH) and a high NADH.H<sup>+</sup>/NAD<sup>+</sup> ratio (due to metabolism of ethanol). There is also diminished removal of L-lactic acid by other organs due to diminished activity of PDH.

## HYPERCHLOREMIC METABOLIC ACIDOSIS

Hyperchloremic metabolic acidosis is characterized by the absence of a rise in the P<sub>anion gap</sub>; hence, it is often also called non-anion gap metabolic acidosis. There are two major groups of causes for this type of metabolic acidosis, the direct loss of NaHCO<sub>3</sub> and the indirect loss of NaHCO<sub>3</sub>. The direct loss of NaHCO<sub>3</sub> may occur via the GI tract—for example, in patients with diarrhea or through the urine in patients at

the early phase of a disease process that causes proximal renal tubular acidosis (pRTA). The indirect loss of NaHCO<sub>3</sub> may be due to a low rate of excretion of NH<sub>4</sub><sup>+</sup> that is insufficient to match the daily rate of production of sulfuric acid from the metabolism of sulfur-containing amino acids (e.g., in patients with chronic renal failure or patients with distal renal tubular acidosis [dRTA], or patients with pRTA in the steady state). Indirect loss of NaHCO<sub>3</sub> may also be due to an overproduction of an acid (e.g., hippuric acid formed during the metabolism of toluene, ketoacids), with the excretion of its conjugate base (e.g., hippurate anions, ketoacids anions) in the urine at a rate that exceeds the rate of excretion of NH<sub>4</sub><sup>+</sup>.

### Concept 19

The expected renal response to chronic metabolic acidosis is a high rate of excretion of NH<sub>4</sub><sup>+</sup>. Based on findings in normal subjects given an acid load of ammonium chloride for several days, the expected normal renal response in a patient with chronic metabolic acidosis, is the excretion of close to 200 mmol of NH<sub>4</sub><sup>+</sup>/day.<sup>45</sup> There is a lag period of a few days, however, before high rates of excretion of NH<sub>4</sub><sup>+</sup> can be achieved. Patients with a hyperchloremic metabolic acidosis that have an appropriately high rate of NH<sub>4</sub><sup>+</sup> excretion generally have diarrhea or an added organic acid the anion of which is excreted in the urine at a rapid rate. These two disorders can be differentiated by assessing the urine anion gap and urine chloride concentration (see [Flow Chart 24.15](#)).

### Concept 20

A low rate of excretion of NH<sub>4</sub><sup>+</sup> in a patient with hyperchloremic metabolic acidosis could be due to a decreased medullary interstitial availability of NH<sub>3</sub> or a decreased net H<sup>+</sup> secretion in the distal nephron.<sup>46</sup>

A low rate of ammoniogenesis has several possible causes. One is intracellular alkalinization of PT cells due to hyperkalemia or a genetic or an acquired disorder that compromises proximal H<sup>+</sup> secretion or HCO<sub>3</sub><sup>-</sup> exit from PT cells (also causing a reduced capacity to reabsorb HCO<sub>3</sub><sup>-</sup> [i.e., pRTA]). The other involves a reduction in the GFR, which diminishes the filtered load of Na<sup>+</sup> because less work is performed in PT cells; thus, the availability of ADP is decreased and the rate of oxidation of glutamine in PT cells is reduced.

The other main cause of a low rate of excretion of NH<sub>4</sub><sup>+</sup> is a **low net secretion of H<sup>+</sup> in the distal nephron**. This could be due to an **H<sup>+</sup>-ATPase defect** (e.g., autoimmune and hypergammaglobulinemic disorders, including Sjögren syndrome), **backleak of H<sup>+</sup>** (e.g., due to drugs such as amphotericin B), or **a disorder associated with the distal secretion of HCO<sub>3</sub><sup>-</sup>** (e.g., in some patients with **Southeast Asian ovalocytosis [SAOI]**). Patients with **medullary interstitial disease** (e.g., due to infections, drugs, infiltrations, precipitations, inflammatory disorders, sickle cell disease) may have **a low rate of excretion of NH<sub>4</sub><sup>+</sup>** because of both diminished accumulation of NH<sub>4</sub><sup>+</sup> in the medullary interstitium and a decreased rate of H<sup>+</sup> secretion in the MCD.

## TOOLS FOR ASSESSING HYPERCHLOREMIC METABOLIC ACIDOSIS

[Table 24.5](#) summarizes the steps for the assessment of patients with hyperchloremic metabolic acidosis.

### Assess the Rate of Excretion of $\text{NH}_4^+$ in the Urine

**Urine Osmolal Gap.** A direct assay for urine  $\text{NH}_4^+$  is not often available in clinical settings. In our opinion, calculation of the  $U_{\text{osm gap}}$  provides the best indirect estimate of the  $U_{\text{NH}_4}$  (Eq. 24.10) because it detects all  $\text{NH}_4^+$  salts in the urine (Fig. 24.7).<sup>47,48</sup>

$$U_{\text{osm gap}} = \text{measured } U_{\text{osm}} - \text{calculated } U_{\text{osm}}$$

$$\text{Calculated } U_{\text{osm}} = 2(U_{\text{Na}} + U_{\text{K}}) + U_{\text{Urea}} + U_{\text{Glu}} \quad (24.10)$$

All values are in mmol/L.

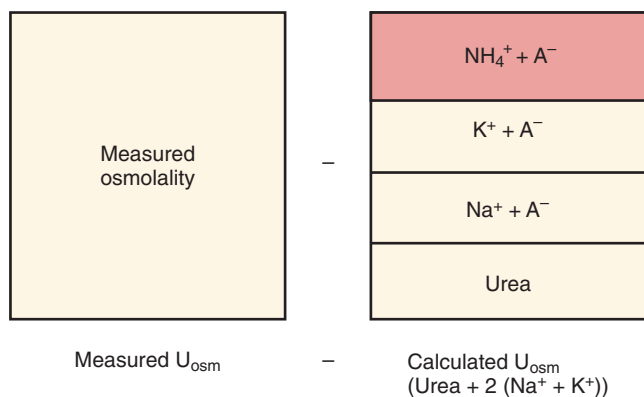
$$\text{Urinary concentration of } \text{NH}_4 = U_{\text{osm gap}}/2$$

We use the  $U_{\text{NH}_4}/U_{\text{Cr}}$  ratio in a spot urine sample to assess the rate of excretion of  $\text{NH}_4^+$ . The rationale is that the rate of excretion of creatinine is relatively constant over a 24-hour period. In a patient with chronic metabolic acidosis, the expected normal renal response is a  $U_{\text{NH}_4}/U_{\text{Cr}}$  ratio higher than 150 mmol  $\text{NH}_4^+$ /g creatinine (higher than 15 if creatinine is measured in mmol).

There are two issues in using the urine net charge (or urine anion gap) to assess the rate of excretion of  $\text{NH}_4^+$  that, in our opinion, limit its utility. First, the calculation of the urine net charge detects high rates of excretion of  $\text{NH}_4^+$  in the urine only when the anion excreted with  $\text{NH}_4^+$  is  $\text{Cl}^-$ . Second, the equation that describes the relationship between  $U_{\text{NH}_4}$  and the urine net charge, which was based on values obtained from 24 hour urine collection, is shown below (Eq. 24.11).

$$U_{\text{NH}_4} = -0.8 (\text{urine anion gap}) + 82 \quad (24.11)$$

where 82 is the difference between the usual rates of excretion of other unmeasured anions and other unmeasured cations



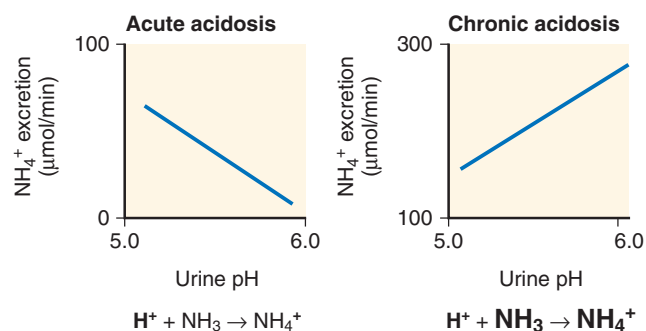
**Fig. 24.7** Indirect assessment of the concentration of  $\text{NH}_4^+$  in the urine using the urine osmolal gap. The essence of the test is that a high concentration of  $\text{NH}_4^+$  (shown in the red shaded region on the right) is detected in the urine from its contribution to the urine osmolality. The urine osmolal gap is the difference between the measured urine osmolality and the urine osmolality calculated from the concentrations (in mmol/L) of the principal usual urine osmoles; urea, double the concentrations of  $\text{Na}^+ + \text{K}^+$  (to account for the concentrations of the usual monovalent anions in the urine) and glucose in a patient with hyperglycemia. The concentration of  $\text{NH}_4^+$  in the urine is the urine osmolal gap divided by 2.  $\text{A}^-$ , Anion. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

in the urine in subjects described as consuming normal diet.<sup>48a</sup> This value however may have large variations depending on dietary intake.

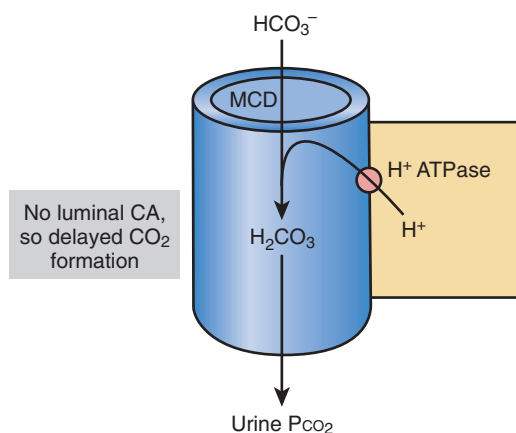
### Determine Why the Rate of Excretion of $\text{NH}_4^+$ Is Low

**Urine pH.** The urine pH is not a reliable indicator for the rate of excretion of  $\text{NH}_4^+$  (Fig. 24.8).<sup>49</sup> On the other hand, the basis for the low rate of excretion of  $\text{NH}_4^+$  may be deduced from the urine pH. A urine pH about 5 suggests that the primary basis for a low rate of excretion of  $\text{NH}_4^+$  is decreased availability of  $\text{NH}_3$  in the medullary interstitial compartment due to a defect in  $\text{NH}_4^+$  production or its transfer in the mTAL of the loop of Henle. A urine pH higher than 7.0 suggests that  $\text{NH}_4^+$  excretion is low because there is a defect in net  $\text{H}^+$  secretion in the distal nephron. Conversely, a urine pH about 6 would suggest a medullary interstitial disease that diminishes both accumulation of  $\text{NH}_4^+$  in the medullary interstitium and  $\text{H}^+$  secretion in the MCD.<sup>50</sup>

**Assess Distal Hydrogen Secretion.**  $\text{H}^+$  secretion in the distal nephron can be evaluated using the measurement of the  $\text{PCO}_2$  in an alkaline urine ( $\text{UPCO}_2$ ) during bicarbonate loading (Fig. 24.9).<sup>51</sup> The patient is given a load of  $\text{NaHCO}_3$  to increase the filtered load of  $\text{HCO}_3^-$  and its delivery to the distal nephron; the urine must be freshly collected under mineral oil to minimize diffusion of  $\text{CO}_2$  and must be analyzed immediately. A  $\text{PCO}_2$  that is about 70 mm Hg in a second-voided alkaline urine implies that  $\text{H}^+$  secretion in the distal nephron is likely to be normal, whereas much lower  $\text{UPCO}_2$  values suggest that distal  $\text{H}^+$  secretion is impaired. Some patients with low net distal  $\text{H}^+$  secretion have a high  $\text{UPCO}_2$ . This occurs if there is a lesion causing a backleak of  $\text{H}^+$  from the lumen of the collecting ducts (e.g., due to the use of amphotericin B)<sup>52</sup> or distal secretion of  $\text{HCO}_3^-$  (as in some



**Fig. 24.8** The urine pH is not a reliable indicator for the rate of excretion of  $\text{NH}_4^+$ . *Left*, During acute metabolic acidosis, the rate of excretion of  $\text{NH}_4^+$  is only modestly higher, and the urine pH is low. This is because distal  $\text{H}^+$  secretion is enhanced, and there is a time lag before the rate of renal production of  $\text{NH}_4^+$  and the availability of  $\text{NH}_3$  in the medullary interstitial compartment are augmented. *Right*, In contrast, during chronic metabolic acidosis, the rate of renal production of  $\text{NH}_4^+$  is so high that the availability of  $\text{NH}_3$  in the medullary interstitial compartment provides more  $\text{NH}_3$  in the lumen of the medullary collecting duct than  $\text{H}^+$  secretion in this nephron segment. Note the much higher  $\text{NH}_4^+$  excretion rate at a urine pH of 6. Note also the different scales on the y-axes. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)



**Fig. 24.9** Use of the  $\text{PCO}_2$  in alkaline urine to assess the distal secretion of  $\text{H}^+$ . The cylinder represents the medullary collecting duct (MCD) and the rectangle on its right side represents an alpha-intercalated cell, which contains an  $\text{H}^+$ -ATPase pump. There are two requirements for using the urine  $\text{PCO}_2$  in alkaline urine to reflect the capacity to secrete  $\text{H}^+$ . First, enough  $\text{NaHCO}_3$  is administered to achieve a second voided-urine sample with a  $\text{pH} > 7.0$ . Second, because the luminal membranes of the MCD lack carbonic anhydrase (CA), the carbonic acid formed is delivered to the lower urinary tract, where it decomposes to  $\text{CO}_2$  and  $\text{H}_2\text{O}$  and thereby elevates the urine  $\text{PCO}_2$ . A high  $\text{PCO}_2$  in alkaline urine (usually to  $\approx 70$  mm Hg) suggests that there is no major defect in  $\text{H}^+$  secretory capacity in this nephron segment. (From Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; A Problem-Based Approach*. ed 5. Philadelphia: Elsevier; 2017.)

patients with SAO who have a mutation in the  $\text{HCO}_3^-/\text{Cl}^-$  anion exchanger that leads to its mistargeting to the luminal membranes of  $\alpha$ -intercalated cells).<sup>53</sup>

### Assessment of Proximal Cell pH

**Fractional excretion of  $\text{HCO}_3^-$ .** In patients suspected of having a disorder that leads to a reduced capacity for the reabsorption of  $\text{HCO}_3^-$  in the PT (pRTA), some clinicians would measure the fractional excretion of  $\text{HCO}_3^-$  after infusing  $\text{NaHCO}_3$  to confirm this diagnosis. In our opinion, this evaluation is not needed. These disorders are recognized clinically by failure to correct the metabolic acidemia after the administration of large amounts of  $\text{NaHCO}_3$ . The defect in  $\text{NaHCO}_3$  reabsorption in the PT may be an isolated defect or part of a generalized PT cell dysfunction (i.e., Fanconi syndrome), in which other  $\text{Na}^+$ -linked transport functions of PT are affected, leading to renal glucosuria, aminoaciduria, and increased rates of excretion of phosphate, urate, and citrate. The most common cause of Fanconi syndrome in the pediatric population is cystinosis, whereas common causes in the adult population are paraproteinemias and use of drugs such as tenofovir and ifosfamide.

**Rate of citrate excretion.** The rate of excretion of citrate is a marker of pH in cells of the PT.<sup>54</sup> The rate of excretion of citrate in children and adults consuming their usual diet is about 400 mg/day ( $\approx 2.1$  mmol/day). The rate of excretion of citrate is very low during most forms of metabolic acidosis because of the effect of intracellular acidosis in PT cells to stimulate the reabsorption of citrate. The absence of hypocitraturia would suggest that the pathophysiology of

the defect in proximal  $\text{H}^+$  secretion is due to a disorder causing an alkaline PT cell pH, such as conditions associated with pRTA or carbonic anhydrase II deficiency (which also impairs distal  $\text{H}^+$  secretion).

### CLINICAL APPROACH TO THE PATIENT WITH HYPERCHLOREMIC METABOLIC ACIDOSIS

The steps in the clinical approach to the patient with hyperchloremic metabolic acidosis are shown in [Flow Chart 24.15](#) and [Flow Chart 24.16](#).

### CLINICAL CASE 13: DOES THIS PATIENT HAVE DISTAL RTA?

A 28-year-old man had been intermittently sniffing glue for the last number of years. Over the past 3 days, he became profoundly weak and had a very unsteady gait. On physical examination, his blood pressure was 100/60 mm Hg and his pulse rate was 110 beats/min when he was lying flat. When he sat up, his blood pressure fell to 80/50 mm Hg and his pulse rate rose to 130 beats/min. Arterial blood pH was 7.20, arterial  $\text{PCO}_2$  was 25 mm Hg, and  $\text{PHCO}_3$  was 10 mmol/L. Venous blood and urine laboratory values were as follows.

| Parameter                                   | Venous Blood                          | Urine      |
|---------------------------------------------|---------------------------------------|------------|
| pH                                          | 7.0                                   | 6.0        |
| $\text{PCO}_2$ , mm Hg                      | 60                                    | —          |
| $\text{HCO}_3^-$ , mmol/L                   | 12                                    | —          |
| $\text{Na}^+$ , mmol/L                      | 120                                   | 50         |
| $\text{K}^+$ , mmol/L                       | 2.3                                   | 30         |
| $\text{Cl}^-$ , mmol/L                      | 90                                    | 5          |
| Creatinine                                  | 1.7 mg/dL<br>(150 $\mu\text{mol/L}$ ) | 3.0 mmol/L |
| Glucose                                     | 63 mg/dL<br>(3.5 mmol/L)              | 0          |
| Urea                                        | BUN: 14 mg/dL<br>(5.0 mmol/L)         | 150 mmol/L |
| Albumin                                     | 60 g/L: (6 g/dL)                      | —          |
| Osmolality,<br>mOsm/kg $\text{H}_2\text{O}$ | 260                                   | 400        |

### Questions and Discussion

What dangers were present on admission?

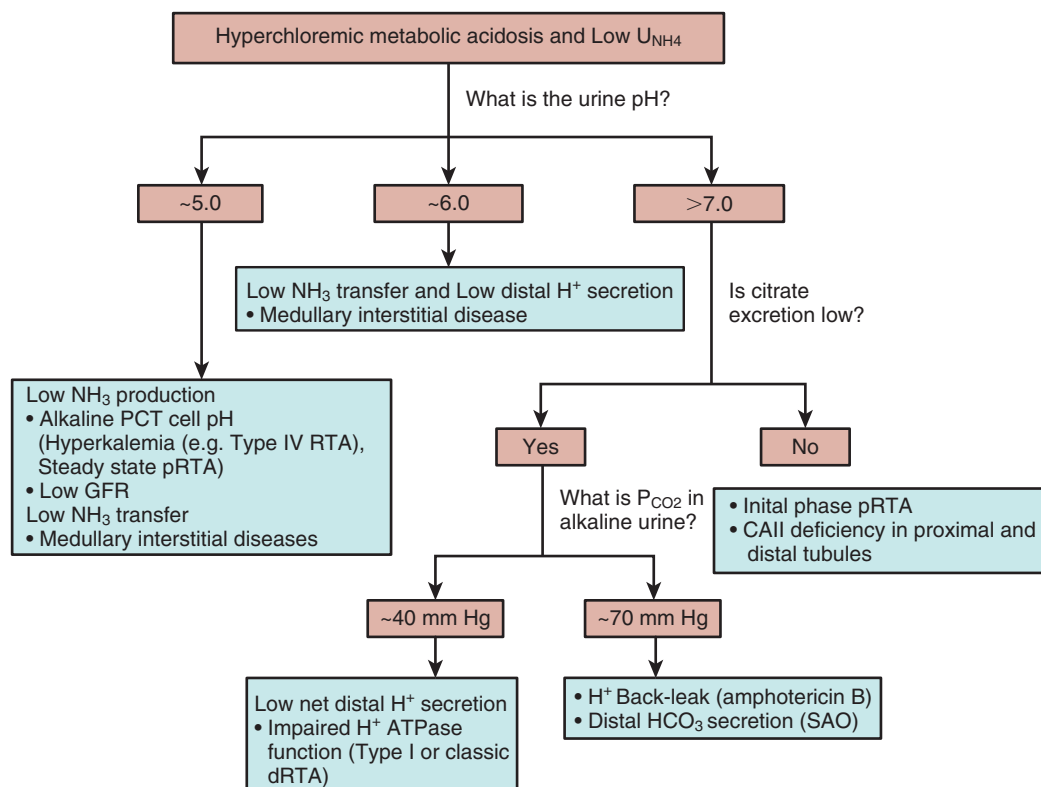
**Hemodynamic instability.** The patient had a marked degree of EABV contraction.

**Severe degree of hypokalemia.** The dangers of a severe degree of hypokalemia are cardiac arrhythmias and respiratory muscle weakness. However, this patient's ECG demonstrated only prominent U waves. Analysis of arterial blood gases showed that the patient had an arterial  $\text{PCO}_2$  of 25 mm Hg, which was appropriate for the fall in  $\text{PHCO}_3$ ; hence, there is no superimposed respiratory acidosis. Therefore, although he had a severe degree of hypokalemia that would require aggressive  $\text{K}^+$  therapy, he did not have an emergency related to hypokalemia on admission.

**Hyponatremia.** Hyponatremia was likely chronic because there were no symptoms that would strongly suggest an appreciable acute component to the hyponatremia nor was there a history of a recent large water intake.

**Binding of  $\text{H}^+$  to proteins in cells.** Because the brachial venous  $\text{PCO}_2$  (60 mm Hg) was considerably higher than the arterial





**Flow Chart 24.16** pRTA, Proximal renal tubular acidosis; GFR, glomerular filtration rate; CA II, carbonic anhydrase type II; dRTA, distal RTA; SAO, South Asain ovalocytosis. (From Kamel KS, Halperin ML. Fluid, electrolytes and acid-base physiology. A problem based approach, ed 5. Philadelphia: Elsevier, 2017.)

$P_{CO_2}$  (25 mm Hg), buffering of  $H^+$  by the BBS in muscle was compromised, and there was a risk of more  $H^+$  binding to proteins in cells of vital organs (e.g., the heart and brain; see Fig. 24.6).

What dangers should be anticipated during therapy?

*A more severe degree of hypokalemia.* Re-expansion of the EABV with the administration of saline can lead to decreased levels of catecholamines. This removes the inhibition of the release of insulin by the binding of catecholamines to pancreatic islet cells  $\alpha$ -adrenergic receptors. The release of insulin could result in a shift of  $K^+$  into cells, with worsening hypokalemia. Administration of  $NaHCO_3$  to correct the metabolic acidemia may also cause a shift of  $K^+$  into cells and a more severe degree of hypokalemia.

*Rapid rise in  $P_{Na}$ .* The  $P_{Na}$  may rise rapidly if water diuresis occurs with re-expansion of the EABV. As  $K^+$  is administered, there will be a shift of  $K^+$  into muscle cells in exchange for  $Na^+$ , which will also lead to a rise in  $P_{Na}$ . Patients who are malnourished and/or hypokalemic are at high risk of osmotic demyelination with a rapid rise in  $P_{Na}$ . It is our opinion that the maximum rise in  $P_{Na}$  in high-risk patients should not exceed 4 to 6 mmol/L in the first 24 hours.

*Further fall in  $PHCO_3$ .* Expansion of the EABV with rapid administration of saline may also lead to a further fall in  $PHCO_3$ . First, there is a dilution effect. Second, with improved blood flow to muscles and the fall in capillary  $P_{CO_2}$ , there will be titration of  $HCO_3^-$  by  $H^+$  ions that are bound to intracellular proteins. The need to give  $NaHCO_3$ ,

however, must be balanced by the danger of creating more severe hypokalemia. We would not administer  $NaHCO_3$  unless there is hemodynamic instability that is not responsive to the usual maneuvers to restore blood pressure and, with a central venous line in place, to give a sufficient amount of KCl rapidly if needed.

**Plan for initial therapy.** On arrival to the emergency department, the patient was given 1 L of intravenous isotonic saline. To correct the hypokalemia while also avoiding a rapid rise in  $P_{Na}$ , the intravenous solution was changed to 0.45% NaCl (77 mmol/L) to which KCl, 40 mmol/L, was added. The concentration of effective cation osmoles of this solution ( $77 + 40 = 117$  mmol/L) is close to his  $P_{Na}$ , and thus should not itself lead to a rapid rise in  $P_{Na}$ . The danger of a rapid rise in  $P_{Na}$  due to enhanced delivery of filtrate to the distal nephron and/or due to eliminating decreased EABV-mediated release of vasopressin is still present. Therefore, the patient was given DDAVP to prevent a water diuresis, and water restriction was imposed. Hemodynamic status,  $P_K$ ,  $P_{Na}$ , arterial pH, arterial  $P_{CO_2}$ , brachial venous  $P_{CO_2}$ , and  $PHCO_3$  were monitored closely.

What is the basis for the metabolic acidosis? Because the  $P_{anion\ gap}$  was not increased, despite the high value for the  $P_{Alb}$ , the metabolic acidosis was judged not to be due to a gain of acids. In fact, the initial diagnosis was type I or classic dRTA which was thought to explain the metabolic acidemia, urine pH of 6.0, and hypokalemia. Calculation of the  $U_{osm\ gap}$  (90 mOsmol/kg  $H_2O$ ), however, revealed a high

urinary concentration of  $\text{NH}_4^+$  (45 mmol/L). Furthermore, the  $\text{U}_{\text{NH}_4}/\text{U}_{\text{Cr}}$  was about 15 (mmol/mmol) and the rate of excretion of creatinine in this patient was estimated to be 10 mmol/day (based on his body weight); hence, the rate of excretion of  $\text{NH}_4^+$  was estimated to be around 150 mmol/day. Therefore, the basis for the patient's hyperchloremic metabolic acidosis was not dRTA.

The anion that was excreted with  $\text{NH}_4^+$  was not  $\text{Cl}^-$ , so diarrhea was not the cause of the hyperchloremic metabolic acidosis. Hence, the patient had an acid gain–type of metabolic acidosis, with a high rate of excretion of its anion in the urine. Because the  $\text{P}_{\text{anion gap}}$  was not elevated, these new anions were more likely to have entered the urine via secretion in the PT, akin to *p*-aminohippurate anions. The major chemical in glue is toluene; it is converted to benzoic acid by cytochrome P450 in the liver, and benzoic acid is in turn conjugated to the amino acid glycine to form hippuric acid.<sup>55</sup>  $\text{H}^+$  titrate  $\text{HCO}_3^-$  resulting in metabolic acidosis, whereas the hippurate anions are actively secreted by PT, so their concentration is very low in plasma and very high in the urine. The rate of excretion of hippurate anions exceeds the rate of excretion of  $\text{NH}_4^+$  in the urine because a limited amount of  $\text{NH}_4^+$  can be made in the PT, so hippurate anions are excreted along with  $\text{Na}^+$ , resulting in the indirect loss of  $\text{NaHCO}_3$  ( $\text{HCO}_3^-$  loss due to titration by added  $\text{H}^+$ ;  $\text{Na}^+$  loss due to excretion in the urine with hippurate anions) and contraction of the EABV. The low EABV, via activation of the RAAS, leads to high levels of aldosterone in plasma, causing electrogenic reabsorption of  $\text{Na}^+$  in the ASDN and thereby a high rate of secretion of  $\text{K}^+$ , leading to hypokalemia.

#### CLINICAL CASE 14: DETERMINE THE CAUSE OF HYPERCHLOREMIC METABOLIC ACIDOSIS

A 23-year-old woman with SAO was referred for assessment of hypokalemia. Her physical examination was unremarkable. The laboratory results in plasma and a spot urine sample are summarized in the following table. The pH and  $\text{PCO}_2$  are from an arterial blood sample, whereas all other blood findings are from a venous plasma sample. The urine was glucose-free.

| Parameter                                | Blood | Urine      |
|------------------------------------------|-------|------------|
| Arterial pH                              | 7.35  | 6.8        |
| Arterial $\text{PCO}_2$ , mm Hg          | 30    | —          |
| $\text{Na}^+$ , mmol/L                   | 140   | 75         |
| $\text{K}^+$ , mmol/L                    | 3.1   | 35         |
| $\text{Cl}^-$ , mmol/L                   | 113   | 95         |
| $\text{HCO}_3^-$ , mmol/L                | 15    | 10         |
| Anion gap, mEq/L                         | 12    | 5          |
| Osmolality, mOsm/kg $\text{H}_2\text{O}$ | 290   | 450        |
| Creatinine, mg/dL                        | 0.7   | 6.0 mmol/L |
| Urea (mmol/L)                            | —     | 220        |
| Citrate                                  | —     | Low        |

#### Questions and Discussion

What is the basis for the hyperchloremic metabolic acidosis? The patient had a low  $\text{U}_{\text{NH}_4}$  because the measured  $\text{U}_{\text{osm}}$  (450 mOsm/kg  $\text{H}_2\text{O}$ ) was very similar to the calculated  $\text{U}_{\text{osm}}$ , 440 mOsm/kg  $\text{H}_2\text{O}$ —that is,  $2(\text{U}_{\text{Na}} [75 \text{ mmol/L}] + \text{U}_{\text{K}} [35 \text{ mmol/L}]) + \text{U}_{\text{urea}}$  (220 mmol/L) +  $\text{U}_{\text{Glu}}$  (0). The rate

of excretion of  $\text{NH}_4^+$  was low because the  $\text{U}_{\text{NH}_4}/\text{U}_{\text{Cr}}$  was very low. Hence, the diagnosis was RTA.

What is the cause of the low rate of excretion of  $\text{NH}_4^+$ ?

**Urine pH.** Because the urine pH is 6.8, and urine citrate was low, the basis for the low rate of  $\text{NH}_4^+$  excretion is a low net secretion of  $\text{H}^+$  in the distal nephron (see [Flow Chart 24.16](#)).

**Assessment of distal  $\text{H}^+$  secretion.** After hypokalemia was corrected,  $\text{H}^+$  secretion in the distal nephron could be evaluated using  $\text{UPCO}_2$  during bicarbonate loading. The  $\text{UPCO}_2$  in alkaline urine was 70 mm Hg (see [Flow Chart 24.16](#)). Because the  $\text{UPCO}_2$  was unexpectedly high and a backleak of an  $\text{H}^+$ -type of defect was unlikely, it was suggested that the defect may be one of increased distal  $\text{HCO}_3^-$  secretion. In some patients with SAO, another mutation in the  $\text{Cl}^-/\text{HCO}_3^-$  exchanger causes it to be targeted abnormally to the luminal membranes of  $\alpha$ -intercalated cells. The  $\text{UPCO}_2$  would be high with bicarbonate loading due to the distal secretion of  $\text{HCO}_3^-$  by alkaline intercalated cells, which increases the luminal fluid pH resulting in release of  $\text{H}^+$  from monovalent phosphate ( $\text{H}_2\text{PO}_4^-$ ) in the luminal fluid, the formation of  $\text{H}_2\text{CO}_3$ , which then dissociates to  $\text{CO}_2$  and  $\text{H}_2\text{O}$ .

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

#### KEY REFERENCES

- Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; a Problem-Based Approach*. 5th ed. Philadelphia: Elsevier; 2017.
- Zhai XY, Fenton RA, Andreassen A, et al. Aquaporin-1 is not expressed in descending thin limbs of short-loop nephrons. *J Am Soc Nephrol*. 2007;18:2937–2944.
- Kamel KS, Halperin ML. The importance of distal delivery of filtrate and residual water permeability in the pathophysiology of hyponatremia. *Nephrol Dial Transplant*. 2012;27:872–875.
- Carlotti AP, Bohn D, Mallie JP, et al. Tonicity balance, and not electrolyte-free water calculations, more accurately guides therapy for acute changes in natremia. *Intensive Care Med*. 2001;27:921–924.
- Napolova O, Urbach S, Davids MR, et al. Assessing the degree of extracellular fluid volume contraction in a patient with a severe degree of hyperglycaemia. *Nephrol Dial Transplant*. 2003;18:2674–2677.
- Kamel KS, Schriber M, Halperin ML. Renal potassium physiology: integration of the renal response to dietary potassium depletion. *Kidney Int*. 2018;93:41–53.
- Kamel KS, Halperin ML. Intrarenal urea recycling leads to a higher rate of renal excretion of potassium: an hypothesis with clinical implications. *Curr Opin Nephrol Hypertens*. 2011;20:547–554.
- Wilson FH, Disse-Nicodeme S, Choate KA, et al. Human hypertension caused by mutations in WNK kinases. *Science*. 2001;293:1107–1112.
- Halperin ML, Scheich A. Should we continue to recommend that a deficit of KCl be treated with NaCl? A fresh look at chloride-depletion metabolic alkalosis. *Nephron*. 1994;67:263–269.
- Gowrishankar M, Kamel KS, Halperin ML. Buffering of a  $\text{H}^+$  load; A “brain-protein-centered” view. *J Am Soc Nephrol*. 2007;18:2278–2280.
- Kraut JA, Madias NE. Serum anion gap: its uses and limitations in clinical medicine. *Clin J Am Soc Nephrol*. 2007;2:162–174.
- Kamel KS, Halperin ML. Acid-base problems in diabetic ketoacidosis. *N Engl J Med*. 2015;372:546–554.
- Dyck R, Asthana S, Kalra J, et al. A modification of the urine osmolal gap: an improved method for estimating urine ammonium. *Am J Nephrol*. 1990;10:359–362.
- Kamel KS, Halperin ML. An improved approach to the patient with metabolic acidosis: a need for four amendments. *J Nephrol*. 2006;65:S76–S85.
- Kamel KS, Briceno LF, Santos MI, et al. A new classification for renal defects in net acid excretion. *Am J Kidney Dis*. 1997;29:136–146.

## REFERENCES

- Kamel KS, Halperin ML. *Fluid, Electrolyte, and Acid-Base Physiology; a Problem-Based Approach*. 5th ed. Philadelphia: Elsevier; 2017.
- Zhai XY, Fenton RA, Andreassen A, et al. Aquaporin-1 is not expressed in descending thin limbs of short-loop nephrons. *J Am Soc Nephrol*. 2007;18:2937–2944.
- Kamel KS, Halperin ML. The importance of distal delivery of filtrate and residual water permeability in the pathophysiology of hyponatremia. *Nephrol Dial Transplant*. 2012;27:872–875.
- Halperin ML, Kamel KS, Oh MS. Mechanisms to concentrate the urine: an opinion. *Curr Opin Nephrol Hypert*. 2008;17:416–422.
- Carlotti AP, Bohn D, Mallie JP, et al. Tonicity balance, and not electrolyte-free water calculations, more accurately guides therapy for acute changes in natremia. *Intensive Care Med*. 2001;27:921–924.
- Kamel KS, Bichet DG, Halperin ML. Studies to clarify the pathophysiology of partial central diabetes insipidus. *Am J Nephrol*. 2001;37:1290–1293.
- Carlotti ACP, St George-Hyslop C, Guerguerian AM, et al. Occult risk factor for the development of cerebral edema in children with diabetic ketoacidosis: possible role for stomach emptying. *Ped Diabetes*. 2009;10:522–533.
- Hoorn EJ, Carlotti AP, Costa LA, et al. Preventing a drop in effective plasma osmolality to minimize the likelihood of cerebral edema during treatment of children with diabetic ketoacidosis. *J Pediatr*. 2007;150:467–473.
- Naplova O, Urbach S, Davids MR, et al. Assessing the degree of extracellular fluid volume contraction in a patient with a severe degree of hyperglycaemia. *Nephrol Dial Transplant*. 2003;18:2674–2677.
- Kamel KS, Magnier PO, Ethier JH, et al. Urine electrolytes in the assessment of extracellular fluid volume contraction. *Am J Nephrol*. 1989;9:344–347.
- Miller TR, Anderson RJ, Linas SL, et al. Urinary diagnostic indices in acute renal failure: a prospective study. *Ann Intern Med*. 1978;89:47–50.
- Weisberg LS. Pseudohyponatremia: a reappraisal. *Am J Med*. 1989;86:315–318.
- Ware JS, Wain LV, Channavajhala SK, et al. Phenotypic and pharmacogenetic evaluation of patients with thiazide-induced hyponatremia. *J Clin Invest*. 2017;127:3367–3374.
- Cheema-Dhadli S, Chong CK, Kamel KS, et al. An acute infusion of lactic acid lowers the concentration of potassium in arterial plasma by inducing a shift of potassium into cells of the liver in fed rats. *Nephron Physiol*. 2012;120:p7–p15.
- Kamel KS, Schriber M, Halperin ML. Renal potassium physiology: integration of the renal response to dietary potassium depletion. *Kidney Int*. 2018;93:41–53.
- Leviel F, Hubner CA, Houillier P, et al. The Na<sup>+</sup>-dependent chloride-bicarbonate exchanger SLC4A8 mediates an electroneutral Na<sup>+</sup> reabsorption process in the renal cortical collecting ducts of mice. *J Clin Invest*. 2010;120:1627–1635.
- Carlisle E, Donnelly S, Ethier J, et al. Modulation of the secretion of potassium by accompanying anions in humans. *Kidney Int*. 1991;39:1206–1212.
- Sansom SC, Welling PA. Two channels for one job. *Kidney Int*. 2007;72:529–530.
- Welling PA, Chang YP, Delpire E, et al. Multigene kinase network, kidney transport, and salt in essential hypertension. *Kidney Int*. 2010;77:1063–1069.
- Welling PA, Ho K. A comprehensive guide to the ROMK potassium channel: form and function in health and disease. *Am J Physiol Renal Physiol*. 2009;297:F849–F863.
- Kamel KS, Halperin ML. Intrarenal urea recycling leads to a higher rate of renal excretion of potassium: an hypothesis with clinical implications. *Curr Opin Nephrol Hypertens*. 2011;20:547–554.
- Berl T, Rastegar A. A patient with severe hyponatremia and hypokalemia: osmotic demyelination following potassium repletion. *Am J Kidney Dis*. 2010;55:742–748.
- Lin SH, Lin YF, Halperin ML. Hypokalaemia and paralysis. *QJM*. 2001;94:133–139.
- Lin SH, Lin YF. Propranolol rapidly reverses paralysis, hypokalemia and hypophosphatemia in thyrotoxic periodic paralysis. *Am J Kidney Dis*. 2001;37:620–624.
- Wu KL, Cheng CJ, Sung CC, et al. Identification of the causes for chronic hypokalemia: importance of urinary sodium and chloride excretion. *Am J Med*. 2017;130:846–855.
- Wilson FH, Disse-Nicodeme S, Choate KA, et al. Human hypertension caused by mutations in WNK kinases. *Science*. 2001;293:1107–1112.
- Hoorn EJ, Walsh SB, McCormick JA, et al. The calcineurin inhibitor tacrolimus activates the renal sodium chloride cotransporter to cause hypertension. *Nat Med*. 2011;17:1304–1309.
- Schambelan M, Sebastian A, Rector FC Jr. Mineralocorticoid-resistant renal hyperkalemia without salt wasting (type II pseudohypoaldosteronism): role of increased renal chloride reabsorption. *Kidney Int*. 1981;19:716–727.
- Harris AN, Grimm PR, Lee HW, et al. Mechanism of hyperkalemia-induced metabolic acidosis. *J Am Soc Nephrol*. 2018;29(5):1411–1425.
- Nishida H, Sohara E, Nomura N, et al. Phosphatidylinositol 3-kinase/Akt signaling pathway activates the WNK-OSR1/SPAK-NCC phosphorylation cascade in hyperinsulinemic db/db mice. *Hypertension*. 2012;60:981–990.
- Halperin ML, Scheich A. Should we continue to recommend that a deficit of KCl be treated with NaCl? A fresh look at chloride-depletion metabolic alkalosis. *Nephron*. 1994;67:263–269.
- Scheich A, Donnelly S, Cheema-Dhadli S, et al. Does saline “correct” the abnormal mass balance in metabolic alkalosis associated with chloride depletion in the rat? *Clin Invest Med*. 1994;17:448–460.
- Rubin SI, Sonnenberg B, Zettl R, et al. Metabolic alkalosis mimicking the acute sequestration of HCl in rats: bucking the alkaline tide. *Clin Invest Med*. 1994;17:515–521.
- DeMars C, Hollister K, Tomassoni A, et al. Citric acidosis: a life-threatening cause of metabolic acidosis. *Ann Emerg Med*. 2001;38:588–591.
- Zalunardo N, Lemaire M, Davids MR, et al. Acidosis in a patient with cholera: a need to redefine concepts. *QJM*. 2004;97:681–696.
- Gowrishankar M, Kamel KS, Halperin ML. Buffering of a H<sup>+</sup> load; A “brain-protein-centered” view. *J Am Soc Nephrol*. 2007;18:2278–2280.
- Halperin ML, Kamel KS. Some observations on the clinical approach to metabolic acidosis. *J Am Soc Nephrol*. 2010;21:894–897.
- Kraut JA, Madias NE. Serum anion gap: its uses and limitations in clinical medicine. *Clin J Am Soc Nephrol*. 2007;2:162–174.
- Emmett M. Anion-gap interpretation: the old and the new. *Nat Clin Pract Nephrol*. 2006;2:4–5.
- Feldman M, Soni N, Dickson B. Influence of hypoalbuminemia or hyperalbuminemia on the serum anion gap. *J Lab Clin Med*. 2005;146:317–320.
- Kamel KS, Cheema-Dhadli S, Halperin FA, et al. Anion gap: may the anions restricted to the intravascular space undergo modification in their valence? *Nephron*. 1996;73:382–389.
- Stewart PA. Modern quantitative acid-base chemistry. *Can J Physiol Pharmacol*. 1983;61:1444–1461.
- Rastegar A. Clinical utility of Stewart’s method in diagnosis and management of acid-base disorders. *Clin J Am Soc Nephrol*. 2009;4:1267–1274.
- Kamel KS, Halperin ML. Acid-base problems in diabetic ketoacidosis. *N Engl J Med*. 2015;372:546–554.
- Shull PD, Rapoport J. Life-threatening reversible acidosis caused by alcohol abuse. *Nat Rev Nephrol*. 2010;6:555–559.
- Simpson DP. Control of hydrogen ion homeostasis and renal acidosis. *Medicine (Baltimore)*. 1971;50:503–541.
- Weiner ID, Verlander JW. Renal ammonia metabolism and transport. *Compr Physiol*. 2013;3:201–220.
- Dyck R, Asthana S, Kalra J, et al. A modification of the urine osmolal gap: an improved method for estimating urine ammonium. *Am J Nephrol*. 1990;10:359–362.
- Kamel KS, Halperin ML. An improved approach to the patient with metabolic acidosis: a need for four amendments. *J Nephrol*. 2006;65:S76–S85.
- Goldstein MB, Bear R, Richardson RMA, et al. The urine anion gap: a clinically useful index of ammonium excretion. *Am J Med Sci*. 1986;292:198–292.
- Richardson RMA, Halperin ML. The urine pH: a potentially misleading diagnostic test in patients with hyperchloremic metabolic acidosis. *Am J Kidney Dis*. 1987;10:140–143.
- Kamel KS, Briceno LF, Santos MI, et al. A new classification for renal defects in net acid excretion. *Am J Kidney Dis*. 1997;29:136–146.

51. Halperin ML, Goldstein MB, Haig A, et al. Studies on the pathogenesis of type I (distal) renal tubular acidosis as revealed by the urinary PCO<sub>2</sub> tensions. *J Clin Invest.* 1974;53:669–677.
52. Roscoe JM, Goldstein MB, Halperin ML, et al. Effect of amphotericin B on urine acidification in rats: implications for the pathogenesis of distal renal tubular acidosis. *J Lab Clin Med.* 1977;89:463–470.
53. Kaitwatcharachai C, Vasuvattakul S, Yenchitsomanus P, et al. Distal renal tubular acidosis in a patient with Southeast Asian ovalocytosis: possible interpretations of a high urine PCO<sub>2</sub>. *Am J Kidney Dis.* 1999;33:1147–1152.
54. Simpson DP. Citrate excretion: a window on renal metabolism. *Am J Physiol.* 1983;244:F223–F234.
55. Carlisle EJJ, Donnelly SM, Vasuvattakul S, et al. Glue-sniffing and distal renal tubular acidosis: sticking to the facts. *J Am Soc Nephrol.* 1991;1:1019–1027.